

TOP^{of} MIND

CRACKS IN PRIVATE CREDIT



After over a decade of rapid growth and relative calm, private credit has come under pressure as high-profile defaults, concerns about inflated valuations, and substantial exposure to a software industry vulnerable to AI disruption have fueled a surge in redemption requests. Could these stresses trigger a systemic crisis? And even if not, could they dampen the outlook for private credit? Oaktree's Howard Marks, Ares' Michael Arougheti, Marathon's Bruce Richards, and GS' Amanda Lynam agree current private credit stresses don't pose systemic risk. But they somewhat disagree on private credit's near-to-medium term outlook. Arougheti and Lynam aren't particularly worried about software exposure and expect credit fundamentals to remain healthy as long as the economy holds up, while Richards is more concerned

about software risk and sees a painful default cycle ahead. But all three remain optimistic about private credit's long-term growth, while Marks expects more cautious investors but a healthier investment environment to ultimately emerge.



The structure of [direct lending investment] vehicles introduces serious concerns around pricing and liquidity... [but] I don't see the same systemic exposure here as was the case in the run-up to the GFC.

- Howard Marks

While defaults will inevitably rise as the cycle matures and some managers will underperform, that doesn't mean the entire credit market will tip over, despite a handful of attention-grabbing defaults.

- Michael Arougheti

3% of the HY bond market is software loans versus 13% of the broadly syndicated loan market and a whopping 23% of the private credit market... So, it's hard to argue that there isn't too much software risk in direct lending.

- Bruce Richards

A key feature of private credit is asset-liability matching. Most capital is locked up, and even in retail-focused structures, liquidity is limited by design. That significantly reduces the risk of a 'run on the bank.'

- Amanda Lynam



WHAT'S INSIDE

INTERVIEWS WITH:

Howard Marks, Co-Founder and Co-Chairman, Oaktree Capital Management

Michael Arougheti, Co-Founder and CEO, Ares Management

Bruce Richards, CEO, Chairman, and Co-Founder, Marathon Asset Management

Amanda Lynam, Chief Credit Strategist, Goldman Sachs

BUSINESS DEVELOPMENT COMPANIES: NAVS VS. PRICES

Shamshad Ali, GS Credit Strategy Research

PRIVATE CREDIT: LIMITED MACRO SPILLOVERS

Manuel Abecasis, GS US Economics Research

ALTS MANAGERS: PRIVATE CREDIT OPPORTUNITIES

Alexander Blostein and Anthony Corbin, GS Equity Research

BANKS AND PRIVATE CREDIT: THE INTERSECTION

Richard Ramsden, GS Equity Research

...AND MORE

Allison Nathan | allison.nathan@gs.com

Jenny Grimberg | jenny.grimberg@gs.com

Ashley Rhodes | ashley.rhodes@gs.com

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

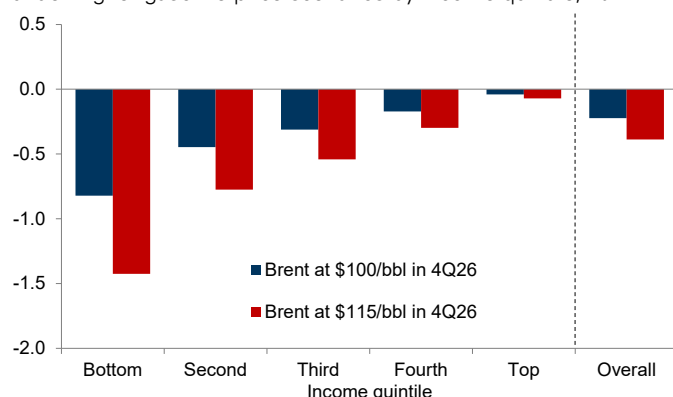
- We raised our Dec 2026 headline/core PCE inflation forecasts to 3.4%/2.6% yoy (from 3.1%/2.5%) given the even higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; longer energy supply disruptions that result in even higher gasoline prices would likely lead to a larger drag on consumer spending.
- Fed Chair transition; we don't expect a major reduction in the Fed's balance sheet under Kevin Warsh as Fed Chair.
- US labor market, which we expect to soften this year due to weaker growth, higher energy costs, and modestly greater reluctance to hire for jobs that can be done by AI.

US: higher gasoline prices bite consumers

Effective change in real PCE spending vs. GS baseline in 2026 under higher gasoline price scenarios by income quintile, %



Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

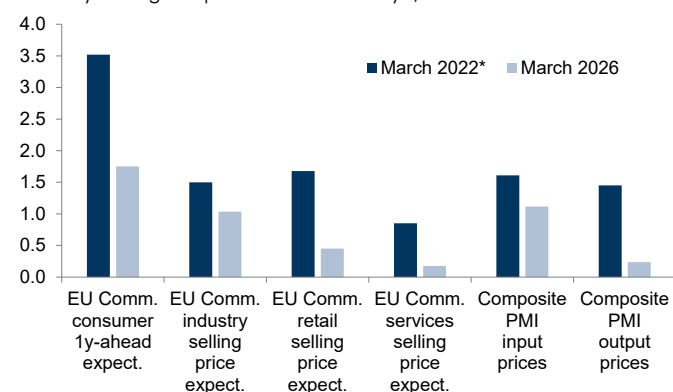
- We raised our 1Q26 UK GDP growth forecast after better-than-expected February GDP data, bringing our full-year 2026 growth forecast to 1.0% (Q4/Q4, vs. 0.6% before).

Datapoints/trends we're focused on

- Iran war; we think the impact of the current energy price shock on European manufacturing should be notably smaller and less persistent than the 2022/23 energy crisis.
- ECB policy; we expect the ECB to hike rates in June and September but view whether the ECB hikes twice or remains on hold this year as a close call.
- UK employment growth, which remains weak.

Euro area: a less painful energy shock than 2022

Monthly change in price-related surveys, std. dev.



*Normalized for size of the change in equally-weighted avg. of oil and gas prices.

Source: Haver Analytics, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

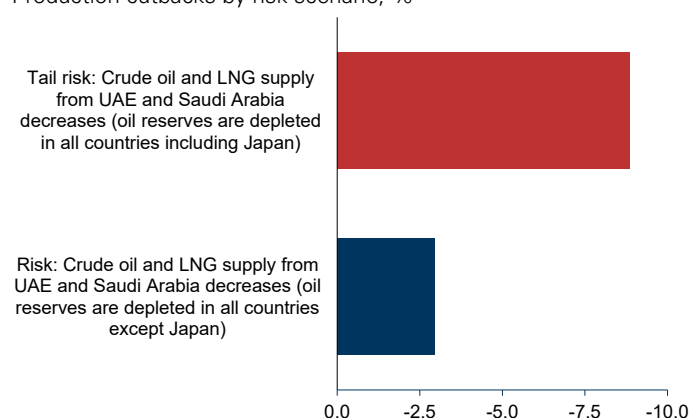
- We lowered our 2026 Japan real GDP growth forecast to 0.3% (from 0.4%) and raised our 2027 core CPI forecast to 2.3% (from 2.1%) given the higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; even with sufficient oil reserves, we think Japan faces potentially significant indirect oil supply constraints as shortages abroad impact global supply chains.
- BoJ policy; we continue to expect the next rate hike in July, though uncertainty about the timing remains high.
- Shunto* spring wage negotiations show a marginally softer base pay rise from 2025, but still high in the mid-3% range.
- Japanese consumer confidence, which deteriorated sharply.

Japan: potential oil shortage impacts

Production cutbacks by risk scenario, %



Source: OECD, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

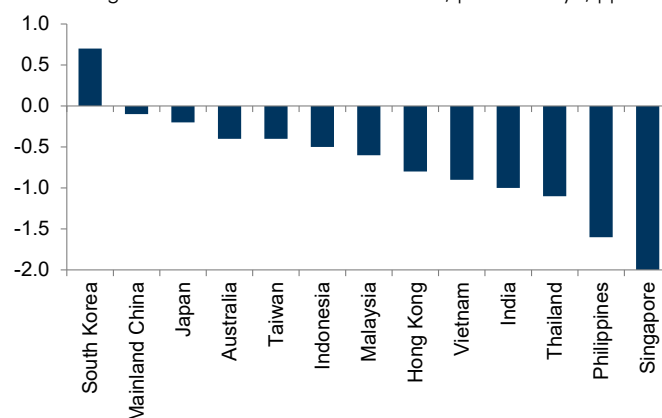
- We raised our 2026/27 China PPI forecasts to 1.2%/0.9% (from 1%/0.6%) given the higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; we now largely expect lower growth and higher inflation across Asia-Pacific from the energy shock, but the impact on inflation should be less persistent than in 2022.
- China growth and activity, which has generally been better than expected this year, implying little urgency for large near-term policy stimulus and adding conviction to our low expectations for stimulus at the April Politburo meeting.
- Hungary elections; opposition party Tisza's decisive win in the parliamentary elections should strengthen EU-Hungary relations and could pave the way for Euro membership.

Asia-Pacific: energy shock weighs on growth

Net change in GS 2026 real GDP forecasts, past 60 days, pp



Source: Goldman Sachs GIR.

Cracks in private credit

After over a decade of rapid growth and relative calm, the private credit market has come under pressure as several high-profile defaults, concerns about potentially inflated valuations, and especially substantial exposure to a software industry vulnerable to AI disruption have fueled a surge in redemption requests (see pg. 16)—drawing [some comparisons](#) to the setup heading into the Global Financial Crisis (GFC). So, could these private credit stresses trigger a systemic crisis? And even if not, could they meaningfully impair the outlook for a market that has become a key source of financing for companies and an increasingly important allocation for investors?

We ask several long-time industry watchers for their perspectives: Howard Marks, Co-Founder and Co-Chairman of Oaktree Capital Management, Michael Arougheti, Co-Founder and CEO of Ares Management, Bruce Richards, CEO, Chairman, and Co-Founder of Marathon Asset Management, and Amanda Lynam, GS Chief Credit Strategist.

Arougheti and Lynam argue that the concerns around private credit stresses seem overdone as they are concentrated in non-traded business development companies (BDCs)—the primary direct lending investment vehicle for retail investors (see pg. 8)—which comprise only a modest portion of the private credit market. And they explain that despite the current worries about elevated redemption requests, the limited liquidity of non-traded BDCs (which Lynam describes as a “feature, not a bug”, though Marks finds some retail investors’ apparent disappointment in this feature problematic) substantially reduces the likelihood of crisis-inducing fire sales.

This, combined with the lack of leverage in private credit funds, well-secured and substantially-protected bank exposure to private credit (see pg. 19 for GS Head of the Financials Group Richard Ramsden’s deep-dive into the evolving intersection between banks and private credit), and GS US Economist Manuel Abecasis’ finding that even a sharp rise in private credit default rates would only moderately impact the flow of new lending to the private sector, all agree: the current private credit stresses don’t pose systemic risk, with comparisons between these stresses and the lead-up to the GFC unwarranted.

But even if private credit doesn’t pose meaningful risk to the financial system or the economy, will the industry itself still experience significant pain? Arougheti and Lynam don’t think so. Arougheti sees no indication of a turn in the credit cycle in private or bank credit portfolios or credit card charge-off data, and doesn’t view last fall’s attention-grabbing defaults as particular cause for worry because they were relatively small exposures and, importantly, mainly funded by bank-led asset-based facilities rather than private credit managers.

Lynam also points out that underlying credit fundamentals appear relatively healthy and expects them to remain so as long as economic growth holds up, which GS Senior Global Credit Strategist Shamshad Ali finds is consistent with the signals from the micro signposts he’s watching (namely, what tends to happen to loans flagged as “PIK-at-restructuring” and the size and profile of loans coming due).

But even if underlying fundamentals appear healthy today, will that remain the case given direct lending’s significant exposure to the software industry, which is ripe for AI disruption?

Arougheti, Lynam, and Marks aren’t particularly concerned, with Arougheti explaining that the software loans in private credit portfolios are senior secured, high up in the capital structure, and extended to large, cash-generative firms, Lynam noting that software exposures are not a uniquely private credit problem and that managers have long underwritten AI disruption risk into their investment criteria, and Marks arguing that even significant losses on software loans wouldn’t be “existential” for most senior lenders.

Richards, however, is very worried about software risk. He points out that 23% of the private credit market is software loans, which he says is an excessive figure by any standard, with leverage at the fund level and especially the company level further magnifying the risk from this exposure. And he doesn’t take comfort from private lenders’ high position in the capital structure given that a significant part of the cushion that sits below these lenders has already been eaten through. So, he expects “incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15% and double-digit default rates in each of those years” and software loan loss rates in the 70-100% range. Lynam, for her part, takes a more nuanced view as she thinks software exposure is more likely to drive performance dispersion among managers than broad-based deterioration in the fundamental outlook, and, absent a sharp downturn in economic growth—which is not GS economists’ base case—doesn’t expect private credit realized losses to rise materially above the longer-term average.

So, the distribution of views on private credit’s near-to-medium outlook is wide. But perhaps the bigger question is whether the recent developments will dent the longer-term growth outlook for private credit, which has become a key financing tool for many firms and growing share of investors’ portfolios. Perhaps surprisingly, Richards doesn’t think so. While he expects the coming software correction to be painful, he expects direct lending—the largest segment of private credit (see pg. 9)—to emerge from it “bigger and stronger than ever” as the correction results in the return of manager discipline.

Arougheti and Lynam are also relatively optimistic, arguing that the current stresses will likely shift market share within private credit rather than slow the industry’s overall growth. In particular, they expect capital to shift from retail-focused strategies to institutional strategies like opportunistic credit, which GS US Financials analysts Alexander Blostein and Anthony Corbin argue should benefit alternative asset managers, though they expect dispersion among them depending on their exposure to the wealth channel.

But Marks is somewhat more cautious, expecting retail investors to be more deliberate and circumspect “now that the tide may have begun to go out on private asset vehicles sold to the public and some of their flaws have been exposed.” That said, he also believes that experiencing a full credit cycle may ultimately lead to the emergence of a healthier investment environment for direct lending and private credit more broadly.

Allison Nathan, Editor

Email: allison.nathan@gs.com
Tel: 212-357-7504
Goldman Sachs & Co. LLC



Interview with Howard Marks

Howard Marks is Co-Founder and Co-Chairman of Oaktree Capital Management. Below, he argues that while nothing is inherently wrong with direct lending, structural aspects of direct lending *investment vehicles* can be problematic for some investors, but don't pose systemic risk.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: You've characterized the rapid expansion of direct lending as a "gold rush." What precipitated that, and is the rapid growth a cause for concern?

Howard Marks: Direct lending—private loans made by non-bank lenders for mid-size private equity deals—has become an integral aspect

of the private equity business, which runs on using borrowed money to lever up equity returns on the purchase of companies. As post-Global Financial Crisis (GFC) regulations curtailed banks' ability to lend for levered transactions in an effort to make banks less risky, private lenders stepped in to fill the void. And because only a few private lenders existed and many private equity firms wanted to borrow money, private lenders were able to demand high interest rates and safety through strong covenants.

But, as in any gold rush, this success led to a flood of new entrants. Available capital increased relative to demand for it, which meant that lenders couldn't demand as much interest or safety, so the lender's advantage was somewhat competed away. But because demand for private equity was growing, demand for financing from private lenders to lever up transactions was also increasing, allowing private lenders to substantially grow their assets under management and generate sizable fees. So, everyone was motivated to hang up a shingle as a direct lender. The result was rapid growth in the direct lending market, which didn't exist in its current form before 2011 and is approaching \$2tn today, with the vast majority of firms engaged in direct lending today not in business 20 years ago. And, as in every upsurge in financial activity that brings heated competition, some participants may have engaged in risky behavior to get in on the gold rush.

Allison Nathan: Is direct lending inherently problematic?

Howard Marks: There's nothing inherently wrong with lending money. But lending can be done well or poorly. Lending carefully to the right companies at an interest rate that appropriately compensates for the default risk can be a profitable and prudent activity. But the opposite is also true. In 1978, I was fortunate to participate in the start of the high yield (HY) bond market. Prior to 1977, a company rated below investment grade (IG) could not issue bonds. The innovation of the time was the concept that a non-IG company should be able to issue bonds if it offered enough interest to compensate for the risk of default. Most people considered this imprudent, calling them "junk" bonds. But, as with early direct lending, because few investors were eager to lend to these companies, those that did could demand high interest rates and strong safety. We have now been in the business of lending money to non-IG companies for 48 years and have produced good returns

with manageable risk. So, there's nothing inherently safe or risky about making a loan, even a non-IG loan.

Allison Nathan: But are there inherent flaws in direct lending investment vehicles?

Howard Marks: The structure of such vehicles introduces serious concerns around pricing and liquidity. On pricing, if investors choose to take their money out of a non-traded business development company (BDC)—the vehicle for retail participation in direct lending—they are paid at the fund's net asset value (NAV), which third parties estimate periodically. But is the NAV accurate? Is it fair? A perception exists that NAVs for private assets—both equity and credit—may be overvalued, which would align with most participants' incentive to show good performance. But since these loans don't trade, it's difficult to determine their fair price. This is problematic because if the NAV is too high, investors that leave are paid too much at the expense of the remaining investors. And if the NAV is too low, investors that leave are penalized unfairly.

It's important to recognize that the whole concept of determining a fair and accurate price is more complicated than many appreciate. For example, should a loan be valued at its "fundamental value"—which is hard to estimate—or at the price an investor could sell it for today, or at the price they could get a month from now after prospective buyers have had the opportunity to conduct due diligence, or at the price an investor would have to pay to buy an equivalent loan? Along the same lines, should a \$1bn loan be valued at the price an investor could get for the whole \$1bn, or for the first \$10mn? Those prices might differ. The point is that an asset does not "have a price", which is true even for traded assets; it depends on market conditions and the size of the trade.

So, valuing assets fairly is a hornet's nest, and especially so for private assets that have no reference point. Private asset investors must therefore accept that there is no correct answer, and no *one* answer, when it comes to a loan's fair value. The only way to avoid the inherent unfairness that the uncertainty around NAVs introduces would be to pay investors leaving the fund the proceeds the manager receives when they sell off the departing investor's share of the assets. But that's not how these vehicles are structured.

Allison Nathan: You also mentioned that liquidity of direct lending vehicles is a concern. Why?

Howard Marks: To avoid managers having to sell assets at unrealistically low prices that would be injurious to their investors and potentially to their business, private credit vehicles are designed either to prohibit investors from taking back their capital, as is the case with traded BDCs, or to allow them to take back only a certain percentage—typically 5% of NAV—per quarter, as is the case with non-traded BDCs. This has recently proven problematic because redemption requests for some non-traded BDC funds have exceeded the 5% cap.

You can imagine how it feels to invest in something and then not be able to get your money back. It can be a problem when a vehicle that invests in private assets promises liquidity.

Allison Nathan: But isn't the limited liquidity of non-traded BDCs a feature—not a flaw—given that such liquidity constraints are embedded in the vehicle's design?

Howard Marks: It's absolutely true that investors participating in a fund that limits redemptions shouldn't be surprised when full liquidity isn't available. But "should" is one of the most interesting words in the English language. If people shouldn't expect liquidity, why are they disappointed when they don't have it? The answer is either that they weren't adequately informed, or they were well informed but didn't understand or take cognizance of the terms. The risk of either of these is highest for retail investors, raising the question of whether these vehicles are suitable for the retail base. But there's nothing inherently wrong with retail investor participation if they know what they're buying.

Allison Nathan: A key concern around private lending is the potential for large redemption requests to force fire sales. But don't the liquidity constraints of non-traded BDCs—well understood or not—prevent that?

Howard Marks: Forced sales are indeed a common cause of crises; I've been around long enough to see several of them. But the redemption limits on non-traded BDCs should prevent such a crash in this market. When people say non-traded BDCs' liquidity limitations are doing their job, that's what they mean. But those limitations are still unpleasant for investors.

Allison Nathan: So, you don't see similarities between what's happening in direct lending today and the GFC?

Howard Marks: No, for three reasons. First, the GFC occurred largely because too much money was invested in highly levered residential mortgage-backed securities (RMBS) that were built out of subprime mortgages, which ultimately had no substance. A great deal of money went into such mortgages because structuring RMBS produced a lot of fees; thus, mortgage brokers were paid to make loans, but not necessarily good loans. Nobody seemed to wonder about the wisdom of lending money to unsuitable borrowers. Today, no inherent lack of substance exists in direct loans to creditworthy companies.

Second, another key contributor to the GFC was banks' high leverage. Structuring RMBS from subprime mortgages was so profitable that the institutions making and securitizing the loans had incentive to keep doing so. But that required somebody to take the riskiest tranche, which often ended up being the institution itself. So, highly levered financial institutions—namely, banks, which were levered up to 32x—held the riskiest tranches. And a 3% decline in the value of the portfolio at an institution that is 32x levered could render it insolvent. Conversely, private credit funds are not highly levered. Some are unlevered, and some are slightly levered—up to perhaps 2x at most. So, they don't face the same type of meltdown risk.

Third, private credit funds aren't interdependent. The years leading up to the GFC saw substantial use of derivatives and hedges contracted in with other institutions—one example is credit default swaps—that created what we called

"counterparty risk", setting up for cascading losses if one institution failed. That type of interconnectedness doesn't exist among private credit funds. So, I don't see the same systemic exposure here as was the case in the run-up to the GFC.

Allison Nathan: Even if there's no systemic risk, are concerns about direct lending's exposure to software firms that are vulnerable to AI disruption warranted, especially since many of these firms were levered up in buyouts?

Howard Marks: It's cause for concern for the funds that loaned money for those buyouts. But the downside is probably manageable in most cases because, if a first-lien loan is made to a software company, the value of the company must decline enough to wipe out the equity as well as the junior lenders before that loan is jeopardized. That type of value destruction is not impossible, but it shouldn't be assumed. And even if the first-lien lender lost half their money in such a case, that wouldn't be too bad in a diversified portfolio. If a quarter of the investments are in software and they all incur such losses, then the lender would lose half their money in a quarter of their investments, or 12.5% of the portfolio. And even if they're levered 1:1, that would amount to a 25% loss, which still isn't existential. There's an old saying that in times of crisis, all correlations go to one. In other words, everybody panics and sees all outcomes as equally probable and equally bad. But losing a quarter of your capital because the software industry loses most of its value isn't abysmal. It's just bad for the fund's investors. And it doesn't jeopardize the financial system.

Allison Nathan: But what could default rates look like?

Howard Marks: I don't make economic predictions. So, I can't say how high default rates could climb. And it's important to note that in private credit, what constitutes a default—or even whether lenders will report defaults—is unclear. Lenders have an incentive to grant the borrower an extension rather than report a default, and nobody may be enforcing the reporting of defaults. Particularly for that reason, I wouldn't put much stock in estimated default rates.

Allison Nathan: Given everything we've discussed, how do you expect the direct lending market to evolve from here?

Howard Marks: It's important to first note that the growth of the private equity industry, which is the main borrower, has slowed. So, the growth in demand for these loans will also likely slow. And, as Warren Buffett famously said, it's only when the tide goes out that you find out who's been swimming naked. Now that the tide may have begun to go out on private asset vehicles sold to the public, and some of their flaws have been exposed, I would expect retail investors to be a bit more deliberate and circumspect. In a recent memo, I ended with a quote from Bob O'Leary, Oaktree's co-CEO, who says direct lending will be okay, but it might take a credit cycle to get there. Only after investors have experienced a full cycle in which it swings from being too easy to borrow money to being too hard are they likely to really achieve the understanding of both the positives and the negatives that enables them to make better investment decisions the next time around. That's ultimately a healthier investment environment, and it may be what lies ahead for direct lending and private credit more broadly. But learning that lesson isn't always pleasant.

Interview with Michael Arougheti

Michael Arougheti is Co-Founder and CEO of Ares Management. Below, he discusses the evolution of private credit and his perspective on the recent concerns around it.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: First, what is private credit, and how has it evolved?

Michael Arougheti: Private credit is self-originated loans that institutional money managers make to private companies, asset owners, and portfolios of assets and cash flows.

The largest segment of private credit is direct lending to small- and medium-sized businesses largely owned by private equity sponsors. It also includes asset-based credit, real estate and infrastructure lending, opportunistic credit, and credit secondaries, where investors trade existing private credit interests. These strategies are generally funded in the sub-investment grade (IG) market. Over the past decade, IG private credit has also grown as insurers have merged with institutional asset managers. So, private credit is a diverse market spanning many strategies, sectors, and geographies.

While there is a perception that the private credit market developed largely in response to post-Global Financial Crisis (GFC) regulation, its origins actually trace back to the 1980s, when the advent of the high yield (HY) market revealed a whole swath of sub-IG borrowers whose funding needs banks and public markets were not meeting. Since then, several structural and capital innovations have expanded the market, including the growth of business development companies (BDCs)—closed-end investment vehicles that lend to small- and mid-sized private companies—increased insurer participation post the GFC, and, most recently, the rise of retail investment. So, private credit has experienced a decades-long evolution.

Allison Nathan: The growth in private credit in recent years seems to be a source of worry. Should it be?

Michael Arougheti: No. Private markets have grown because banks and public markets have shrunk, leaving them increasingly unable to meet firms' capital needs. The number of US banks has fallen by roughly half over the last 25 years. The share of bank balance sheet assets that represent commercial and industrial (C&I) loans—a proxy for loans made in private credit portfolios—has also declined from ~30% in the 1980s to <15% today. At the same time, the US has half as many publicly traded companies today as 25 years ago, with only ~2,100 firms listed on the NYSE and 3,400 on the NASDAQ, a tiny fraction of the 18 million companies in the US today.

Private credit growth also hasn't been extraordinary. Private market AUM has grown at a 13% compound annual growth rate (CAGR) over the past decade, comparable to private credit's 14% CAGR, with private credit growing only slightly faster than the overall private market over the last three years as the Silicon Valley Bank (SVB) crisis disrupted private equity flows and capital shifted toward opportunistic credit. Capital raised in institutional private credit has also declined for the last four years. And direct lending dry power remains modest relative to private equity dry powder, at roughly \$300-400bn vs.

over \$1tn. More broadly, leveraged credit—non-IG private credit, bank C&I loans, and HY and leveraged loans—as a share of US nominal GDP has remained steady at around 23% since 2015. So, private credit growth hasn't been excessive, and many parts of the market still don't have enough capital to meet demand.

Allison Nathan: Even if the growth doesn't warrant worry, aren't private credit markets taking risks that banks won't?

Michael Arougheti: No. Private markets are taking credit risks that banks *can't*. It's underappreciated that banks are 10-15:1 leveraged while private credit funds are only 0-1:1 leveraged. Leverage amplifies returns but also risks. Banks' inherent asset-liability mismatch and regulatory capital frameworks mean that whole swaths of loans don't make sense for banks to extend. Private credit hasn't created demand for credit that otherwise didn't exist. It's simply an alternate capital provider.

Allison Nathan: That said, are the worries about several high-profile defaults warranted, or overblown?

Michael Arougheti: They're overblown. Downturns in credit cycles are a normal part of the business. But the last two cycles—which spanned the GFC and the pandemic—were super cycles dotted with a series of mini cycles around the 2013 Taper Tantrum, 2022 Gilt Crisis, and 2023 SVB collapse. So, even some long-time investors have never experienced a credit cycle downturn and so are reacting strongly to defaults even though fundamentals remain generally solid. Looking at private and bank credit portfolios and credit card charge-off data, nothing indicates a turn in the credit cycle. While defaults will inevitably rise as the cycle matures and some managers will underperform, that doesn't mean the entire credit market will tip over, despite a handful of attention-grabbing defaults last fall that were relatively small exposures and, notably, mainly funded by bank-led asset-based facilities.

Allison Nathan: Could the increased use of payment-in-kind (PIK) features—which allow borrowers to preserve cash by shifting interest payments to the loan's principal balance—be masking deteriorating credit quality?

Michael Arougheti: To answer this, PIK should be thought about in three baskets. First is "PIK at the outset", where the instrument is underwritten with PIK because the lender wants to earn a higher return and the borrower wants more flexibility in deploying their cash. That's a structural choice, not a distress measure. Second is "good PIK", where a performing company wants to divert cash from debt service to growth. This is attractive to institutional investors because the borrower is willing to pay the lender more, driven by a belief that the return from redeploying cash into the business exceeds the cost of capital. Third is "bad PIK", where a company can't service its debt because operations are deteriorating. Bad PIK is rising, which warrants watching. But not all PIK is bad PIK.

Allison Nathan: How concerning is private credit exposure to the software industry amid worries about AI disruption?

Michael Arougheti: Some concern may be warranted, but the fears around software exposure also seem overdone. Software names in private credit portfolios are typically the biggest US private software companies, which are largely institutionally equity-backed. These loans are senior secured, high up in the capital structure, and extended to firms with high free cash flow. For context, at Ares, software is ~9% of our private credit portfolio. These are large businesses, with average EBITDA of ~\$350mn. The average loan-to-value is ~38%, meaning 62% of the capital structure sits below us, owned by brand-name private equity sponsors. So, the bulk of the risk is in the private equity market. These firms are also generally growing at low double-digit rates. And the loans typically have around 3.5 years of remaining maturity, which aligns with the timeframe of expected revenues. These structural mitigants provide meaningful downside protection. That's not to say some software firms won't be disrupted and losses won't occur. But for every firm AI disrupts, another will likely benefit.

Allison Nathan: Fueling these concerns is a perception that private credit lacks transparency. What's your response?

Michael Arougheti: The question is, transparency to *whom*? Institutional investors in private credit funds have perfect transparency into what they own and how it's performing on a monthly basis, along with access to company management and private credit portfolio teams. That's far more transparency than public market investors have given reporting mandates and regulatory limitations on public company disclosures. Third-party firms also evaluate and value institutionally-managed private credit portfolios, and public accounting firms review that. The banks providing leverage to private credit portfolios also frequently review the underlying credit exposures. It's true that the Fed doesn't have perfect transparency into private credit portfolios because they aren't bank-chartered or highly leveraged. Even so, the Fed meets with us regularly and sees our portfolios through the regulation of our bank partners. So, I struggle with the idea that private credit isn't transparent.

Allison Nathan: How concerning is the recent surge in redemption requests, and could they trigger fire sales?

Michael Arougheti: It's not particularly concerning owing to the structure of non-traded BDCs, which is where redemption requests are concentrated. Key aspects of these structures are often misunderstood. First, so-called "redemption gates" aren't actually gates. Before non-traded BDCs existed, investors could either have daily liquidity in a traded BDC or illiquidity in a 10-12 year private commingled fund. The innovation of non-traded BDCs was to allow investors to own illiquid assets while providing a structured path to liquidity of 5% of net asset value per quarter or 20% per year, with the structure self-amortizing so investors could be made whole without forced asset liquidations. The 5%/20% limits weren't pulled out of thin air but rather designed to track the weighted average life of the underlying loan portfolio. So, meeting the contractual 5%/20% is exactly what these structures were designed to do.

Second, no "asset-liability mismatch" or "run on the bank" dynamic exists in non-traded BDCs. Unlike a bank with short-term deposits and long-term assets, these funds align liquidity with the five-year amortization profile of their loans. They also typically hold 20-30% of assets in liquid securities like loans,

bonds, or high-grade investments that can be sold to generate cash, and managers can draw on loan facilities to meet redemption requests. Those features exist precisely to avoid having to liquidate private assets below their intrinsic value.

Third, scale matters. The non-traded BDC market is ~\$300bn, representing less than 10% of the nearly \$4tn private credit market. Even if every non-traded BDC met its maximum redemption limit, that would amount to ~\$5bn of loan sales out of their tradable bucket every quarter, which is small relative to the ~\$85bn of loans that trade quarterly in the syndicated loan market. More broadly, adequate dry powder exists in the market to resolve private credit portfolios in an orderly fashion without any disruption to pricing.

Lastly, the current volatility in non-traded BDCs is a technical, not fundamental, issue driven by narrative and positioning. Case in point: our non-traded BDC fund saw no credit underperformance and zero non-accruals as of 4Q2025, and over 95% of the investors didn't request their money back in the most recent redemption period. That said, some managers have and may continue selling assets to demonstrate confidence in their marks to the retail base, which may benefit their franchise but not necessarily be in stakeholders' best interest. So, unwinds could occur. But that would play out over years and in an orderly fashion.

Allison Nathan: What do you make of comparisons between private credit worries and the lead-up to the GFC?

Michael Arougheti: The setups differ dramatically. First, while the Iran conflict has generated significant uncertainty, the US economy remains relatively strong. Second, credit today is owned by a much broader swath of investors and with much less asset-liability mismatch, leading to a significant decline in volatility in the traded credit markets. If anything, private credit has become a buffer to market volatility and risk as opposed to a conduit for them. Third, banks' exposure to BDCs and private credit today is ~0.8% compared to their nearly 20% exposure to subprime mortgages in the lead-up to the GFC. The assets themselves are also fundamentally less risky. A CLO security rated single-A or higher—which is where banks typically invest—has never defaulted. And as regulators seek more transparency, they will discover that banks have offloaded risk and now hold only the safest loans. Lastly, regulators have much more transparency into the financial system today than in the lead-up to the GFC, when off-balance sheet structures and subprime loan portfolios weren't visible to them.

Allison Nathan: Will recent developments impact the growth outlook for private credit?

Michael Arougheti: The recent developments will likely shift market share within private credit rather than slow growth overall. The current narrative is creating noise, but also opportunity for those able to cut through that noise. In practice, that likely means slower growth in non-traded BDCs, with capital reallocating toward strategies like opportunistic credit, credit secondaries, and direct lending. Over time, as investors see how these structures actually performed, capital will likely flow back. In the meantime, private credit will continue to find ways to fund the markets' needs, creating significant opportunity for managers who are structured the right way.

Private credit, explained

WHAT IS PRIVATE CREDIT?

Private credit refers to debt financing provided by non-bank financial institutions directly to companies. The rise of private credit was largely fueled by stricter regulatory frameworks following the Global Financial Crisis, including Dodd-Frank and Basel III, that forced traditional banks to tighten lending standards and reduce risk. This created a funding gap that non-bank lenders stepped in to fill. Private credit's growth was further accelerated by a prolonged period of low interest rates, which encouraged both institutional and retail investors to seek the higher yields and illiquidity premiums available through private debt.

Borrowers

Borrowers in the private credit space tend to be small and mid-sized businesses, although the borrower profile has evolved from a niche group of small businesses into a broad spectrum of corporates and entities. Borrowers can also be the portfolio companies of private equity sponsors who borrow funds to finance acquisitions or buyouts.

Lenders

The major non-bank lenders in the private credit space include alternative asset managers (many of which also participate in private equity), specialized credit funds, business development companies (BDCs)—which are SEC-regulated investment vehicles—and occasionally large individual investors.

Banks

Banks have increasingly become involved in the private credit ecosystem in recent years, largely as indirect participants, with banks acting as key capital providers to private credit funds. Banks also provide leverage to private credit funds through subscription lines and fund financing. Some banks have also increasingly taken a more proactive role in the market by establishing their own internal private credit capabilities.

Investors

Investors in private credit funds include institutional investors like pension funds, family offices, endowments, insurance companies, as well as retail investors (see pg. 9 for a more detailed ownership breakdown).



What types of private credit strategies exist?



1 Direct Lending

Direct lending is the most common (and 'traditional') form of private credit. Private lenders provide financing directly to corporate borrowers. Such loans are typically secured and involve corporate recourse, offering borrowers flexible solutions tailored to their specific financial needs.



2 Asset-Based Finance (ABF)

Asset-based private credit refers to a type of private credit in which lending is backed by specific collateral such as real estate, equipment, or financial assets. The loan amount depends on the value of the pledged assets, which can reduce credit risk for lenders. ABF is common among businesses with strong asset bases but limited cash flow or short-term financing needs.



3 Mezzanine, Second-Lien Debt, and Preferred Equity

These three forms of credit, known as junior capital, provide borrowers with subordinated debt. These instruments are not secured by assets and rank below senior loans for repayment in the event of a default or bankruptcy. Junior capital often comes with equity "kickers" like warrants. Borrowers use mezzanine financing to fund growth or acquisitions when senior debt alone isn't sufficient, and investors accept higher risk in exchange for higher returns.



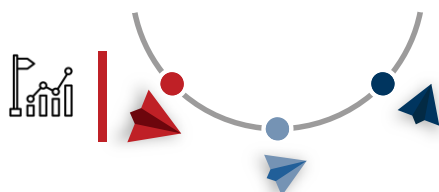
4 Distressed and Opportunistic Credit

These are investments in companies under financial stress or those that target market dislocations. Distressed debt refers to loans or bonds from companies experiencing financial difficulties or facing bankruptcy. Private credit investors in this space seek to profit by purchasing the debt at a discount and working toward a turnaround or restructuring that increases its value.

How can investors access private credit?

BDCs are one way investors, particularly retail investors, can access private credit strategies. They are investment vehicles that invest in small and mid-sized private businesses. They generate income mainly through interest payments and distribute income to shareholders, typically in the form of dividends. The BDC market is comprised of three main structures that offer a different level of accessibility to investors.

Publicly traded BDCs trade daily on major exchanges and are available to any investor with a brokerage account.



Non-traded BDCs (perpetual/evergreen funds) are often sold through RIAs/broker-dealers and don't trade on an exchange. Instead, they provide liquidity through periodic share repurchase programs. Most non-traded BDCs limit total repurchases to 5% of net asset value (NAV) per quarter, though the cap is discretionary. If redemption requests exceed 5%, the fund has the ability to repurchase shares on a pro-rata basis.

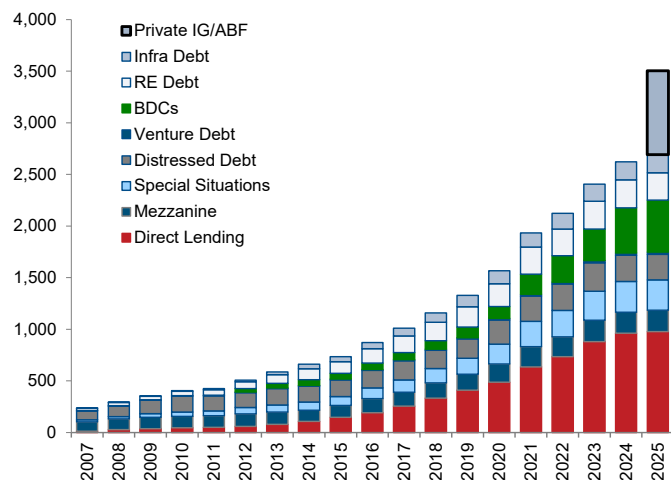
Private BDCs (drawdown funds) are closed-end investment vehicles where investors commit a capital amount upfront but provide the cash only when the fund manager calls for it to fund specific loans. These funds generate returns through income and capital appreciation, with profits distributed to investors as the fund matures. Capital is locked up for the life of the fund, and access is restricted to accredited investors and institutions.

Source: Federal Reserve, Blackstone, Carlyle, KKR, various news sources, Goldman Sachs GIR.

Sizing the private credit market

The private credit market has grown significantly since 2007 to around \$3.5tn in assets under management (AUM) today, with direct lending accounting for the largest slice of the market

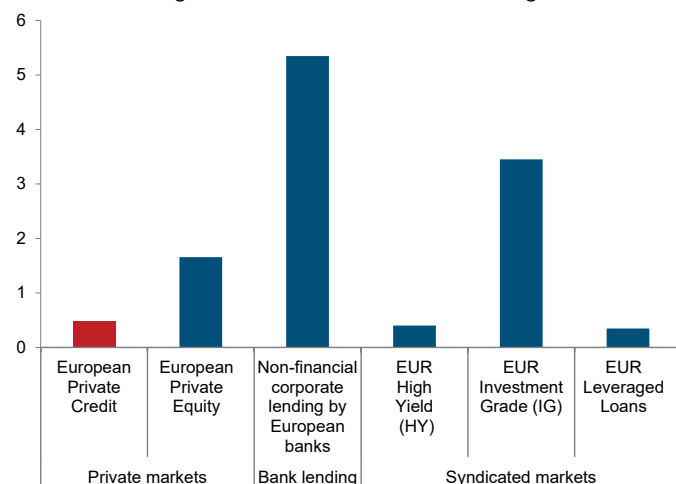
Global private lending AUM by strategy, \$bn



Source: PitchBook LCD, Preqin, LSEG Data and Analytics, BDC Collateral, Goldman Sachs GIR.

...and the European private credit market is relatively tiny

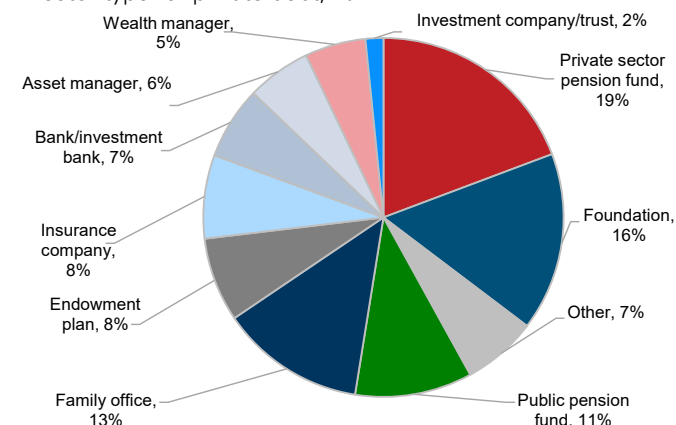
Amt outstanding across various Euro area lending markets, €tn



Source: Preqin, ECB, Haver, Bloomberg, PitchBook LCD, Morningstar, GS GIR.

...with the ownership structure of private credit funds largely skewed toward liquidity-agnostic investors

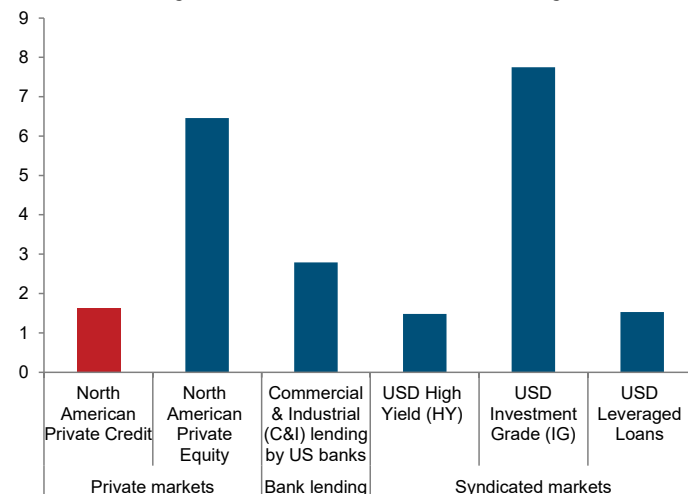
Investor type for private debt, %



Source: Preqin, Goldman Sachs GIR.

Despite this growth, the private credit market is modest in size relative to other North American corporate financing markets...

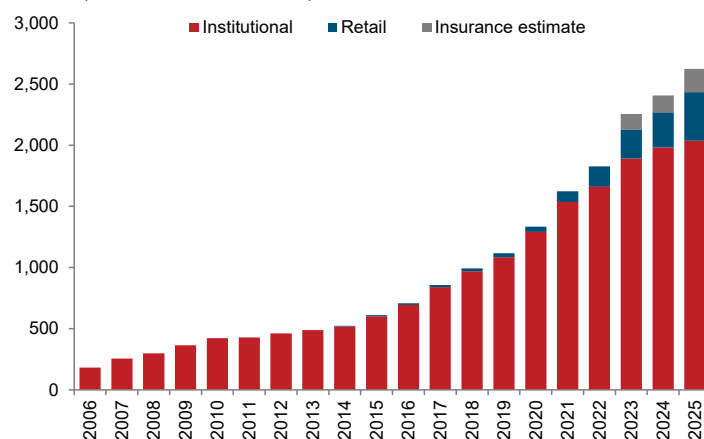
Amt outstanding across various N. American lending markets, \$tn



Source: Preqin, Cliffwater, Federal Reserve Board, Haver, Bloomberg, PitchBook LCD, Morningstar/LSTA, Goldman Sachs GIR.

While retail investors are playing a growing role in the private credit market, institutional investors continue to dominate...

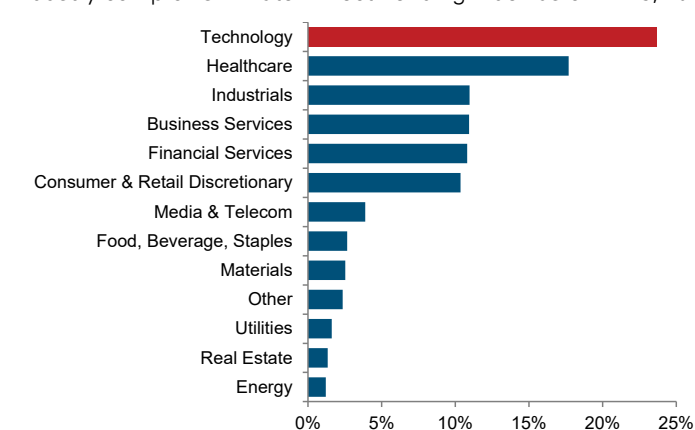
Global private credit AUM by investor channel, \$bn



Source: PitchBook LCD, Goldman Sachs GIR.

At nearly a quarter of loans, the technology sector accounts for the largest part of the US direct lending market

Industry comp of Cliffwater Direct Lending Index as of YE25, %



Source: Cliffwater, Goldman Sachs GIR.

Special thanks to GS Credit Research team (Amanda Lynam and Shamshad Ali) and Asset Managers Equity Research team (Alexander Blostein and Anthony Corbin) for data. Exhibit 1 originally appeared in Alexander Blostein's October 19, 2025 Americas Asset Managers note. Exhibits 2-6 appeared in the Credit team's April 13, 2026 and July 29, 2025 Private Credit Monitors (see latest Monitor for more details on chart calculations).

Interview with Amanda Lynam

Amanda Lynam is Chief Credit Strategist at Goldman Sachs. Below, she explains the role of private credit in the financial system and why many concerns about the space are misplaced.



Jenny Grimberg: What role does private credit play in the financial system, and how has that evolved?

Amanda Lynam: Private credit is one of the three core pillars of the financing continuum alongside the bank lending and public debt markets. While the private credit market was initially reserved for smaller companies

that lacked access to the public debt markets, it has significantly expanded since the Global Financial Crisis (GFC), and especially since the pandemic, to include borrowers who could otherwise access public debt markets or, in many cases, have already done so and are refinancing their debt into the private credit market. So, over time, the boundaries between the three markets have become increasingly fluid.

Jenny Grimberg: So, it's not the case that companies who use the private credit market do so simply because they lack alternatives?

Amanda Lynam: No; that's a common misperception. Firms with access to public markets could still opt for private credit for several reasons. First and foremost is certainty of execution, especially during periods of market volatility when public markets may be less accessible. For example, amid recession fears in 2022-2023, the leveraged loan and high yield (HY) bond markets weren't widely available for lower-rated borrowers due to lower risk appetite and structural constraints. In particular, the predominant buyers in the leveraged loan market are Collateralized Loan Obligations (CLOs), rating-sensitive vehicles that can only hold a small amount of triple-C rated debt, which restricts their lending to lower-rated firms in times of economic uncertainty. Private credit doesn't face such constraints, making it a more reliable source of capital in choppy markets.

Second is the availability of customized and bespoke financing solutions. Public markets tend to work best for borrowers that fit neatly within standardized credit frameworks. For unique companies—like those in early growth stages, with high customer concentration risk, or with binary success outcomes—private credit can offer tailored solutions the public markets may not be comfortable providing. And third is higher barriers to entry in the public markets. As companies stay private for longer, public market deal sizes have become extraordinarily large. The average deal size in the HY market today is well above \$700mn. This makes public issuance less practical for firms seeking relatively modest amounts of capital.

Jenny Grimberg: Is private credit riskier than public credit?

Amanda Lynam: Not necessarily. The additional spread that private market financing commands over public market financing—known as the illiquidity premium—is often mischaracterized as purely a reflection of higher risk. In many instances, it just reflects compensation for the fact that private credit lenders commit to owning a loan for the entirety of its life. Borrowers are also paying extra for the customization and

greater certainty of execution that private credit offers. And while private credit tends to serve smaller or less mature companies than public debt markets, lenders manage this risk through thorough due diligence, the use of covenants, and active oversight, including the ability to renegotiate deal terms early if a borrower's financial health deteriorates.

Jenny Grimberg: The current worries about private credit are focused on non-traded business development companies (BDCs). What are these structures, and how important is their role within private credit?

Amanda Lynam: Non-traded, or evergreen, BDCs are vehicles for retail investors to access private credit strategies. They offer a faster path to investment with lower minimum denominations and more simplified tax reporting than institutional private credit vehicles. Non-traded BDCs represent ~15% of "traditional" private credit AUM (per PitchBook LCD), which doesn't include asset-based finance or investment grade (IG) private credit. So, despite the significant focus on them, non-traded BDCs comprise only a small fraction of the private credit universe. The vast majority of private credit AUM is institutional capital that is locked up in long-term vehicles with fundamentally different structures.

Jenny Grimberg: So, are the concerns around the surge in non-traded BDC redemption requests overdone?

Amanda Lynam: Let me first say that despite the attention-grabbing headlines about major private credit firms capping redemptions at 5% of net asset value (NAV) per quarter, these redemption "gates" aren't new, even if some investors seemed previously unaware of them, and are a feature, not a bug, of these structures. Such features are designed to safeguard investors' capital by preventing managers from being forced to sell assets at depressed prices to meet short-term liquidity demands. The current stresses in these retail vehicles don't exist in the institutional market, because institutional credit vehicles were never structured to offer redemptions in the first place. So, concerns about these redemptions, and potential for spillover to other vehicles, seems misplaced.

Jenny Grimberg: Are recent defaults in the space cause for concern?

Amanda Lynam: While private credit default rates have risen somewhat, these statistics can paint a misleading picture of fundamental health. Some people compare defaults in private credit to defaults in public credit, but they are not actually comparable. Maintenance covenants—periodic financial metrics borrowers must meet—are much more common in private credit than in the broadly syndicated leveraged loan market. Because of that, a covenant breach in private credit can be classified as a covenant default, which counts in the default statistics. By contrast, roughly 92% of the broadly syndicated leveraged loan market is "covenant-lite", meaning the loans don't have regular maintenance covenants. Consequently, the first sign of default is often a distressed exchange or a missed payment, which can quickly translate into a monetary loss.

For that reason, realized losses are a better gauge of the health of the market than default rates. As of end-2025—the latest available data—realized losses in private credit, which we track through the Cliffwater Direct Lending Index that comprises over 20k US middle-market loans representing \$550bn in AUM, were around 65bp, below the long-term average of 100bp and generally in line with the public credit markets. And the yield on that index is 9.9%, so the income is more than offsetting the losses. So, underlying market fundamentals appear relatively healthy, and we expect that to remain the case as long as the economic backdrop remains resilient.

Jenny Grimberg: Concerns have also grown about the use of payment-in-kind (PIK), which allows borrowers to defer cash interest payments, and non-accrual loans, which are delinquent in payment. What are you actually observing, and how worrying is it?

Amanda Lynam: PIK is another metric that requires nuance because not all PIK is created equal. “Good” PIK is PIK embedded at loan origination, often for high-growth borrowers seeking to preserve liquidity for investment or acquisitions and comes with guardrails around duration and magnitude. “Bad” PIK is PIK introduced after origination when a borrower is experiencing financial stress. “Bad” PIK modestly rose after the Fed embarked on its hiking cycle, but not to outsized levels. And total PIK has remained around 7–8% of BDC total income over the past several quarters, well below the 2020 peak. Non-accrual rates have also remained in a tight range over that period. So, directional indicators of potential credit deterioration aren’t sending concerning signals right now.

Jenny Grimberg: How concerned should investors really be about private credit’s software exposure?

Amanda Lynam: It’s something to watch, but software exposures aren’t uniquely a private credit problem—while software is the largest sector in private credit, it’s also the largest in the broadly syndicated leveraged loan market. So, if AI does disintermediate a significant share of software firms, both the public and private markets would be impacted. In such an environment, private credit could actually be well-positioned to deploy dry power because the public markets would likely be choppy, and opportunistic credit strategies could see capital inflows. AI disruption risks have also long been on many private credit managers’ radars even if the pace of disruption may be occurring faster than they anticipated. So, their investments have focused on software firms with moats like proprietary data or deeply embedded customer relationships. I largely view the software exposure in both private and liquid markets as a catalyst for additional dispersion, not widespread market disruption.

Jenny Grimberg: Given all that, how likely is private credit to pose a systemic risk?

Amanda Lynam: Very unlikely, in my view. While there is absolutely potential for performance dispersion and losses at the manager and fund levels, the tentacles through which those losses could amplify stress across the broader financial system seem largely contained. A key feature of private credit is asset-liability matching. Most capital is locked up, and even in retail-focused structures, liquidity is limited by design. That

significantly reduces the risk of a “run on the bank.” Leverage at the fund level is also modest, particularly compared to the banking system, and the private credit market is small relative to other parts of the financial ecosystem. In the US, the Bloomberg USD IG Corporate Bond Index is roughly \$8tn in size while private equity is six times larger than private credit. More broadly, unlike during the GFC—when hidden correlations and excessive leverage in structured securities triggered a domino effect—losses in private credit would largely be borne by the investors in those funds. While that would undoubtedly be painful, it wouldn’t be systemic.

It’s also worth mentioning that rather than being a source of systemic risk, private credit has the potential to help stabilize markets in times of stress. For example, during the 2023 US regional banking crisis, private lenders stepped in to fill some of the financing void when banks and public debt markets stepped back. As a result, corporate default rates didn’t rise nearly as much as they otherwise could have during that period.

Jenny Grimberg: But couldn’t the interconnectedness between private credit firms and banks act as a transmission channel for systemic shocks?

Amanda Lynam: Loans to non-depository financial institutions comprise 15% of overall bank lending, much less than other types of loans, and private credit accounts for only around a quarter of that, with the loans largely held by large banks with over \$10bn in assets. A recent Federal Reserve report found that bank loans to private credit vehicles have low loss rates. So, the interconnectedness isn’t a particular concern.

Jenny Grimberg: Conversely, could upcoming regulatory changes like Basel III Endgame boost lending capacity in the banking system and slow the growth of private credit?

Amanda Lynam: A regulatory rollback is unlikely to materially dampen private credit’s growth given the size of the addressable market. And for banks with excess capital, it is not obvious to us that it would be redeployed into middle-market lending given the many competing uses for capital, including shareholder returns, M&A, market-making activities, or lending to private credit firms themselves. So, banks and private credit firms will likely remain strategic partners, with banks providing the origination networks and leverage, while private credit firms serve as the ultimate holders of the loans and associated risk.

Jenny Grimberg: More broadly, will the current stresses in private credit dent its growth?

Amanda Lynam: Noise will likely persist over the near term, which may slow growth in some areas of the market, particularly retail-focused strategies. However, over the longer term, these challenges will likely prompt private credit to deploy capital in areas beyond traditional direct lending, such as opportunistic credit, mezzanine financing, and private asset-based finance. And the North American private credit market is far from saturated, especially given the ongoing AI-driven capex buildout. Private credit also has significant room to capture a larger share of overall lending in Europe and Asia, where financing remains relatively bank-dependent. So, while growth is unlikely to be linear, the scope for private credit to continue expanding over the longer term remains very much intact.

Interview with Bruce Richards

Bruce Richards is CEO, Chairman, and Co-Founder of Marathon Asset Management. Below, he argues that private credit is meaningfully overexposed to the software sector, which he expects will result in significant defaults and losses, though not a systemic crisis.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Are the concerns around software exposure in private credit portfolios justified, or overblown?

Bruce Richards: Let's put perspective on software exposure within private credit by focusing on three key numbers: 3, 13, and 23. 3% of the high yield bond market is software

loans versus 13% of the broadly syndicated loan market and a whopping 23% of the private credit market. For the top 10 private lenders, that number climbs to 26%. Given the roughly 20 investable industry sectors that exist, exposing 26% of a portfolio to software is excessive, especially considering that software is just a subset of the Technology, Telecom, and Media (TMT) sector. So, it's hard to argue that there isn't too much software risk in direct lending.

Allison Nathan: But concerns about technology concentration risk are nothing new and are as much a feature of the public equity market as the private market. So, why the particular concern here?

Bruce Richards: Comparing portfolio exposure to market cap weight is a mischaracterization of how portfolio managers manage risk, whether it be in a bank lending, asset-based or direct lending program. The cardinal rules of risk management for any lender running a diversified portfolio are to limit the maximum amount of exposure to any one company and industry. And software companies with anywhere from \$10 to \$100mn of revenues represent only around 1% of all private companies in the US. That number is around 7% across the public markets, whether in the NASDAQ, S&P 500, or Russell 2000. So, software exposure is outsized any way you slice it.

Allison Nathan: But many of these software companies are well-established and profitable. And while agentic AI technology may disrupt some of them, it may enhance others. Does that give you comfort that some wins will offset some losses?

Bruce Richards: There will be tremendous wins. I'm somewhat constructive on software in the public markets because public companies have, net of cash, half of one turn of leverage or debt to EBITDA of 0.5. Many companies have strong margins, solid earnings, and substantial free cash flow, which will allow them to make the operating investment required to reposition themselves as AI-first software firms.

But many companies are not in that position, with leverage of 8 to 10 times debt to EBITDA. Those elevated leverage levels are my primary concern because they leave companies with very little to no free cash flow after debt service. In fact, many companies are actually burning cash on an operating basis after debt service. Without adequate free cashflow, these highly

leveraged software companies will have little ability to make the upfront investment necessary to transition into AI-first firms.

Allison Nathan: Some have argued that private credit concerns are overdone because these funds aren't highly leveraged. What is that argument missing?

Bruce Richards: Let me explain with a simple example. \$1bn of capital with one turn of leverage on the balance sheet amounts to \$2bn of assets. \$2bn of assets with 26% of the portfolio in software means that the software exposure on \$1bn of investor capital is \$520mn—or 52% of the capital. So, the only one turn of leverage that private credit funds or business development companies (BDCs) have still amounts to exceptionally high software exposure. At the same time, the underlying software firms typically have upwards of 8 to 10 turns of leverage in the direct lending business and a minimum of 4 to 5 turns of leverage in the broadly syndicated loan market. So, it's important to distinguish between the leverage of the fund versus the amount of debt on the specific company the fund is lending to. Both metrics are important to fully understand the risk.



The only one turn of leverage that private credit funds or BDCs have still amounts to exceptionally high software exposure. At the same time, the underlying software firms typically have upwards of 8 to 10 turns of leverage in the direct lending business."

Allison Nathan: Some have also argued that even if software companies are disrupted, private lenders' position in the capital structure protects them relatively well. What's your view?

Bruce Richards: It's true that private—and public—equity investors have more downside exposure. But they also have all of the upside exposure. There will be winners and losers, and even the best software investors in the private equity space will undoubtedly see some zeros in the 2020-2025 investment vintages. But they'll also see some companies with higher multiples than their original underwriting assumed as AI adoption drives valuations higher. So, it's possible that some equity investors will end up doing quite well overall across those vintages, even in the event of elevated defaults. Conversely, lenders can't afford the zeros because they only make par back; they don't have the upside that the private and public equity markets do.

At the same time, a substantial part of the cushion that sits below private credit in the capital structure has already been eaten through given the collapse in software valuations across

public and private markets. Valuations for mid-sized software companies are generally down 50% or more from peak levels.

And when a software company goes bankrupt, even senior loans will take a sizable hit. It's important to understand that software bankruptcies look different than bankruptcies in other industries because enterprise customers won't tolerate uncertainty. A CTO or CIO needs to know that their firm's software is reliable and can't gamble with operational or reputational issues that arise in a bankruptcy process. So, keeping a software business intact through a bankruptcy process is exceptionally challenging. As a result, while defaults in other industries typically result in recovery rates for high yield bonds, broadly syndicated loans, and direct lending of 60 to 80 cents on the dollar, recovery rates will likely prove considerably worse in software bankruptcies. Given the collapse in valuations, heavy debt loads, and substantial uncertainty about which software companies will be successful amid the AI transition—as well as likely limited capital availability until that becomes clearer—software default recovery rates will likely be in the 0 to 30 cent on the dollar range for direct lenders.

Allison Nathan: So, what do you expect for software default and loss rates?

Bruce Richards: I expect incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15% and double-digit default rates in each of those years given the valuation reset this year and likely reluctance of the broadly syndicated loan and direct lending markets to extend new credit to highly leveraged, below-investment grade software companies. Direct lending recovery rates of 0 to 30 cents would amount to 70-100% losses on those loans.

“I expect incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15%... Direct lending recovery rates of 0 to 30 cents would amount to 70-100% losses on those loans.”

Allison Nathan: Is there a historical analogue for such an industry-specific default cycle?

Bruce Richards: The energy sector's huge technological transformation in 2013-2014 owing to the rise of horizontal drilling and hydraulic fracking, which caused a significant shift in the price structure for energy, comes to mind. Energy had been the number one sector for the high yield bond and broadly syndicated loan markets, with capital rushing in. But the technological shift led to the collapse of several companies, leaving creditors with substantial losses. Debtor-in-possession (DIP) financing came in senior to broadly syndicated loans in the restructuring process, and default rates for these loans hit 15 to 20% at their peak and remained at double digits for three successive years.

While Oil & Gas and Software are very different industries, the common thread between the two periods is a technological transformation driving a change in pricing structure that highly

leveraged companies are unable to sustain, leading to exceptionally high default rates and low recovery rates. Very little capital was available in the Oil & Gas sector in the ensuing years to refinance debt other than the new capital that came in senior to existing debt during the bankruptcy process. I believe the software sector will see the same lack of capital in the coming years, especially given the elevated leverage levels already in the sector today.

Allison Nathan: Could the losses in the software sector that you expect pose systemic risk?

Bruce Richards: No. During the Global Financial Crisis (GFC), banks became ensnared given their exposure to collapsing home prices. And consumers were highly exposed both at the high end, given that the majority of high-income households' net worth resided in housing, and at the low end, given the loss of over 10mn jobs. So, banks failed, businesses were wiped out, and consumers were punished.

Today, banks are stronger than they've ever been. So, there is no systemic risk in the financial system. And there is no systemic risk stemming from consumers because consumers don't pay for software, companies do. Consumer net worth today is \$180tn—an over-threefold increase from the GFC—and consumers are benefitting from near-maximum employment, high housing prices, and record-high equity levels and valuations. And, while the AI transformation will undoubtedly result in software winners and losers, as we discussed, the sector represents only around \$300bn of capital on a \$31tn economy. So, I am not at all concerned that the troubles I foresee for the software sector will lead to a broader crisis.

“Ultimately, while I am very concerned about the coming wave of software defaults and losses, I am very constructive on direct lending.”

Allison Nathan: Even if you don't expect a broader crisis, could software losses dent private credit's growth outlook?

Bruce Richards: No. Direct lending is a critical business for capital formation that will emerge from this correction bigger and stronger than ever. It's worth noting that private credit has grown 18-fold over 18 years to \$1.8tn today and has never experienced a correction until now. This reset will result in the return of manager discipline with respect to maximum exposure levels to any one company or industry, tighter documentation, and stricter covenants. So, the next vintage will likely be very strong. And European direct lending carries materially less of the software exposure I am concerned about than its US counterpart, given that software constitutes a far larger share of both the US economy and US private equity activity than it does in Europe. Ultimately, while I am very concerned about the coming wave of software defaults and losses, I am very constructive on direct lending, which has become an essential financing solution for many companies and an integral and invaluable part of our financial system that will only continue to grow.

BDCs: NAVs vs. prices

Shamshad Ali finds that BDC portfolio fundamentals and signposts of future portfolio health look less concerning than public BDC equity prices suggest

Since the fall of 2025, business development companies (BDCs) have become a key barometer of private credit health and sentiment as retail participation in middle-market lending has grown. Concerns about software exposure in private credit funds and a surge in non-traded BDC redemption requests on the back of worries about forward returns, underwriting standards, and broader secular disruption from AI drove a significant selloff in public BDC equities since June 2025. While they have bounced off their March lows, public BDCs still trade at a nearly 20% discount to their net asset value (NAV), raising a key question: is the NAV right, or is the price?

Current BDC portfolios show some signs of deterioration in credit quality at the margin, but little evidence of broad-based distress. As such, the fundamentals don't seem to justify the price. But fundamentals are backward-looking. Prices, which are forward-looking, may be reflecting signs of trouble ahead that aren't yet baked into portfolio loan values, which tend to adjust with a lag going into credit distress. That said, while the macro outlook will hold the key to how BDC loans ultimately perform, we don't think the current signposts validate the extent of concerns embedded in the price.

BDCs are trading at a nearly 20% discount to their NAV

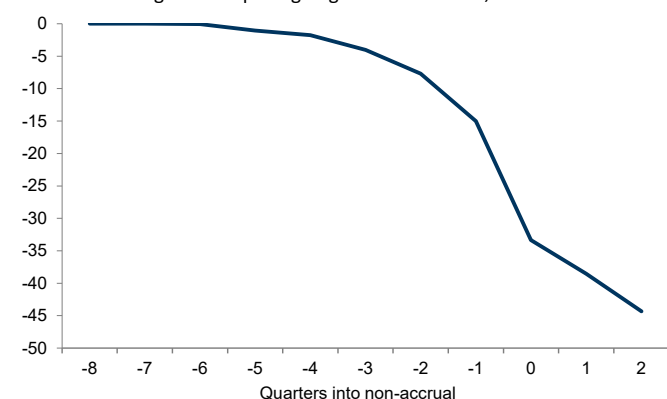
BDC share price/NAV per share, ratio



Source: Bloomberg, Goldman Sachs GIR.

Loan valuations tend to adjust with a lag going into credit distress

Cumulative change in loan price going into non-accrual, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

Little sign of broad-based stress in portfolio fundamentals...

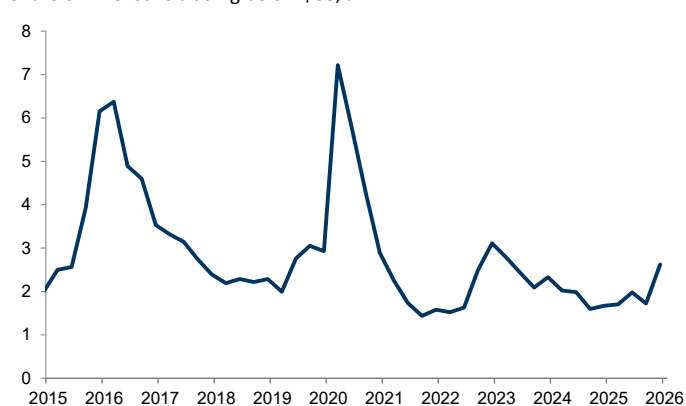
BDC portfolio fundamentals generally remained healthy through 4Q25. The share of first- and second-lien loans trading below \$80, an important indicator of credit portfolio health as a senior secured loan trading below \$80 signals credit distress, has remained relatively flat over the past few quarters, while other common measures of delinquency have moved modestly higher.

The use of payment-in-kind (PIK), which allows borrowers to defer interest payments by adding them onto the loan's principal balance rather than paying them in cash, is one such measure. PIK can be baked in at origination ("PIK-at-origination"), often when lending to growing businesses with long ramps to profitability, or structured into an existing loan if a borrower has difficulty making interest payments in cash ("PIK-at-restructuring"). While the latter is more concerning, both types of PIK have risen in recent quarters, suggesting borrowers are seeking greater flexibility.

Non-accrual loans—loans where a lender has stopped recording interest income as it's earned because it's doubtful that the interest will be collected—is another such measure. The share of such loans has risen to just below 1% of fair value, still comfortably below historical averages. Taken together, these metrics suggest incremental deterioration in BDC fundamentals, not broad-based impairment.

The share of BDC loans trading <\$80 has remained relatively flat

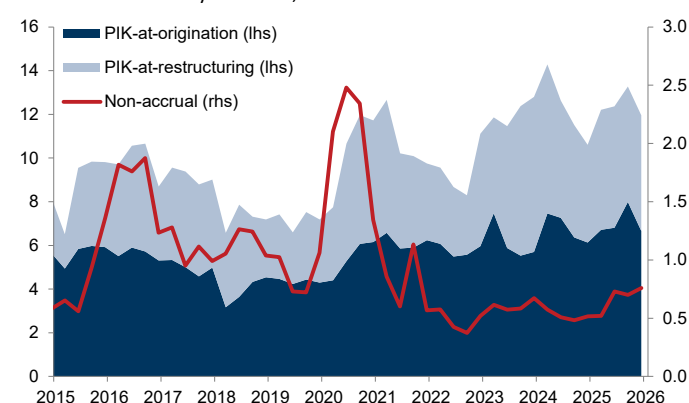
Share of BDC loans trading below \$80, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

While the share of PIK has risen, over half is at origination, and non-accruals remain below historical averages

PIK and non-accrual by fair value, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

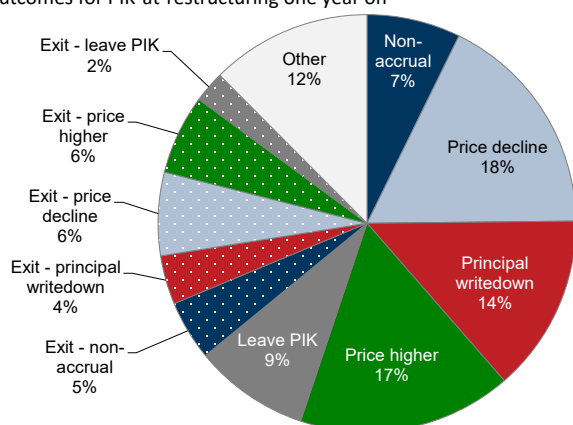
...and the current signposts aren't particularly concerning

But even if fundamentals don't look concerning today, could they in the future? While the macro outlook ultimately remains the main driver of loan health, two signposts leave us generally constructive.

Although investors often treat loans that are flagged as PIK-at-restructuring or non-accrual as reliable markers of future credit impairment, in practice the outcomes are more mixed than that characterization implies. Looking at over 3,000 loans tagged as PIK-at-restructuring, we find that 53% experienced negative outcomes, including principal writedowns, lower prices, or transition into non-accrual status, four quarters later. But over a third *improved*, either by exiting PIK status or increasing in price. So, while less favorable outcomes are more likely, negative outcomes aren't a certainty.

Loans tagged as PIK-at-restructuring tend to see negative outcomes, but not always

Outcomes for PIK-at-restructuring one year on*



*"Exit" refers to positions that left the portfolio before the four quarters and prior to their maturity month. We use the values prior to exit to categorize these loans. "Other" refers to loans which left portfolios immediately after the PIK toggle.

Source: PitchBook LCD, Goldman Sachs GIR.

Given the prospect of non-performing loans, we also consider the size and profile of loans coming due, known as the maturity wall. In our view, refinancing risk is relatively benign. Near-term maturities are relatively manageable, with loans maturing by 2027 accounting for only 10% of the total loan balance, and the loans coming due generally having higher weighted average coupons, which improves refinancing incentives given lower prevailing yields—in the absolute or relative to later maturities.

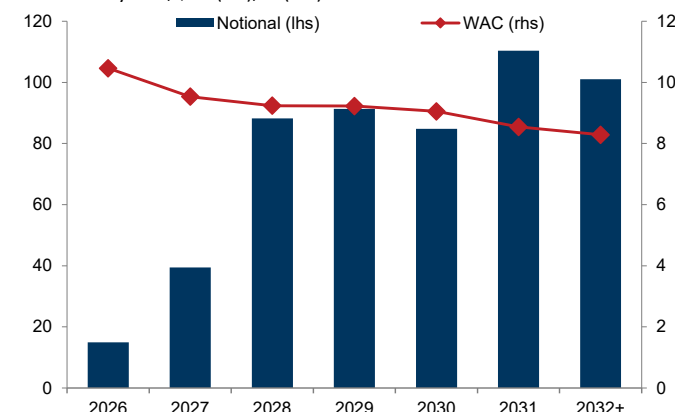
Another concern is redeployment risk—the risk that managers may not be able to reinvest the proceeds from maturing loans into new loans with comparable risk-return profiles. However, we see no indication that near-term loans are healthier than longer-maturity loans when looking at prices or delinquency status, with upcoming maturities also fairly well diversified by origination vintage.

So, while a prolonged slowdown in investment activity remains a key concern for the viability of BDCs and capital access for middle-market companies, the maturity wall does not strike us as an immediate source of stress. We also see potential for expanded deployment opportunities over the medium- to long-term, especially outside of traditional direct lending.

Taken together, we think that NAVs could be optimistic, but not overly so. We would continue to flag that dispersion across BDCs will likely increase as underwriting remains in focus.

The maturity wall is not elevated over the near term, and loans coming due generally have higher weighted average coupons

BDC maturity wall, \$bn (lhs), % (rhs)*



*Includes first- and second-lien and subordinated loans.

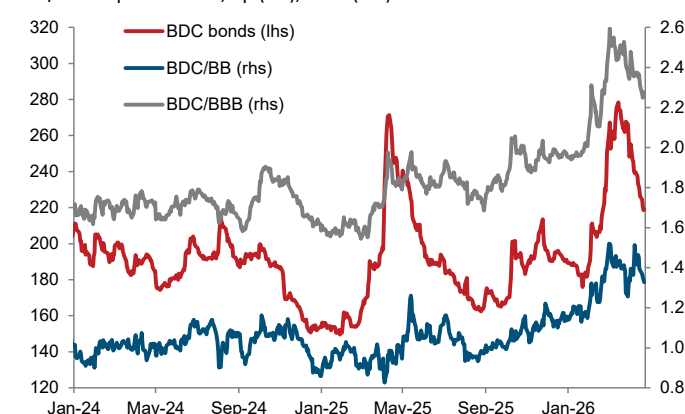
Source: PitchBook LCD, Goldman Sachs GIR.

Look to credit spreads

Looking ahead, we view BDC credit spreads as a better barometer for NAV risk than BDC equity prices, as equity prices will likely continue to reflect an elevated liquidity premium following the recent redemption surge and lower future dividend prospects for floating rate lending. BDC credit spreads appear to have turned a corner. Despite continued headline risk and a recent credit rating downgrade, secondary spreads have tightened by 60bp from their local wides in March and primary market activity has resumed. While we think the fundamentals should provide a level of support for senior capital in BDCs, we would be cautious of some of the recent exuberance. Since mid-March, the most distressed BDCs have seen their credit spreads narrow the most, by 93bp vs. only 44bp for the median bond spread. As a result, while we see value in BDC spreads at current levels, especially for investors comparing spreads to BB bonds, we think manager selection will be critical.

BDC credit spreads have tightened by 60bp since mid-March

BDC/index spread ratios, bp (lhs), ratio (rhs)



Source: iBoxx, Bloomberg, Goldman Sachs GIR.

Shamshad Ali, Senior Global Credit Strategist

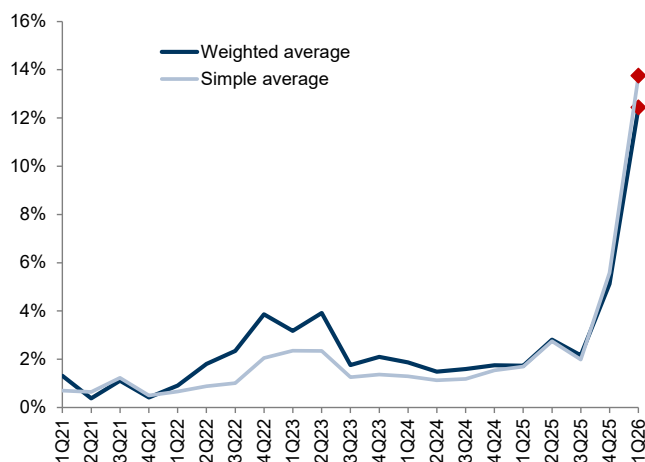
Email: shamshad.ali@gs.com
Tel: 212-902-6712

Goldman Sachs & Co. LLC

Private credit concerns, in pics

Recent private credit concerns have prompted a rise in retail investor redemption requests...

Evergreen/non-traded BDC private credit fund redemption request rates, %



Source: Company data, Goldman Sachs GIR.

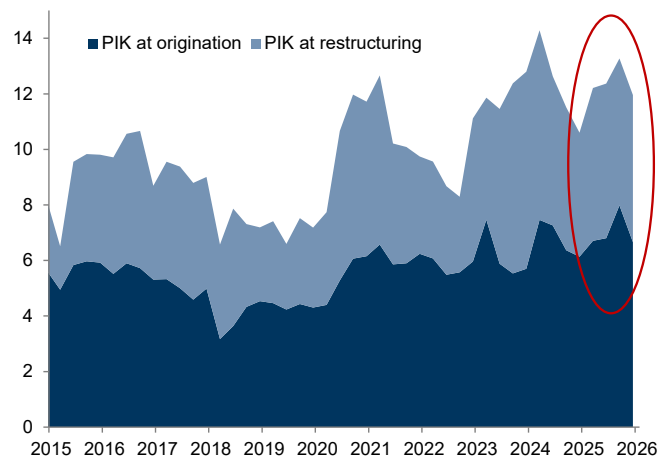
BDCs are trading at a substantial discount to their net asset value (NAV)...

BDC share price/NAV per share, ratio



Source: Bloomberg, Goldman Sachs GIR.

Payment-in-kind (PIK) usage, which can be linked to financial stress of a borrower, rose last year but not to outsized levels
Share of PIK by fair value, %

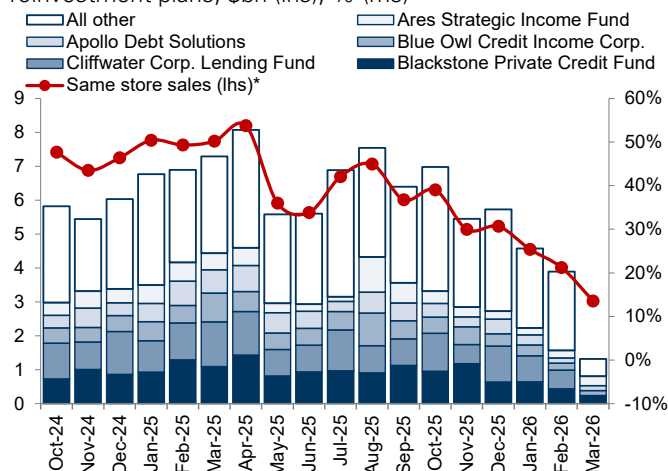


Note: Includes first- and second-lien loans in BDC portfolios. Captures data through December 31, 2025.

Source: PitchBook LCD, Goldman Sachs GIR.

...as well as a sharp reduction in gross flows into retail private credit products in recent months

Semi-liquid private credit monthly gross flows ex. dividend reinvestment plans, \$bn (lhs), % (rhs)



*Percent of prior NAV, annualized.

Source: Company data, Goldman Sachs GIR.

...and the share of underperforming private credit loans increased slightly in 2H2025

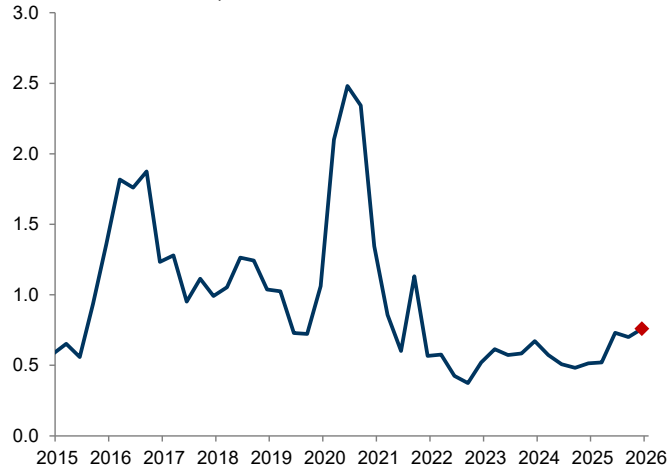
Share of underperforming private credit loans, %



Source: Company data, LSEG Data & Analytics, BDC Collateral, GS GIR.

Non-accrual rates increased only slightly and are not signaling broad-based impairment in the BDC complex

Non-accrual rates by fair value, %



Note: Includes first- and second-lien loans in BDC portfolios. Captures data through December 31, 2025.

Source: PitchBook LCD, Goldman Sachs GIR.

Special thanks to GS Global Credit Strategist Shamshad Ali and US Financials Equity Research Analyst Anthony Corbin for charts.

Private credit: limited macro spillovers

Manuel Abecasis argues that private credit stress will generate limited macro spillovers

Recent private credit concerns have driven a pickup in retail investor redemptions as well as growing worries about broader risks to financial stability and the economy given private credit's increased interconnectedness with other financial institutions (see pg. 19) as well as its significant software exposure.

Despite the worries, we think that private credit stress is unlikely to generate large macroeconomic spillovers, for a few reasons. First, private credit remains a relatively small part of the financial system, with direct lending accounting for only around 4% of all credit to the private non-financial sector (or roughly \$1.7tn) compared to the 45% and 30% that residential and securitized mortgages comprised in the run-up to the Global Financial Crisis (GFC), respectively. While the modest size of direct lending doesn't imply that broader contagion is impossible, it raises the bar for private credit stress to become macroeconomically significant.

Second, private credit faces less of a funding mismatch than other financial institutions like banks. Third, leverage at private credit funds remains modest. And fourth, while banks' exposure to private credit has risen in recent years, it remains modest overall, is senior to private credit fund investors' stakes, and the banking system is well-capitalized. As a result, we estimate that private credit stress will likely lead to only small GDP losses, even in relatively extreme default scenarios.

Limited economic losses from private credit

To gauge how higher losses in the private credit sector might affect economywide lending, we estimate the economic impacts under several potential loss scenarios¹. For each loss scenario, we consider three cases: (1) losses are proportional across all funds but don't spill over into banks; (2) losses are concentrated in a small number of funds and spill over into the banks that lend to them; and (3) these concentrated losses see lower recovery rates than even our conservative baseline assumption (30% vs. our assumption of 40%), to capture the potential risks from lower recoveries on software loans.

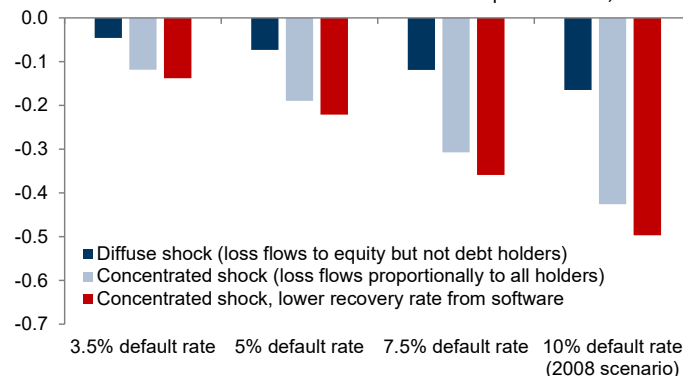
A scenario in which private credit default rates rise from around 1% in 2025 to 3-4% (the lower end of the range for leveraged loans in prior credit cycles) would result in around \$45bn in additional defaults, which would generate a small drag of 0.2% or less on the stock of loans, or roughly 1.5% or less of the gross flow of new lending, even after accounting for a pullback in lending from other financial institutions. That, in turn, would result in a small GDP drag of around 0.1%.

A more severe scenario in which private credit default rates rise to 10% (the top of the historical range for leveraged loans) in a concentrated shock would result in around \$150bn in additional defaults, leading to a 5-6% pullback in the gross flow of new lending to the private sector. That would translate to a GDP drag of 0.4-0.5%. So, even if defaults rose sharply, the macroeconomic impact would be moderate and much smaller than the pullback in new lending in both the 1990 and 2008 recessions (of around 30% and 55%, respectively).

These scenarios focus only on spillovers from stress that is specific to the private credit sector. A broader tightening in financial conditions or a deterioration in corporate balance sheets and the macroeconomic backdrop would pose larger risks than those we estimate here.

Private credit stress will likely lead to only small GDP losses

Estimated GDP effects of downside stress scenarios for private credit, %



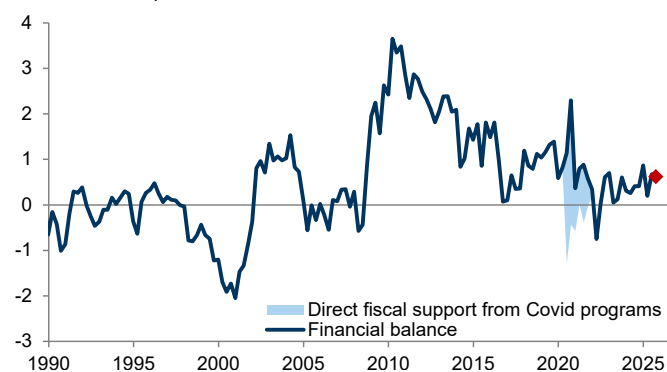
Source: Goldman Sachs GIR.

A supportive macro backdrop

While private credit lending will likely tighten in coming months, corporate balance sheets are very healthy, bank lending to businesses has picked up significantly over the last year, and increased AI-related investment demand will likely act as a tailwind to credit growth over the next several quarters. Taken together, this should further limit the impact of a pullback in private credit on the broader economy.

Corporate balance sheets are healthy...

Nonfinancial corporate financial balance, % of GDP



...and bank lending has accelerated recently

Bank commercial and industrial loans, % chg., year ago



Source: Federal Reserve, Goldman Sachs GIR.

Manuel Abecasis, US Economist

Email: manuel.abecasis@gs.com
Tel: 212-902-8357

Goldman Sachs & Co. LLC

¹ These estimates leverage our representation of the financial system's architecture based on Fed data and insights from our equity analysts.

Alts managers: private credit opportunities

Alexander Blostein and Anthony Corbin assess the implications of private credit concerns for Alternative Asset Managers

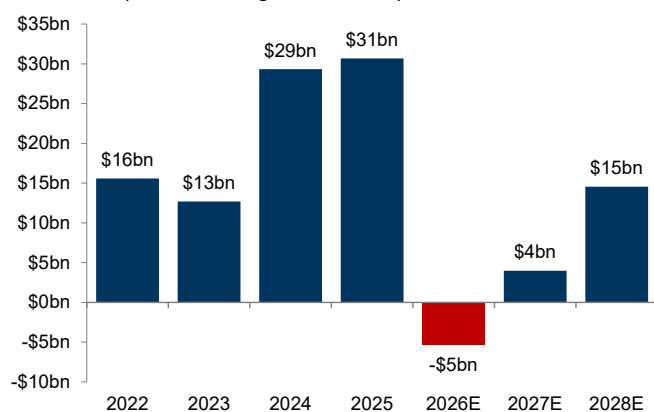
Private credit has been a significant driver of management fee growth for Alternative Asset Managers, accounting for roughly 50% of the growth from 2023-2025. Accordingly, mounting concerns about private credit have weighed heavily on the group, with stocks down over 25% YTD and valuations now 20% below historical averages at 18x NTM P/E. In particular, headlines around redemption requests in retail vehicles have sparked worries about the durability of the wealth channel, which contributed nearly 30% of the group's management fee growth over 2023-25. While we think concerns about the wealth channel are largely overblown, we expect its growth to moderate and a widening dispersion across firms based on their exposure to the wealth channel. However, beyond wealth, private credit opportunities remain compelling, and we expect accelerated institutional deployment to support healthy management fee growth for the group.

Headwinds in the evergreen credit channel

While investors largely expected slower growth from evergreen credit funds—also known as non-traded business development companies (BDCs)—heading into this year given lower base rates/returns, intense media scrutiny has amplified investor anxiety and driven a surge in redemption requests from the retail investor base. These requests reached 12% in 1Q26 (~50% annualized), while gross flows into BDCs in March were over 50% lower than their 2025 monthly run-rate, resulting in a 4% (16% annualized) gross inflow rate (compared to ~50% last year). In response, most asset managers have capped redemptions at 5% of net asset value (NAV) per quarter.

We expect net outflows in retail credit products in 2026

Net flows into public alt manager retail credit products, \$bn



Source: Company data, Goldman Sachs GIR.

We expect these dynamics to persist, with outflows from evergreen funds likely to continue for some time as the industry works through redemption queues. Based on March's gross inflow run-rate and a 20% maximum annual redemption rate, we estimate a 4% annualized pace of net outflows, or approximately \$5bn of outflows across the public alt managers in 2026 before flows turn marginally positive in 2027. We expect a wide dispersion among managers based on underlying

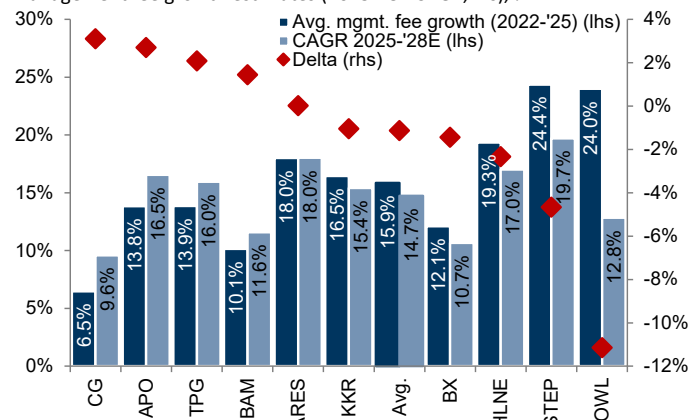
credit quality and vintage, with weaker-performing or later-vintage vehicles facing more acute outflow pressure. This will likely remain a drag on growth for several firms, but we ultimately do not expect retail redemption requests to trigger a "run on the bank" dynamic that would lead to fire sales.

Other compelling private credit growth opportunities

While retail private credit vehicles face near-term headwinds, the institutional channel offers significant growth opportunities. Indeed, despite the recent focus on the wealth channel, over 80% of alts managers' private credit-related management fees are derived from institutional sources compared to the roughly 7% of fees-related revenues from evergreen credit vehicles (though BX, OWL, and HLNE are relatively more skewed toward the wealth channel). And though institutional fundraising has declined in recent years largely owing to tighter credit spreads and faster deployment from non-traded BDCs, we expect faster institutional deployment supported by significant institutional available capital (\$419bn in direct lending/opportunistic unlevered available capital) as wealth pulls back and spreads widen. Substantial loan maturities through 2028 should further boost institutional fundraising and investment activity, particularly for opportunistic, mezzanine, and special situations strategies, which have been relatively dormant in recent years.

We expect Alt managers' management fees to grow at a healthy 15% CAGR through 2028 on average

Historical average management fee growth (organic, ex. catch-up fees, lhs) vs. management fee growth estimates (2025-28E CAGR, lhs), %



Source: Company data, Goldman Sachs GIR.

The path forward: wider dispersion

Looking ahead, we expect Alt managers' management fees to grow at a healthy 15% compound annual growth rate (CAGR) through 2028 (vs. 16% from 2024-25) on average. However, we expect wider dispersion among managers depending on their exposure to the wealth channel—a dynamic that has not yet been reflected in relative stock performance and valuation multiples, presenting potential opportunities.

Alexander Blostein, Senior US Financials Equity Research Analyst

Email: alexander.blostein@gs.com
Tel: 212-357-9976

Goldman Sachs & Co. LLC

Anthony Corbin, US Financials Equity Research Analyst

Email: anthony.corbin@gs.com
Tel: 212-357-7512

Goldman Sachs & Co. LLC

Banks and private credit: the intersection

Richard Ramsden discusses the evolving intersection between banks and private credit

The rapid expansion of the private credit industry over the past two decades has reshaped the lending landscape. The post-Global Financial Crisis (GFC) regulatory environment made it increasingly difficult for banks to compete in certain lending categories, and regulatory constraints around lending to corporates above specified leverage thresholds have pushed some borrowers toward non-bank financing providers that can offer greater flexibility on structure and terms, including covenant-lite and payment-in-kind (PIK) features (see pgs. 10-11).

But while banks have ceded share to the private credit industry, particularly in commercial lending, they have also benefitted from the industry's growth and have become increasingly interconnected with it, providing direct financing opportunities to private credit funds and partnering with private credit firms to serve customers whose financing needs are beyond banks' risk or balance sheet constraints. Amid the current stresses in private credit, this interconnectedness has raised concerns about systemic risk, which we view as overdone.

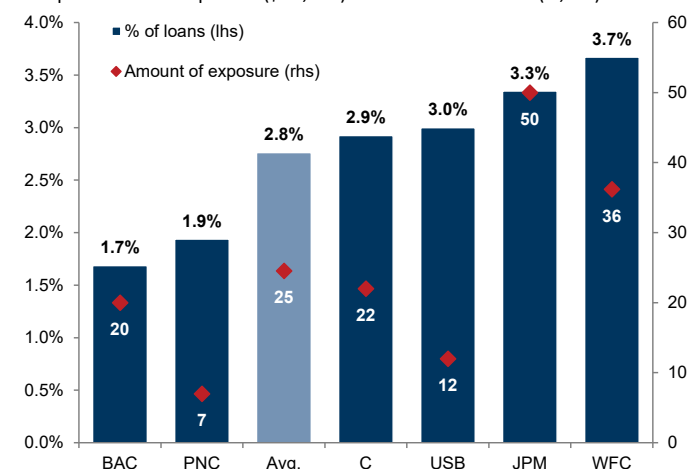
Meaningful linkages...

Currently, banks provide financing to private credit managers in the form of working capital facilities and fund-level leverage. These facilities are typically secured by diversified loan portfolios, which generally have relatively conservative advance rates or loan-to-value ratios (LTVs). Based on disclosures from large banks, advance rates are generally in the 60-70% range, suggesting that defaults would need to rise materially before banks would incur losses. Banks are typically insulated from first-loss risk within their private credit exposure through both underlying operating company equity and a meaningful layer of fund investor capital. Banks also actively monitor collateral values and, in some cases, have reduced the leverage they provide private credit funds over the last six months. As a result, loss rates on banks' private credit portfolios haven't materially risen, and we expect risk-adjusted returns to remain healthy in the near term.

Lending to private credit firms has also become an important source of lending growth for the banking industry, though it remains a relatively small portion of bank balance sheets. Loans to private credit currently account for 3% of bank loan portfolios, having grown at double-digit compound annual growth rates (CAGRs) over the last five years. This exposure sits within the broader non-depository financial institution (NDFI) lending category that represents around 15% of total bank lending. However, as private credit firms experience net outflows, the demand for these working capital loans will likely decline, leading to slower growth—or even some contraction—in bank lending to private credit.

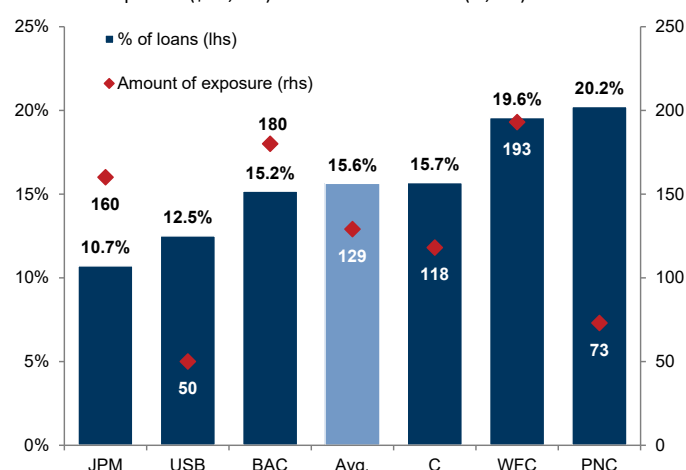
Loans to private credit account for 3% of total bank loans...

Bank private credit exposure (\$bn, rhs) and as share of loans (%), lhs



...with broader NDFI lending accounting for 15%

Bank NDFI exposure (\$bn, rhs) and as share of loans (%), lhs



Note: WFC, USB, and PNC as of 1Q26, others as of 4Q25; chart excludes MS.

Source: Company data, Goldman Sachs GIR.

Banks have also increasingly partnered with private credit firms to serve customers whose financing needs may fall outside a bank's risk appetite or balance sheet constraints. These partnerships enable banks to distribute risk, helping diversify portfolio exposure while allowing banks to offer clients a broader range of lending options than they could provide on a standalone basis, as they can now sell and offload loans that do not fit their underwriting criteria. So, while a pullback in private credit may allow banks to regain some share in loan origination, it could also reduce the range of lending options banks can offer clients, particularly in situations where banks are unwilling to hold the loans on their own balance sheet.

...but limited systemic risk

Taken together, banks' exposure to private credit is real but structurally buffered, with direct exposures generally secured, conservatively leveraged, and protected by substantial first-loss capital. As a result, while ongoing monitoring is warranted, particularly if private credit stresses continue, private credit seems unlikely to pose a risk to the traditional financial system.

Richard Ramsden, Head of the Financials Group

Email: richard.ramsden@gs.com
Tel: 212-357-9981

Goldman Sachs & Co. LLC

Summary of our key forecasts

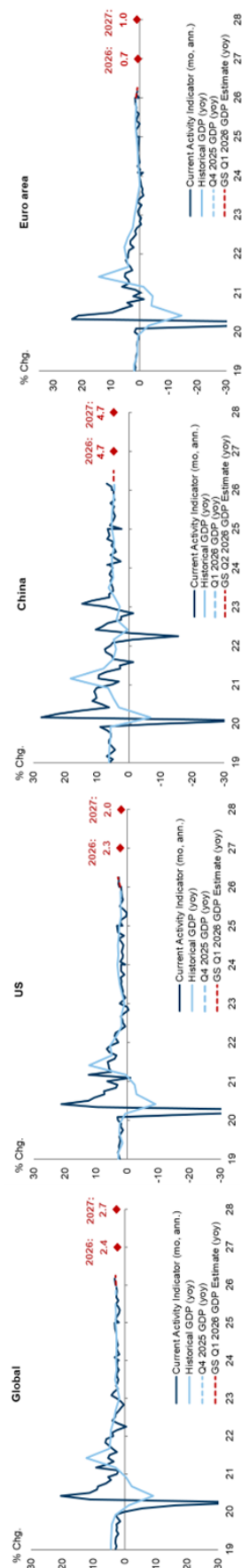
GS GIR: Macro at a glance

Watching

- **WATCH THE IRAN WAR.** While the US and Iran have agreed to extend the ceasefire, the situation remains fluid and oil flows through the Strait of Hormuz remain very low, with risks to economies and markets growing the longer these supply disruptions continue.
- **Globally,** we expect real GDP growth of 2.4% yoy in 2026 amid growing headwinds from higher energy prices owing to the Iran war. We expect global core inflation to decline to 2.3% by the end of 2026, reflecting a fading tariff boost and further normalization in shelter and wage inflation but a boost from higher energy prices.
- **In the US,** we expect real GDP growth of 2.1% on a Q4/Q4 basis in 2026, reflecting a drag from higher oil prices, tighter financial conditions, and elevated geopolitical risk. We expect core PCE inflation to fall to 2.6% yoy by December 2026, as the tariff boost fades and wage and shelter inflation continue to cool, though higher oil prices provide some offset. We expect the unemployment rate to end 2026 at 4.6%.
- **We expect the Fed** to deliver two 25bp cuts this year in September and December for a terminal rate range of 3-3.25%.
- **In the Euro area,** we expect below-potential real GDP growth of 0.6% on a Q4/Q4 basis in 2026, reflecting higher energy prices and tighter financial conditions owing to the Iran conflict. We expect core inflation to rise to a peak of 2.5% yoy in 3Q26 amid higher energy prices.
- **We expect the ECB** to deliver two 25bp hikes this year in June and September to a peak policy rate of 2.5% before cutting back to 2% in 2027, although we view whether the ECB hikes twice or holds rates unchanged this year as a close call.
- **In China,** we expect real GDP growth of 4.7% yoy in 2026 as resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn offset headwinds from higher energy prices and sluggish domestic demand. We expect CPI/PPI inflation to rise to 1.0%/1.2% yoy this year owing to higher global commodity prices, government policy efforts, and a low base.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.

Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics	Markets										Equities									
	GDP growth (%)					Interest rates 10Yr (%)					FX					Commodities				
	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030
Global	2.2	2.4	2.7	2.7	2.6	US	4.31	4.10	4.15	EUR/US	1.17	1.14	1.20	Price	7.600	US	2.7	4.6	2.1	4.4
US	2.1	2.0	2.3	2.2	2.2	Germany	3.00	3.00	3.00	GBP/US	1.35	1.33	1.33	EPS	\$309	US	2.9	6.3	2.1	6.2
China	4.6	4.5	4.7	5.0	4.6	Japan	2.44	2.00	2.00	\$/JPY	159	160	155	Growth	12%	China	1.0	--	1.0	--
Euro area	0.6	0.9	0.7	1.4	1.0	UK	4.80	4.40	4.25	\$/GBP	6.82	6.85	6.70	Credit	4.00	China	1.0	--	1.0	--
Policy rates (%)	2026					2027					2028					2029				
	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS
	3.13	3.54	3.13	3.37	3.13	Crude Oil, Brent (\$/bbl)	105	100	90	USD	IG	79	95	90	US	2.7	4.6	2.1	4.4	3.7
	2.50	2.51	2.00	2.46	2.00	Nat Gas, NYMEX (\$/mmBtu)	2.52	3.50	3.50	USD	IG	79	95	90	US	2.9	6.3	2.1	6.2	--
	1.40	1.50	1.30	--	1.30	Nat Gas, TTF (EUR/MWh)	44.89	50	40	EUR	IG	92	120	114	China	1.0	--	1.0	--	--
Japan	1.00	1.20	1.50	1.73	1.50	Copper (\$/lb)	13.247	12.700	12.500	EUR	IG	92	120	114	China	1.0	--	1.0	--	--
						Gold (\$/troy oz)	4,712	4,865	5,308	USD	HY	286	375	360						

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of April 24, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

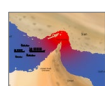
Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

Top of Mind archive



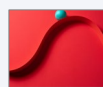
Issue 147
Iran conflict: how long, and how bad?
March 20, 2026



Issue 132
Post-election economic policies
October 21, 2024



Issue 146
Will AI eat software?
March 12, 2026



Issue 131
Is the Fed behind the curve?
September 3, 2024



Issue 145
Americas First
January 26, 2026



Issue 130
How investable is Europe?
August 1, 2024



Special Issue
2025: 4 themes in charts
December 15, 2025



Issue 129
Gen AI: too much spend, too little benefit?
June 25, 2024



Issue 144
The US-China tech race
December 4, 2025



Issue 128
Central bank divergence room to run?
May 21, 2024



Issue 143
AI: in a bubble?
October 22, 2025



Issue 127
Weighing the GLP-1 market
April 12, 2024



Issue 142
Data (un)reliability
September 11, 2025



Issue 126
Global transit & trade: in rough waters
March 12, 2024



Issue 141
Stablecoin summer
August 19, 2025



Issue 125
2024: The year of elections
February 1, 2024



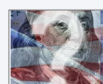
Issue 140
US fiscal worries: Is this time different?
June 12, 2025



Issue 124
Middle East risks
December 5, 2023



Issue 139
Fed independence: how concerning?
May 15, 2025



Issue 123
US outperformance: at a turning point?
October 30, 2023



Issue 138
Tariff-induced recession risk
April 17, 2025



Issue 122
Commercial real estate risks
October 9, 2023



Issue 137
Europe's shifting security landscape
March 13, 2025



Issue 121
Corporate credit concerns
August 10, 2023



Issue 136
Trump tariffs: mostly talk, or big action?
February 26, 2025



Issue 120
Generative AI: hype, or truly transformative?
July 5, 2023



Issue 135
High bond yields: here to stay?
February 5, 2025



Issue 119
Daunting debt limit dynamics
May 22, 2023



Special Issue
2024: 4 themes in charts
December 18, 2024



Issue 118
US-China: more decoupling ahead?
May 1, 2023



Issue 134
Will China's policy stimulus be enough?
December 11, 2024



Issue 117
All about bank(panic)s
April 3, 2023



Issue 133
Market concentration: how big a worry?
November 25, 2024



Issue 116
(Japanese) Bonds, Bonds, Bonds
February 23, 2023

Source of photos: www.gettyimages.com.

Disclosure Appendix

Reg AC

We, Allison Nathan, Jenny Grimberg, Ashley Rhodes, Amanda Lynam, CPA, Shamshad Ali and Manuel Abecasis, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

We, Richard Ramsden, Alexander Blostein, CFA and Anthony Corbin, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Rating and pricing information

Ares Management Corp. (Buy, \$115.23) and StepStone Group (Buy, \$51.82).

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Distribution of ratings: See the distribution of ratings disclosure above. **Price chart:** See the price chart, with changes of ratings and price targets in prior periods, above, or, if electronic format or if with respect to multiple companies which are the subject of this report, on the Goldman Sachs website at <https://www.gs.com/research/hedge.html>.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor

grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other

circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

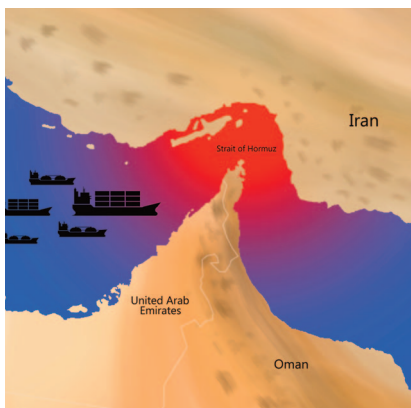
Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

TOP^{of} MIND

IRAN CONFLICT: HOW LONG, AND HOW BAD?



The unprecedented US and Israeli coordinated attack on Iran has resulted in the largest energy supply disruption in history. With seemingly no end to the conflict in sight, its potential duration, impact on global energy supplies, and economic and market implications are Top of Mind. Chatham House's Sanam Vakil and long-time US Middle East advisor Dennis Ross agree that an end to the conflict doesn't look imminent owing to (the lack of) incentives on Iran's side and its chokehold on the Strait of Hormuz, which stands in the way of the US declaring victory. To that end, Vice Admiral Kevin Donegan, USN (ret) explains that military convoys would be capable of ensuring safe transit through the Strait, but not of restoring oil flows to anywhere near normal levels. So, we explore the growing implications of these

disruptions for energy prices, the global economy, and markets, finding that assets are pricing an inflationary shock, but not a growth shock, which could be the next shoe to drop.



For now, all evidence suggests that the conflict will persist—Iran isn't yet signaling that it wants out, and the US has yet to achieve an outcome it can plausibly claim as a victory.

- Sanam Vakil

The key to the US being able to walk away is the Strait of Hormuz... [Trump] likely can rightly claim today that Iran won't be able to pose a conventional threat to its neighbors for at least five years. But he can't say he's won and stop as long as Iran controls who can and can't export oil and sail through the Strait.

- Dennis Ross

I'm confident the US and its allies have the capability to escort commercial ships through the Strait... [but] it is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels.

- Kevin Donegan



WHAT'S INSIDE

INTERVIEWS WITH:

Vice Admiral Kevin Donegan, USN (ret), Former Commander of the US Navy's Fifth Fleet and Commander of the 32 Nation Combined Maritime Forces in the Middle East

Ambassador Dennis Ross, Fellow, The Washington Institute for Near East Policy, Former Director of Policy Planning, Special Middle East Coordinator, and National Security Council senior director for the Central Region

Sanam Vakil, Director of the Middle East and North Africa Programme, Chatham House

IRAN WAR: US POLITICAL PRESSURES

Alec Phillips, GS US Economics Research

SIZING THE ENERGY SHOCK

Samantha Dart, Daan Struyven, Alexandra Paulus, GS Commodities Research

IRAN CONFLICT: GROWING ECONOMIC IMPACTS

Joseph Briggs, GS Global Economics Research

MARKETS: GROWTH, THE NEXT SHOE TO DROP?

Kamakshya Trivedi, GS Markets Research

MENA ECONOMIES: BIG HIT, POSSIBLE SCARS

Farouk Soussa, GS MENA Economics Research

...AND MORE

Allison Nathan | allison.nathan@gs.com

Jenny Grimberg | jenny.grimberg@gs.com

Ashley Rhodes | ashley.rhodes@gs.com

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

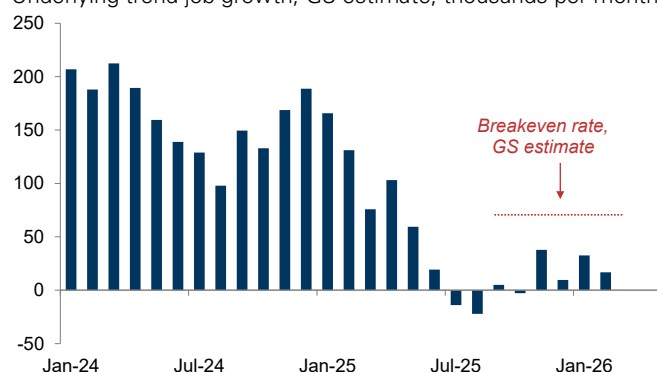
- We lowered our 2026 US growth forecast and raised our 2026 US inflation forecast to reflect higher oil prices amid the Iran war, with growth risks skewed further to the downside and inflation risks to the upside.
- We pushed back our forecast for the next Fed cut from Jun to Sep, with a second cut in Dec to the same terminal rate of 3-3.25% given the higher inflation path we now expect.

Datapoints/trends we're focused on

- Labor market, which we expect to soften further.

Labor market softening concerns: alive and well

Underlying trend job growth, GS estimate, thousands per month



Note: We estimate underlying trend job growth as $0.75 \times 3\text{-month average payroll growth} + 0.25 \times 9\text{-month average household employment growth}$.

Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

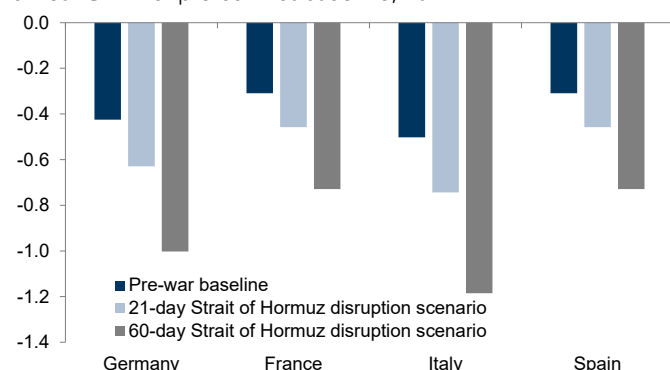
- We further lowered our 2026 Euro area real GDP growth forecast and raised our 2026 headline inflation forecast to reflect more prolonged oil disruptions amid the Iran war.
- We now expect the BoE to remain on hold for longer and expect Bank Rate to remain at 3.75% through 2026 (vs. quarterly cuts from July previously) given the growing energy shock.

Datapoints/trends we're focused on

- ECB policy; we continue to expect the ECB to remain on hold this year, although the likelihood of hikes has risen given the continued upward pressure on energy prices.

Europe: energy price shock to growth

Model-implied peak effect of energy price shock on the level of real GDP vs. pre-conflict baseline, %



Source: Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

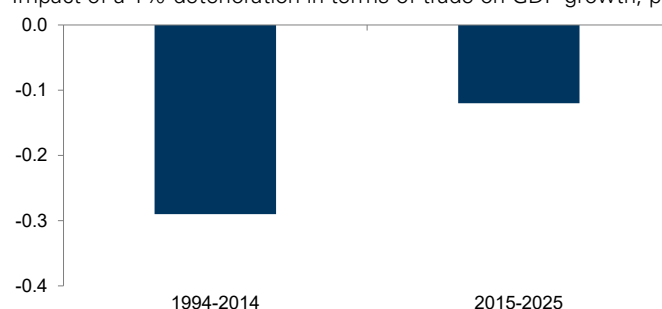
- We lowered our CY26/27 real GDP growth forecasts and raised our 2026 core CPI inflation forecast to reflect higher energy prices, though the sensitivity of growth to rising oil prices is smaller than in the past.

Datapoints/trends we're focused on

- BoJ policy; we continue to expect the next hike in July, but the uncertainty around timing remains high as inflationary pressures from higher energy prices could prompt an earlier hike while rising growth concerns could delay further hikes.
- 2026 *shunto* negotiations; the first round of collected data will likely confirm high wage hike momentum this year.

Higher oil prices: less shocking than in the past

Impact of a 1% deterioration in terms of trade on GDP growth, pp



Note: Impulse response from a VAR model with terms of trade (logarithmic difference) and GDP (logarithmic difference) as endogenous variables, and the real effective exchange rate (logarithmic difference) and overseas GDP (logarithmic difference) as exogenous variables.

Source: Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

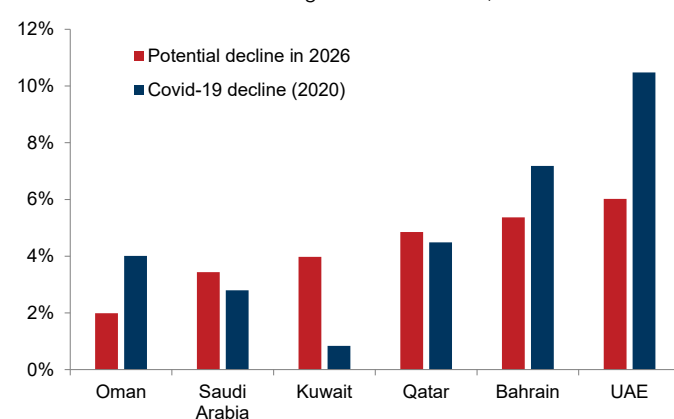
- We raised our 2026 GDP growth forecasts in Brazil and Ecuador but lowered them in Mexico, Chile, and Peru while revising inflation higher across the region amid the Iran war.
- We lowered our 2026 GDP growth forecast for China and most other Asian economies while revising inflation up across the region to reflect higher energy prices.

Datapoints/trends we're focused on

- GCC economies, some of which could see a larger decline in non-oil GDP amid the Iran war than during the pandemic.
- Hungary elections, in which Viktor Orban's Fidesz party faces its strongest challenge since returning to office.

GCC: potentially bigger growth hit than Covid

Decline in non-oil GDP among GCC economies, %



Source: Haver Analytics, Goldman Sachs GIR.

Iran conflict: how long, and how bad?

The unprecedented US and Israeli coordinated attack on Iran has resulted in the largest energy supply disruption in history (see pg. 9), as shipping traffic in the Strait of Hormuz—through which 20% of global oil supplies typically flow—has ground to a near halt and regional energy infrastructure has become the latest target in the conflict. While the economic and market impacts have so far remained relatively muted, the longer the conflict and the resulting chokehold on the Strait of Hormuz and global energy supply last, the less likely that will remain the case. With seemingly no end to the conflict in sight as it continues to escalate, its potential duration, impact on global energy supplies, and economic and market implications are Top of Mind.

To gain insight on when and how the conflict could end, we first turn to Sanam Vakil, Director of the Middle East and North Africa Programme at Chatham House, Ambassador Dennis Ross, Fellow at The Washington Institute for Near East Policy, and Alec Phillips, GS Chief US Political Economist.

Vakil, a longtime Iran watcher, argues that Iran views the conflict as nothing less than an “existential struggle” and is prepared to use its ample asymmetric drone capabilities for as long as is required to guarantee the Islamic Republic’s long-term survival, which likely entails sanctions relief. “Until Iran sees a credible path to those guarantees,” she says, “it has little incentive to end this war.”

Ambassador Ross, who served as a Middle East advisor in five US administrations, contends that President Trump likely would have already declared victory and ended the war were it not for Iran controlling transit through the Strait of Hormuz, which has the potential to inflict substantial pain on the global economy. Until that chokehold is broken, he says, the US cannot unilaterally end the war. Ross sees mediation—potentially facilitated by Russian President Putin—as the quickest way to end the war if the US can’t wrest control of the Strait, though he doesn’t see the conditions to enter a mediation in place at this point and says it isn’t clear who would even mediate on the Iranian side now that Former Speaker of the Parliament Ali Larijani has been killed. So, Ross seems to agree that an end to the war doesn’t look imminent.

While Phillips takes no view on when the war will end, he argues that US political considerations don’t seem to be much of a constraint on the war’s duration for now, but that could change if gasoline prices continue to rise. While he explains that the Trump Administration has some tools in its toolbox to lower energy prices—several of which it has already deployed—these may not be enough to completely offset the price shock.

Given the critical role the Strait of Hormuz is playing in the conflict and its potential outcome, we then speak with Vice Admiral Kevin Donegan, USN, Former Commander of the US Navy’s Fifth Fleet, to better understand the potential capability and capacity to restore flows through the Strait. He’s confident US and allied military convoys would be capable of ensuring

safe transit through the Strait, explaining that the US and many other nations have long rehearsed how to keep vital waterways in the region open. But he clarifies that convoys would never be able to restore oil flows to normal levels, noting that potentially only 20% of flows could be restored. Returning oil flows to pre-conflict levels, he says, would ultimately require Iran to stop harassing ships, which is a matter of incentives and leverage on all sides. But, even then, he argues that restoring flows would not be a matter of simply flipping a switch, because recovering the trust that transit would indeed be safe would take time.

With a quick end to the conflict seeming increasingly elusive and global energy prices already high, GS Commodities Strategists Samantha Dart, Daan Struyven, and Alexandra Paulus then lay out potential energy supply disruption scenarios and the implications for prices, with a gradual recovery in oil supplies from here allowing Brent prices to decline close to \$70/bbl in 4Q26 and a ramp-up in Qatari LNG production starting in early April allowing TTF prices to gradually decline to €30-40/MWh in 2H26.

But, they estimate that longer disruptions could present substantial upside risk to prices, with Brent likely to exceed its 2008 all-time high if depressed flows keep the market focused on the risk of lengthier disruptions. And even if LNG supplies begin to recover in the short term, they warn that the damage to Qatari LNG infrastructure is likely to delay the significant pressure on global LNG prices that they’ve expected from “the largest ever LNG supply growth wave” from 2027.

So, how hard might the global economy be hit? Joseph Briggs, GS Senior Global Economist, provides some useful rules of thumb: each 10% increase in oil prices lowers global GDP by a bit more than 0.1%, raises global headline inflation by 0.2pp—with larger impacts in Asia ex-China and Europe—and leads core inflation 0.03-0.06% higher, depending on the country. For now, these rules imply moderate hits to growth and boosts to inflation, but those impacts could quickly grow should energy flows remain disrupted for a prolonged period. (Farouk Soussa, GS Senior MENA Economist, then zeros in on the hit to MENA economies in particular, finding a significant near-term hit and possible long-term scars).

And what about markets? Kamakshya Trivedi, GS Chief FX and EM Strategist, explains that markets so far have generally priced the conflict and resulting spike in energy prices as an inflationary shock, suggesting that growth downside could be the next shoe to drop if disruptions continue. If it is, he sees equities and relatively resilient cyclical assets facing increased pressure and the recent hawkish repricing in front-end yields reversing, creating an environment for a key safe haven to shine: the Japanese Yen.

Allison Nathan, Editor

Email: allison.nathan@gs.com

Tel: 212-357-7504

Goldman Sachs & Co. LLC



Interview with Sanam Vakil

Sanam Vakil is Director of the Middle East and North Africa Programme at Chatham House. Below, she argues that Iran views the current conflict as an existential struggle and will not seek an end until it receives guarantees that ensure the Islamic Republic's long-term survival.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Ashley Rhodes: Fundamentally, what is this conflict about for Iran?

Sanam Vakil: Tehran sees the US and Israel as waging a regime change war. So, what remains of the Iranian regime is fighting for its very survival. Its strategy is to spread the costs of the war as widely as possible, so that when off-ramps emerge, the war will

have become so costly for everyone involved that Iran can obtain guarantees that it will not be repeated again whenever external powers see fit. This strategy reflects lessons from the Twelve-Day War in June 2025, which Iran felt it conceded too early on, signaling weakness and inviting future challenges. So, Iran isn't seeking a quick de-escalation but rather responding to what it perceives as an existential struggle.

Ashley Rhodes: But Iran is also bearing a tremendous cost in this conflict. How long can it continue to do so?

Sanam Vakil: In conventional terms, the US and Israel are the dominant military players and will likely succeed in destroying/degrading Iran's ballistic missile capabilities, its defense industrial complex, and the remnants of its nuclear program. But Iran isn't fighting this war on conventional terms. Instead, it's playing for time—dragging the conflict out, spreading out the costs, and forcing the US, Israel, and regional countries to drain their resources, thereby putting them on the defensive. Iran's strength lies in asymmetric capabilities like low-cost drones that can continue to inflict damage even as the US and Israel make conventional military gains. That Iran is still able to do so three weeks into the conflict underscores that it can continue to endure significant costs.

Ashley Rhodes: How has Mojtaba Khamenei's selection as the new Supreme Leader influenced Iran's strategy?

Sanam Vakil: Mojtaba Khamenei's appointment as Supreme Leader in the wake of his father's death has had little to no impact. His elevation is largely symbolic and likely reflects the regime's desire to preserve the status quo. The Islamic Republic is often presented as a personalistic theocracy. In reality, it's a bureaucratic, institutional system in which power is distributed among various factions. The Islamic Revolutionary Guard Corps (IRGC) is particularly important given its responsibility for the war effort and internal security, but the importance of Speaker of the Parliament Ghalibaf and President Pezeshkian shouldn't be discounted.

The system is prepared for leadership losses and continues to operate even with Mojtaba reportedly incapacitated. The IRGC is managing the day-to-day defensive response through a decentralized, mosaic command structure, while the technocratic civilian government runs the country. And should Mojtaba be removed, another leader would emerge, underscoring the regime's resilience and cadre of loyalists. The

Islamic Republic has endured for 47 years because its leadership unites in times of crisis. This time is no different.

Ashley Rhodes: How and who could end this war?

Sanam Vakil: This war won't end simply because the US or Israel decide it should. The lesson Iran learned in the Twelve-Day War is that this conflict must end in a conclusive way that guarantees the Islamic Republic's longer-term survival, which entails securing an agreement to that effect. Iran is currently making extreme zero-sum demands, including that US military bases not be used to strike it. But Iran ultimately needs sanctions relief, which will require negotiations in which both it and the US make concessions on longstanding issues—including Iran's nuclear program and broader regional security. Until Iran sees a credible path to those guarantees, it has little incentive to end this war.

Ashley Rhodes: How likely is a domestic uprising that brings an end to the conflict?

Sanam Vakil: Historically, sustained protests or mobilizations are rare during wartime, and this conflict has allowed the regime to intensify internal security, restrict movement, and suppress dissent under the banner of national defense. In that sense, the war actually works *against* the emergence of an uprising, which no doubt factors into the regime's motivations for continuing the war, particularly with the brutal way the massive January protests have been handled on people's minds.

While most Iranians remain deeply dissatisfied with the system and fearful that the war will further entrench the Islamic Republic, they lack the organizational and military capacity to mobilize. Revolutions require unity and planning—qualities absent in the fragmented opposition. And even if the regime emerges from the war militarily weakened, it will likely remain capable of suppressing internal dissent. So, I don't see a scenario where domestic unrest hastens the end of the war.

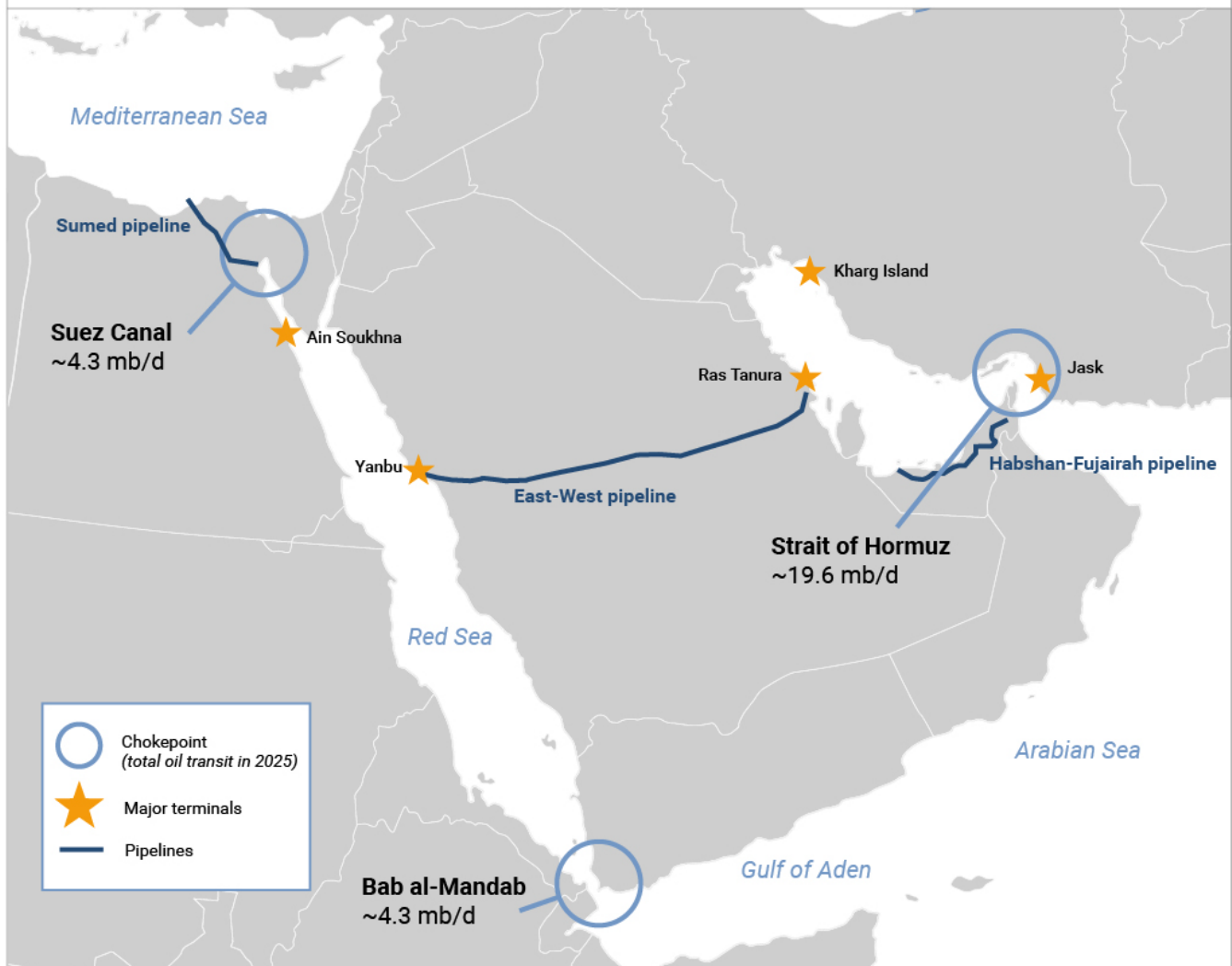
Ashley Rhodes: What are you watching most closely to determine when and how this conflict could end?

Sanam Vakil: First and foremost, I'm watching Iran's appetite for further escalation. Although Iran initially focused its efforts on disrupting Gulf economic activity and effectively closing the Strait of Hormuz, it recently broadened its approach by launching direct attacks on regional energy infrastructure.

I'm also watching for any Iranian statements that indicate interest in an off-ramp and, importantly, the specific terms it asks for. A softening of Iran's current out-of-left-field demands may indicate a shift in Tehran's cost-benefit calculus and reveal that Iran is ready to enter serious negotiations. But for now, all evidence suggests that the conflict will persist—Iran isn't yet signaling that it wants out, and the US has yet to achieve an outcome it can plausibly claim as a victory. Until one or both of those dynamics meaningfully changes, we should expect this war to continue.

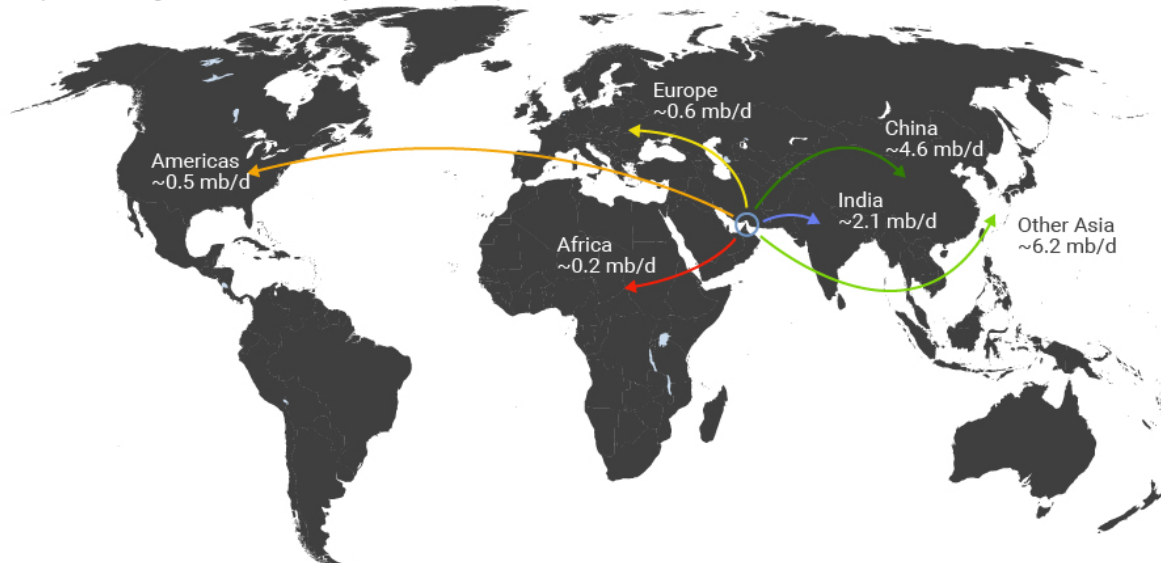
Middle East oil flows, mapped out

Although the Saudi East-West Pipeline and the UAE Habshan-Fujairah route can redirect some oil flows amid disruptions to the Strait of Hormuz, their estimated spare capacity remains low...



...and is likely insufficient to compensate for the large volume of oil normally transported through the Strait to the rest of the world

Crude oil exports transiting the Strait of Hormuz by destination (2025)



Note: Top map does not show all energy infrastructure in the region.

Source: Bloomberg, IEA, Kpler, Library of Congress, Goldman Sachs GIR.

Special thanks to GS Commodities Research analyst Alexandra Paulus for data.

Interview with Dennis Ross

Ambassador Dennis Ross is Fellow at The Washington Institute for Near East Policy. Previously, he served as Director of Policy Planning under President H.W. Bush, Special Middle East Coordinator under President Clinton, and National Security Council Senior Director for the Central Region under President Obama. Below, he argues that control of the Strait of Hormuz will remain key in determining how and when the conflict ends.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: There are conflicting narratives about how the US and Israel came to go to war together for the first time. What's your view?

Ambassador Dennis Ross: On both the far-right and the far-left of the US political spectrum there is a narrative that Israel dragged the US into this war. Destroying Iranian capabilities that threaten Israel and the broader region has been a strategic priority for Israeli Prime Minister Netanyahu for 30 years. I refer to it as his "Winston Churchill issue." And he has undoubtedly made the case for this to President Trump. So, it's possible that Netanyahu influenced Trump's thinking. During Trump's first term, Israel, in a covert operation in Tehran, seized 55,000 documents—the "nuclear archive"—that detailed Iran's nuclear program. This archive demonstrated that, contrary to its denials, Iran had been developing nuclear weapons—with plans for at least five such weapons. So, perhaps Netanyahu was able to influence Trump's thinking with the intelligence that Israel provided in the past and more recently.

But Trump made the decision to go to war. It was Trump who built up a huge armada in the region and said in response to the Iranian protests in January that the US was "locked and loaded" and "help was on the way." In fact, the day after US Secretary of State Rubio explained the imminent need for the US to strike Iran because Israel was planning to do so and the US would be a target for Iranian retaliation, so better for the US to strike first, Trump disagreed, saying instead that he probably forced Israel's hand. I've never seen anybody who determines what President Trump does other than himself.

If anything, it's Netanyahu who cannot say no to Trump. Trump has all the leverage in the world on Netanyahu. He has put Trump on a pedestal for all the help and support he has provided to Israel over the years, including a readiness to attack Iran last June, which no other US president had been willing to do. So, when Trump decides, Netanyahu responds. For example, when Iran fired missiles at Israel just after a ceasefire in the Twelve-Day War was supposed to go into effect, and Israel launched 50 planes in response, Trump told Netanyahu to turn those planes around, and he did.

Allison Nathan: Iran poses an existential threat to Israel, but not to the US. So, why now for the US to strike Iran?

Ambassador Dennis Ross: I don't speak for the Administration and am humble about predicting Trump. But my interpretation is that the Administration saw this as a moment of opportunity. Although Trump said the US had obliterated Iran's nuclear capability last June, intelligence suggested that the Iranians

were beginning to clean out some of their nuclear weapon sites, probably with an eye to reconstituting them. That created a perceived need to do something. But it was the absence of any significant air defense in the wake of the Twelve-Day War, which left Iran militarily vulnerable, that created a sense of opportunity. There was also the sense that the Iranian regime might be teetering given the massive January protests and the brutality with which the regime responded, which was seen as a sign of weakness. So, the Administration seemed to assume—wrongly—that taking out the Supreme Leader would lead to regime collapse and seized the opportunity to strike.

Allison Nathan: Some divergence seems to exist between the US and Israel's war strategy/goals. Is that true?

Ambassador Dennis Ross: I see more convergence than divergence between their war aims right now. Israel absolutely aims to end the current Iranian regime. That might be Trump's preference too, but this war will probably end without the regime collapsing, even if it has been terribly weakened. So, US and Israeli goals likely diverge somewhat in that sense.

But their goals converge in the shared conviction that Iran's military capabilities must be damaged sufficiently to avoid the need for another war anytime soon. As I mentioned, Iran was already beginning to return to its nuclear sites and had also recovered most of its ballistic missile production capability since the Twelve-Day War. Money and weapons were flowing back to Iran's proxy network, even as the proxies have proven largely ineffective yet expensive to support. So, the Iranians were acting as if the dramatic weakening of the proxies and the Twelve-Day war had never happened; whatever the cost, they had learned no lessons and were keeping the same strategy. Both Trump and Netanyahu wanted the Iranians to understand that strategy would end one way or the other. As such, the US and Israel's key shared objective in this conflict is to weaken Iran to the point that even if it wanted to reconstitute its proxies, nuclear program, and ballistic missile and drone capability, the cost would be simply too great. Bottom line: the goal is to ensure Iran cannot practically pose a threat to Israel and others in the region for at least five years, if not longer.

Allison Nathan: Iran seems to be pursuing a strategy of inflicting as much pain as possible in the hopes of securing a better outcome for itself. Is there a point at which the costs of the war will be too great for the US/Israel to bear?

Ambassador Dennis Ross: This is an area where I see potential for a real divergence between the two countries. The Israelis can continue this conflict for as long as it takes. While Netanyahu doesn't have strong support in the country writ large, he has almost 100% support for dealing with the Iran threat. Remember, Ayatollah Ali Khamenei had a clock ticking

down to the end of the Israeli state in downtown Tehran. The Israeli public has been sleeping in shelters every night for fear of missile strikes from Iran or its proxies. The attitude is do enough that this won't be necessary again for a long time.

That's not the case in the US. This conflict marks a rare instance of the US going to war without the support of the majority of the public. So, Trump isn't benefiting from a "rally around the flag" effect, which may owe partly to the lack of any serious effort on the Administration's part to explain what it's doing and why. That said, the President likely feels he has some license to continue for now. I'm not sure that those around him who are worried about affordability given surging energy and fertilizer prices, which will lead to higher gasoline, airline, food prices, and beyond, agree. But my sense is that the President will care more about the inflationary implications of the conflict as the midterm elections approach. So, he has some license but probably also some endpoint in mind.

Allison Nathan: Can the US/Israel end the war unilaterally?

Ambassador Dennis Ross: The key to the US being able to walk away is the Strait of Hormuz. On his own, Trump probably would have already declared victory, pointing to the amount of damage the US and Israel have done to Iran's key targets, naval, missile, and drone capabilities, and nuclear weapons program. He likely can rightly claim today that Iran won't be able to pose a conventional threat to its neighbors for at least five years. But he can't say he's won and stop as long as Iran controls who can and can't export oil and sail through the Strait.

At this point, the Iranian threat has been severely diminished; it's firing a relatively small number of ballistic missiles and even drones every day. Some of this is Iran holding back to preserve an ongoing capability to fire missiles and drones to try to outlast the US. But Iran also doesn't have the capability to fire much more in no small part because the US can act preemptively once launchers emerge. Given the amount of firepower the Iranians still retain, they are likely capable of continuing to fire at their current rate for at least several more weeks, if not longer. So, Iran's capacity to threaten hasn't disappeared, but has been severely reduced. If this, combined with a convoy system that helps protect ships, enables the US to wrest control of the Strait away from Iran, Trump would be able to declare an end to the war. And even if the Iranians don't agree, there wouldn't be much they could do about it. Of course, many questions remain about whether the US can achieve such control. It may not be possible to stop Iran from threatening ships without seizing territory along its coast. A US Marine Expeditionary Unit is currently en route to the Middle East, which may open the door for island seizures, and the US and some allies have seemingly stepped up their efforts to reopen the Strait.

Allison Nathan: If the US can't gain control of the Strait, would a quick end to the war still be possible?

Ambassador Dennis Ross: Barring wresting control of the Strait, the only way the conflict resolves soon is through mediation. Russian President Putin seems well-placed to serve as the mediator since Iran doesn't really have any other friends, so it has incentive to respond to him, and Putin also has a direct line into Trump, who also seems to be somewhat responsive to Putin. Both sides have something to offer in a negotiation. Iran

could cede control of the Strait and pledge to not close it again and hand over its 440 kilograms of highly enriched uranium—or more likely allow the Russians to excavate it from its current position, buried deep under rubble, making it a massive project for anyone—which would allow Trump to say he's put an end to the Iranian nuclear program for a very long time. In return, Iranians—as well as the Russians, if they do help—would seek guarantees of no new attacks on Iran as long as they continue to comply with the agreement as well as some sanctions relief.

But the conditions to enter a mediation don't seem to be in place at this point as both sides further harden their positions. My sense is Trump would still need to show a changed reality in the Strait before any mediation could occur to avoid looking like the one who needs a way out. He is sending even more forces to the region in the hopes of gaining the whip hand to that end. But Iran also wants to show it's not the one who needs to end the war, even as the costs continue to grow. So, mediation seems more likely than a unilateral end as long as the Strait is in play, but it doesn't seem imminent.

Allison Nathan: Who would Putin mediate with in Iran?

Ambassador Dennis Ross: If one person embodied the leadership in the wake of Ali Khamenei's death, it wasn't Mojtaba Khamenei but Former Speaker of the Parliament Ali Larijani, who was recently also killed. Larijani had relationships with every regime faction, including the Islamic Revolutionary Guard Corps (IRGC). If anyone could orchestrate a move to end the war through mediation, it would have been him.

Although Mojtaba Khamenei has received much attention, he seems unlikely to be able to speak for the regime. His physical state remains unclear. Even if we assume he is physically ok, he lacks authority. He was his father's son but had no public persona or religious credentials that would have made him an ayatollah. He ran the Supreme Leader's office, which had significant power and tentacles in every institution and facet of Iranian society, but he was just an implementer. President Pezeshkian also doesn't have the same access as Larijani did to the different factions nor his authority or credibility. Perhaps Mohammad Bagher Ghalibaf, the current Speaker of the Parliament, could provide some weight, but that's unclear. So, a hole currently exists in the Iranian leadership lineup, which reduces the prospect of a mediated outcome in the near term.

Allison Nathan: So, what are you watching from here?

Ambassador Dennis Ross: I am watching the Russians and the Chinese. As we discussed, the Russians can play a major role as mediator. The Chinese, though, can pressure the Iranians. The Iranian position has always been that if they can't export, nobody can. But right now, the Iranians are saying that they can export and will determine who else can. The minute the US takes the stand that Iran won't be allowed to export if it continues to prevent others from doing so, China would likely strongly object given its dependency on Middle Eastern energy supplies and would pressure the Iranians to change course. Again, at the end of the day, this all comes down to the Strait of Hormuz. Trump cannot end the war as long as Iran controls it. So, either the US does something dramatic to deny Iran that leverage or waits until conditions shift enough to allow for a mediated outcome. That's what we should be watching for.

Iran war: US political pressures

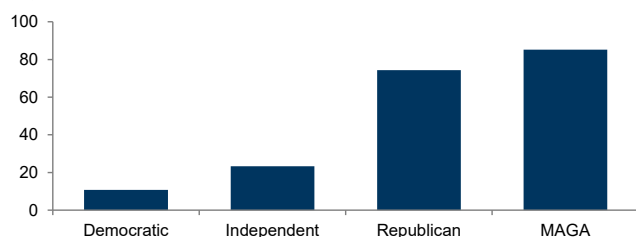
Alec Phillips explores US political pressures amid the Iran war and the Administration's response to the energy price shock

President Trump and his cabinet have consistently indicated the Iran war will likely last 4-6 weeks. On March 15, Energy Secretary Wright said it would end "in the next few weeks" and on March 17, Trump described the timetable as "a couple more weeks, not much longer." While US military objectives—elimination of Iran's nuclear program, missile capabilities, and naval assets—will likely drive US decisions, the economic and political costs of the war could become constraints.

So far, the lack of broad public backing for the war does not appear to be weighing on support for the Administration or congressional Republicans. Polls conducted before and after the first US strikes show Trump's approval rating holding steady at 42.6%, while support for Republicans in this year's midterm election is also roughly unchanged (Democrats still lead by 4pp). Importantly, President Trump's base appears to broadly support the war, with self-reported MAGA supporters viewing the war even more favorably than Republicans overall.

President Trump's base appears to broadly support the war

Support for the war/strikes on Iran, %



Source: YouGov, NBC, CNN, Angus-Reid, Goldman Sachs GIR.

But public opinion may become a constraint if retail gasoline prices, currently hovering around \$4/gallon (national average), rise further. Financial and commodity markets could also play a role. On March 17, President Trump indicated that markets influenced his thinking on striking Iran, citing record equity market prices as providing an opening to enter the conflict.

A soft check on the war's duration might come from the War Powers Resolution of 1973, which requires the president to obtain congressional authorization for the use of military force within 60 days. While unlikely to force a change in military strategy, nearing the 60-day mark could raise public pressure to clarify objectives and endpoints.

Trump also signaled that his planned early-April meeting with Chinese President Xi would be postponed by around six weeks. With this meeting seemingly unlikely to occur while the conflict remains ongoing, the timing might also provide some insight into how Trump is thinking about the duration of the conflict. In the meantime, the White House has taken a few steps to lower energy prices:

- **Release from the Strategic Petroleum Reserve (SPR):** In coordination with IEA members, the US announced a draw of 172mb over 120 days, or an average of 1.4mb/d. In theory, the total amount or daily rate could increase, as the SPR holds 415mb with a theoretical maximum draw rate of around 4mb/d for 30 days and around 2mb/d after 60 days. On March 19, Treasury Secretary Bessent suggested US

SPR releases could exceed the recently-announced amounts. However, we expect the Administration to be cautious for now, as US SPR stocks are already below 60% of capacity following the 2022-2023 drawdowns and would drop to 33% by mid-year under the recently-announced plan.

- **Sanctions relief:** Treasury waived sanctions on purchases of Russian crude and refined products on the water as of March 12 (estimated at 130mb). On March 19, Bessent suggested the US might similarly waive sanctions on Iranian oil on the water (estimated at 105mb). Treasury also provided a general license to purchase Venezuelan oil that, unlike the Russia-related waiver, is not explicitly temporary and applies to new production as well as to oil on the water.
- **Jones Act waiver:** The 1920 Jones Act prohibits shipping between US ports by foreign-built/flagged/owned vessels. The Administration waived this rule for 60 days, which could ease oil and refined products transport from the Gulf Coast (including the SPR) to other US regions (particularly the East Coast). This might incrementally reduce fuel prices and also signals the Administration's willingness to temporarily break with policies it supports to address higher energy costs.

Further steps are possible. The Administration seems likely to waive limits on gasoline's ethanol content in summer, allowing for a greater share of lower-cost ethanol in gasoline blends.

The President also likely has the authority under IEEPA to ban the export of crude and refined products and might become more inclined to use it if disagreements with allies reemerge over military assistance in and around the Persian Gulf. That said, Wright said on March 12 that an export ban was not under consideration, and media reports indicate White House officials reiterated this to oil executives on March 19. It is also far from clear that a ban would benefit US consumers, and it could discourage domestic shale oil production. So, while very plausible, we don't view such a ban as the base case.

Other policy changes would require congressional support. The most obvious would be a suspension of the \$0.18/gallon federal gas tax, which raises around \$30bn/yr. We think the odds lean against it, as a good chance exists that prices would have started to fall by the time Congress could agree. While we expect a few states to consider suspending their gas taxes, which collectively raise \$42bn/yr, as some did in 2022, a broad suspension seems unlikely as states no longer have the fiscal cushion they did following Covid-related federal fiscal relief.

Beyond fuel taxes, the recent developments look likely to restart talk of a second party-line fiscal package passed through the reconciliation process, but we remain skeptical that major legislation along these lines will pass this year. Trump looks likely to soon request \$100-200bn in supplemental defense funding, but Democrats will likely object and thin Republican majorities make passage of a party-line reconciliation bill difficult. However, if such a spending package gains political momentum, it could become a vehicle for other items like a gas tax holiday or affordability-focused measures. For now though, we do not expect such a package to pass this year.

Alec Phillips, Chief US Political Economist

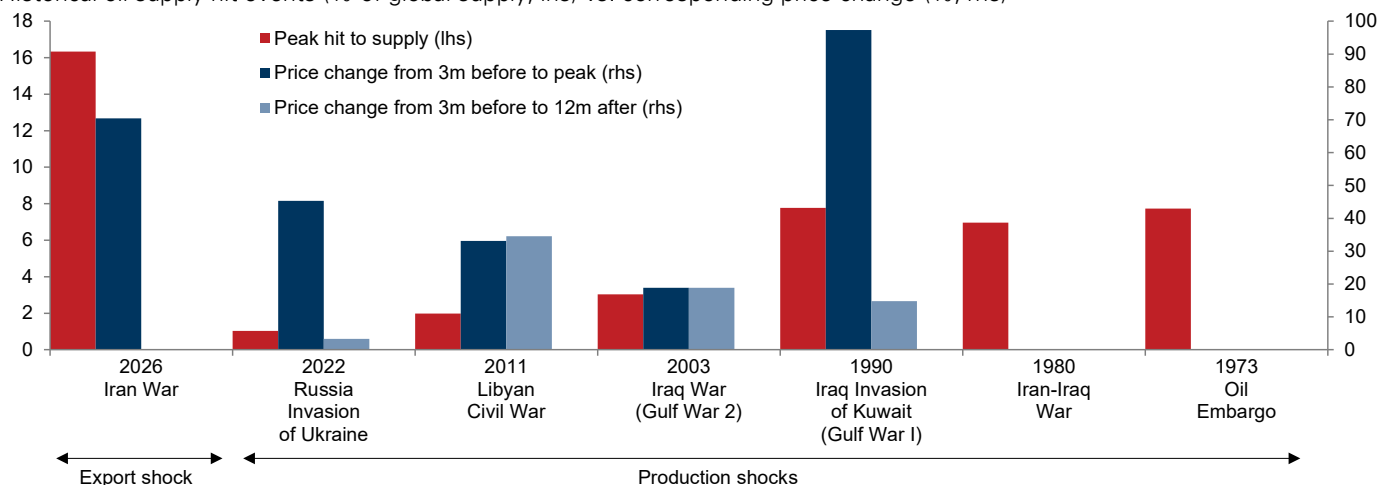
Email: alec.phillips@gs.com
Tel: 202-637-3746

Goldman Sachs & Co. LLC

The oil supply disruption, visualized

The size of the current hit to oil flows from the Persian Gulf marks the largest oil supply shock in history...

Historical oil supply hit events (% of global supply, lhs) vs. corresponding price change (% , rhs)

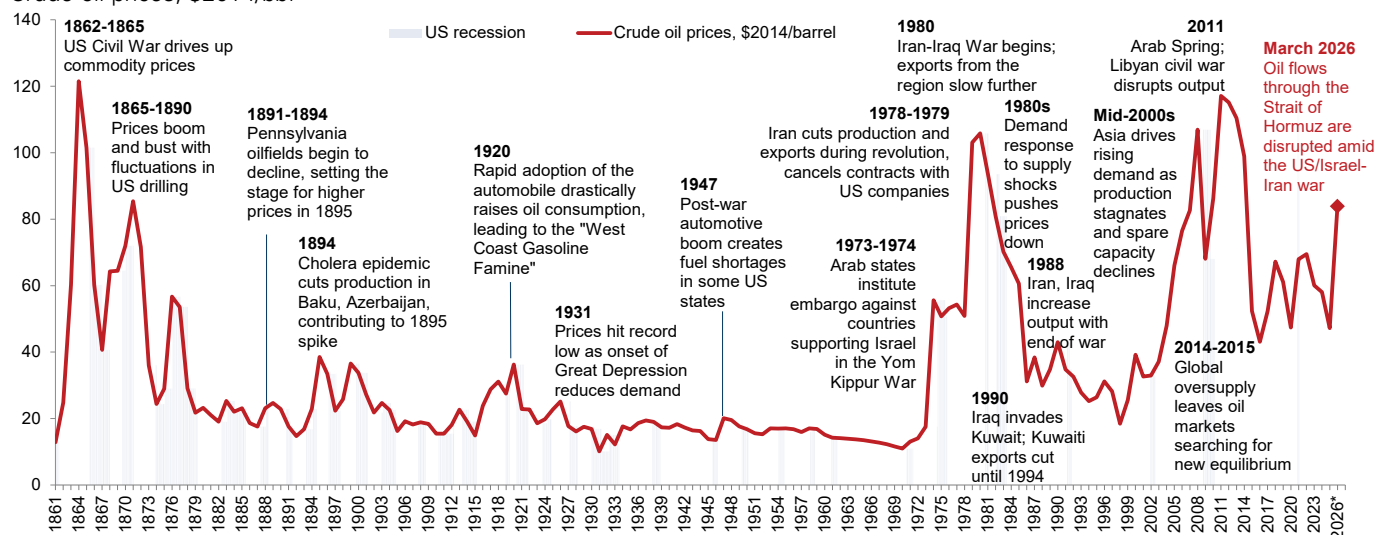


Note: Peak price is the highest daily close price. For the 2026 Iran War, peak hit to supply is defined as the estimated latest hit to Persian Gulf oil exports based on reported vessel count; peak price is taken to be the highest realized daily close price since February 28.

Source: IEA, ICE, Goldman Sachs GIR.

...which has led oil prices sharply higher, although the oil price response is so far smaller than during the 2003 Iraq War and 1980 Iran-Iraq war...

Crude oil prices, \$2014/bbl

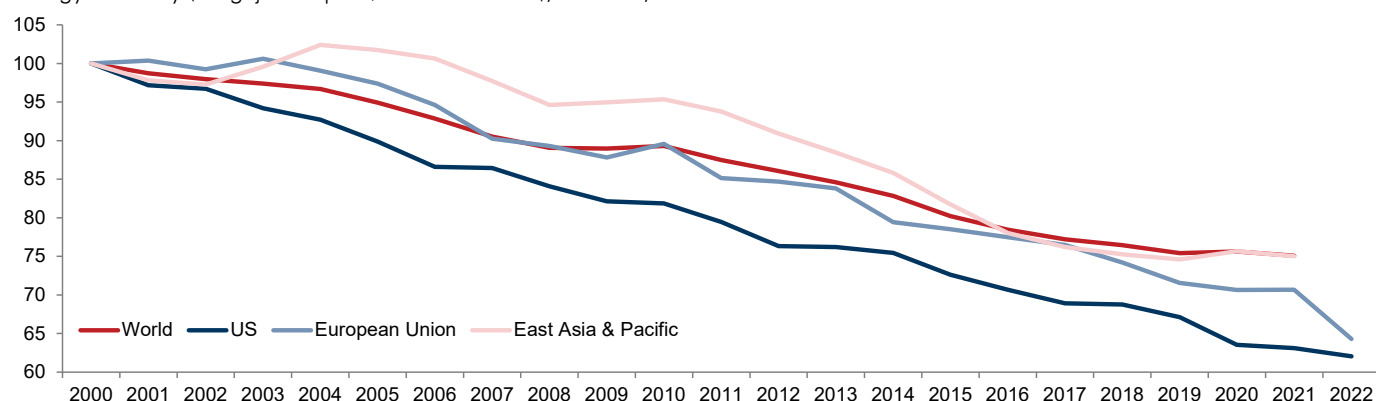


Note: 2026 is year-to-date data.

Source: Bloomberg, BP, Federal Reserve, Haver Analytics, various news sources, Goldman Sachs GIR.

...potentially as the world is much less vulnerable to oil price shocks now than it was in prior decades amid declining energy intensity

Energy intensity (megajoules per \$2021 PPP GDP), indexed, 2000=100



*Data as of 2022 for EU and US; data as of 2021 for World and East Asia & Pacific.

Source: World Bank, Goldman Sachs GIR.

Special thanks to GS Commodities Research analyst Alexandra Paulus for data.

Interview with Kevin Donegan

Vice Admiral Kevin Donegan, USN (ret) served as Commander of the US Navy's Fifth Fleet, Commander of the 32 Nation Combined Maritime Forces in the Middle East, and Director of the Navy Staff and Deputy Chief of Naval Operations for Operations, Strategy and Planning at the Pentagon. Below, he argues that the US has made significant progress in achieving its military objectives in the Iran conflict, but restoring Strait of Hormuz oil flows to normal levels won't be achieved through military convoys alone.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: What are the US military objectives in the Iran conflict?

Vice Admiral Kevin Donegan:

Despite all the political back and forth, the US military campaign's objectives have always been clear and stable: destroy the Iranian national security apparatus' ability to project power

outside their borders by degrading Iran's extensive ballistic missile and attack drone capabilities and the naval capabilities used to hold the economic chokepoint of the Strait of Hormuz hostage. The US has unsurprisingly done a good job in defanging Iran and degrading those capabilities. The US' targeting of Iran's mobile launchers and command-and-control nodes and use of Low-cost Uncrewed Combat Attack Systems (LUCAS) has significantly reduced Iran's ability to launch ballistic missiles and drones—Iran went from launching 1,000 missiles on day one of the conflict to a few hundred each day after during the first week to now only around several dozen missiles a day. The US Navy has also sunk or damaged almost all Iranian Navy vessels.

Iran has retaliated in several ways, most of which were expected—the US always knew Iran would hold the Strait of Hormuz hostage and attack regional US military bases and Gulf allies' territory. What has been unexpected is Iran's willingness to attack Gulf nations' regional civilian infrastructure—hotels, airports, etc.—in an effort to drive the Gulf states to pressure the US to end the conflict. But the Gulf states haven't done so, either publicly or behind the scenes, and are instead more worried about the prospect of the US ending its military campaign too early. The last thing they want is for the conflict to end with a half-defanged Iran.

“Despite all the political back and forth, the US military campaign's objectives have always been clear and stable: destroy the Iranian national security apparatus' ability to project power outside their borders.”

Jenny Grimberg: Can the US ultimately “finish the job” without putting boots on the ground?

Vice Admiral Kevin Donegan: Yes. The US undoubtedly has enough firepower to accomplish its military objectives. The US initially used advanced at-range weapons like Tomahawk Land

Attack Missiles (TLAMs) and extended-range missiles to gain air superiority over Iran. Now that it has air superiority and can conduct land, sea, and air operations without Iranian interference, it has shifted to using its ample arsenal of lower-tech and lower-cost weapons. The US' ability to achieve its objectives using its military hardware has never been in doubt.

What's likely impossible to achieve without boots on the ground is regime change. The first wave of US/Israeli strikes changed the regime's leadership, not the regime itself, which remains even after successive waves of strikes. The Islamic Revolutionary Guard Corps (IRGC), the core pillar of the regime, is deeply woven into Iranian society, from internal security to banking institutions to construction companies. It also has an internal security apparatus that oppresses the population and its own intelligence arm. And it controls Iran's missiles, drones, and sea mining capabilities. So, the IRGC isn't a traditional military, and the US would have to dismantle most of the Iranian government system to remove the IRGC, which would require ground forces. But again, regime change has never been a US military objective. President Trump and the Israelis have mentioned it, but the Commander of US Central Command (CENTCOM) Admiral Brad Cooper and Chairman of the Joint Chiefs of Staff General Dan Caine were never given this as an objective.

“The US has unsurprisingly done a good job in defanging Iran and degrading [Iran's] capabilities.”

Jenny Grimberg: Even if the US has gone a long way toward achieving its military objectives, won't Iran's asymmetric capabilities—its low-cost drones and sea mines—present ongoing challenges?

Vice Admiral Kevin Donegan: While the threat from ballistic missiles will likely be constrained as Iran remains leery of firing them for fear of exposing launch and storage sites, prolific weapons will remain a threat. Iran manufactures attack drones by the tens of thousands. This drone threat will be severely degraded but will never be zero. As a result, we can expect this threat, which is most effective against soft targets, to linger even after the US military operation ends.

Iran also has plenty of sea mines, ranging from the simple to the sophisticated, that it can threaten to lay in vital waterways. But the US and Israeli military campaign has destroyed many Iranian minelaying vessels and storage bunkers. So, Iran can no longer deploy sophisticated, anchored mines in specific

locations, instead having to resort to deploying drifting mines from non-military vessels like fishing boats. These mines are unpredictable because they are a slave to the strong currents in the Strait of Hormuz and may not stay in the areas that are problematic for ships.

Laying sea mines in the Strait of Hormuz is also ultimately counterproductive to Iran's needs because the vast majority of Iran's oil exports flow through it and up to 60% of Iran's government budget comes from oil exports, mostly to China. For that reason, the rhetoric that has come out of Iran is not that it has closed the Strait, but that it *controls* the Strait.

“While the threat from ballistic missiles will likely be constrained as Iran remains leery of firing them for fear of exposing launch and storage sites, prolific weapons will remain a threat.”

Jenny Grimberg: If Iran wants to continue causing trouble in the Strait of Hormuz, is it practical to think that convoys could successfully transit through it?

Vice Admiral Kevin Donegan: I'm confident the US and its allies have the capability to escort commercial ships through the Strait. The US has long rehearsed how to keep the Strait of Hormuz and the Bab al-Mandab Strait open, and under much more challenging conditions than exist today in terms of Iranian capabilities. The most recent such exercises took place in early 2025 with 30 nations participating.

The US can obviously deploy meaningful capabilities, but so can many other countries, whether it's ships, staff, or experience. The US 5th Fleet Commander knows exactly what each nation can contribute because this exercise has been rehearsed annually for the past 15 years. France and Australia have very capable ships, and France has experience owing to its fight against the Houthis in the Red Sea. Greece has also received substantial training because of the Houthis in particular. The UK, Germany, Japan, South Korea, Turkey, Italy, Spain, and Egypt are also equipped to participate to varying degrees.

The basic idea would be to create a mobile sanctuary of security as the ships transit the Strait, which would include air cover, ships, drones, satellite assets, and intelligence. Securing a convoy is actually more achievable today than what's been planned for because of the US' air superiority over the Iranian coast and the severe degradation of Iranian military capability. And while Iran, even with no radar capabilities, will be able to locate the convoys, which would certainly be targets, the US will likely have little tolerance for hostile boats in the water near the convoys and is well-equipped to deal with the residual Iranian capabilities launched from shore to attack the convoys.

Whether the US and other countries have the political will and desire to participate in a convoy is a different question, and one we're currently seeing play out. The Trump Administration said that restoring the free flow of commerce through the Strait of

Hormuz should be everybody's job. While many countries initially refused or were noncommittal, the leaders of the UK, France, Germany, Italy, the Netherlands, and Japan have since issued a joint statement expressing their “readiness to contribute to appropriate efforts to ensure safe passage through the Strait.” Caine has also said that some allies are using Apache attack helicopters to “handle one-way attack drones” as part of a stepped-up effort to reopen the Strait. How this issue evolves bears close watch.

Jenny Grimberg: Even if convoys could allow some ship passage through the Strait, could they do so at the scale required to restore the flows the global economy needs?

Vice Admiral Kevin Donegan: No. It is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels. They don't scale well because ships must be escorted out as well as back in.

So, convoys may only be able to restore potentially 20% of normal oil flows, with overland pipelines adding another 15-20%, though this will depend significantly on how the conflict ends and how Iran reacts to the first few escorts. The bottom line is that no on-off switch exists here—the conflict won't stop one day and oil flow normally the next. A negotiated ceasefire would be the fastest path to restoring normal flow. And even then, shipping, insurance, tanker, and LNG companies would need time to gauge Iran's behavior and gain trust before resuming transit.

“I'm confident the US and its allies have the capability to escort commercial ships through the Strait... [but] it is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels.”

Ultimately, restoring oil flows to pre-conflict levels will require Iran to stop harassing ships in the Strait. It may not be incentivized to do so now, especially if the IRGC is currently making decisions as opposed to more seasoned political figures like the late Ali Larijani that tend to think more about the longer-term picture. But my sense is that eventually the Iranian regime will, because it needs cash and sanctions relief to appease its population and wants to ensure its survival, which is its singular most important goal.

The US also has cards to play. It has the capability to hold crucial Iranian flows from Kharg Island hostage. The US could also stop the “shadow fleet” carriers on either side of the Gulf that are moving Iranian oil. And regional Gulf nations like Oman and Qatar can also apply pressure. So, restoration of flows is not so much a question of convoys but of incentives and leverage on all sides, and no switch exists to just turn on flows when the conflict ends.

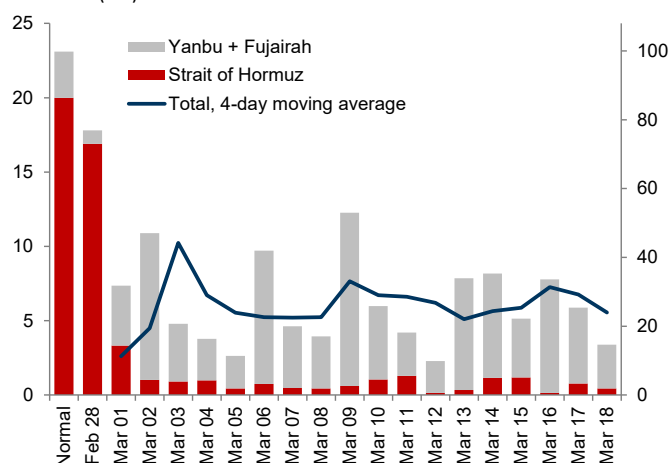
Sizing the energy shock

The GS commodity research team explores energy disruptions amid the Iran war

As the world continues to digest the largest shock to energy supplies in history, we assess the impact on prices, finding that price risks are skewed to the upside both in the short term and over the longer term given the potential for lengthier disruptions and persistent supply losses.

The estimated hit to oil flows from the Persian Gulf is 17.6mb/d, making today's shock the largest in history

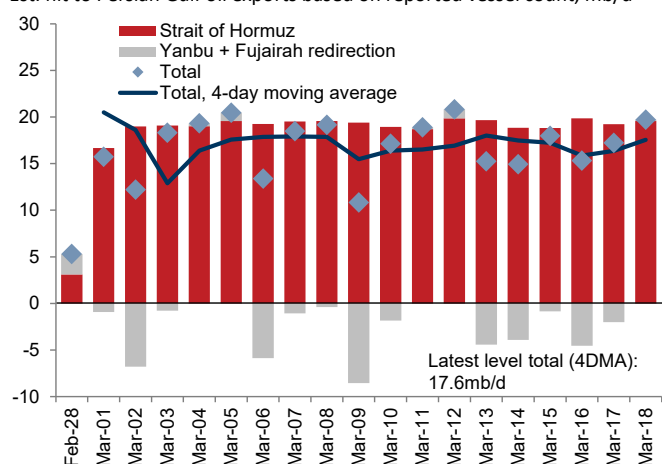
Est. oil flows from Persian Gulf based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Normal flows are assumed to be 20mb/d for Strait of Hormuz and 2025 average for Yanbu (1.4mb/d) + Fujairah (1.7mb/d).

Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Est. hit to Persian Gulf oil exports based on reported vessel count, mb/d



Note: Hit is calculated relative to normal flows.

Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Current state of the oil shock: the largest in history

We estimate that the hit to total oil flows from the Persian Gulf stands at 17.6mb/d (17% of global supply), making today's shock the largest in history and 18 times larger than the peak April 2022 hit to Russian oil production. Oil flows through the Strait of Hormuz are down 97% from their normal levels (0.6mb/d versus 20mb/d normal) as the physical risks of crossing the Strait remain very high, with the latest confirmed attack on an oil tanker occurring three days ago. Some flows are being redirected to pipeline-adjacent ports outside of the Persian Gulf, most notably Yanbu (Red Sea, Saudi Arabia) via the Saudi East-West pipeline and Fujairah (Gulf of Oman, UAE)

via the Habshan-Fujairah pipeline, but the extent of redirection is capped by the combined spare capacity of the two pipelines—our latest estimate of net redirections stands at just 1.8mb/d.

Near-term oil prices: high risk premium required

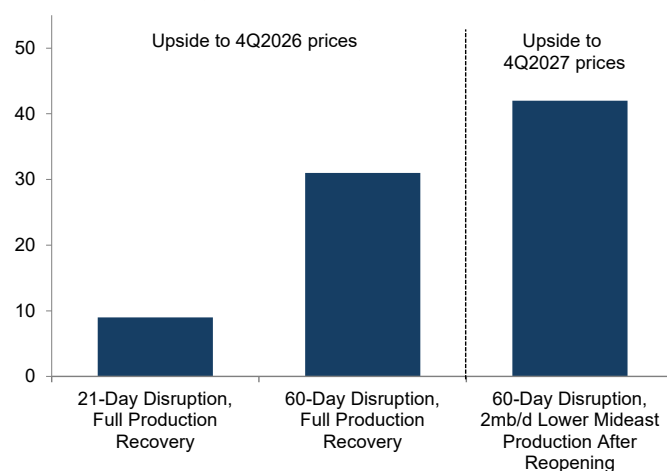
We expect oil prices to continue trending higher in the short term, until the market gains more certainty about how long the disruption will last, especially in light of increasing strikes on energy infrastructure. This risk premium is required to generate enough precautionary demand destruction to hedge against the risk that commercial oil inventories might fall to critically low levels. Brent is likely to exceed its 2008 all-time high if depressed flows keep the market focused on the risk of lengthier disruptions. Any rise in market-perceived risks of US export restrictions would likely further widen the Brent-WTI price gap.

Medium-term oil prices: the longer, the higher

Given the fluidity of the situation, we consider three scenarios to assess the potential direction of oil prices over the medium term.

Given uncertainty around the longevity of the conflict and associated oil disruptions, we consider three potential scenarios

GS estimated impact on 4Q2026/4Q2027 Brent oil prices under different Strait of Hormuz disruption and recovery scenarios, \$/bbl



Source: Goldman Sachs GIR.

For the first two scenarios, we estimate 2026Q4 prices, relative to a no-Iran shock counterfactual of \$62/bbl, as a function of 1) cumulative hits to production and commercial oil inventories, 2) global policy responses, and 3) a boost from positioning.

In a (most optimistic) first scenario where flows through the Strait of Hormuz gradually recover from here and return to pre-war levels within a month, we would expect

prices to average \$71/bbl in 4Q2026. This supply shock would translate into a 6% (617mb) hit to global commercial oil inventories. The release of strategic petroleum reserves (SPR) by [IEA member states](#) and China, as well as the uptake of Russian oil on water, would likely offset 50% of the hit, translating into \$6/bbl upside to fair value prices. We assume positioning would provide an additional \$3/bbl boost, bringing our estimate of 4Q2026 price upside to \$9/bbl.

In a second scenario where flows remain disrupted for 60 days (through April 28th), we think 4Q2026 prices would average \$93/bbl, as the supply shock would lead to a

significantly larger nearly 20% (1816mb) hit to global commercial oil inventories. Even with increased releases of global SPR (although constrained by logistical speed limits) and Russian oil on water, policy responses would likely only offset about 30% of the hit, translating into upside of \$26/bbl. Combined with an additional \$5/bbl boost from positioning, we estimate 4Q2026 price upside at \$31/bbl.

Finally, in a third, more extreme scenario where Hormuz flows remain very low for 60 days and significant scarring to long-run supply occurs (specifically, that Middle Eastern production remains 2mb/d lower after the Strait opens), we estimate that prices would reach about \$110/bbl in 4Q2027, consisting of over \$40/bbl of upside relative to a no-Iran shock counterfactual of roughly \$70/bbl.

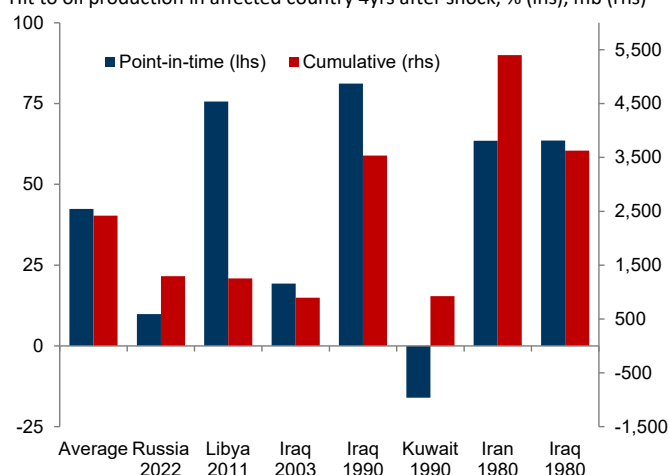
Risks to long-term oil prices

We also consider four risks to long-term oil prices—two on the supply side and two on the demand side.

On the supply side, the primary long-term risk to prices is the potential for oil supply to remain low for longer if production potential is damaged, which would pose upside price risk. The five prior largest supply shocks (Iran-Iraq War, Gulf War I, Gulf War II, Libyan Civil War, and Russian invasion of Ukraine) suggest that production four years after the shock remains over 40% below normal levels, largely driven by damage to oil infrastructure and low investment. This risk could be particularly significant to today's shock given that Iran and the seven other Persian Gulf countries collectively produced 30% of the world's crude oil in 2025. Additionally, roughly 25% of the region's crude oil is produced offshore, where engineering and geological complexity requires significantly longer ramp-up times than for onshore production.

The five prior largest supply shocks suggest that production four years after the shock remains over 40% below normal levels

Hit to oil production in affected country 4yrs after shock, % (lhs), mb (rhs)



Note: We estimate point-in-time production hit as percent decrease in a country's production four years after the shock relative to the pre-shock oil production. We estimate cumulative production hit as a total decrease in a country's production over the four years of the shock relative to the pre-shock oil production.

Source: IEA, EIA, Goldman Sachs GIR.

But oil supply could be higher than expected if OPEC deploys spare capacity, i.e., production that can be brought online quickly and sustained for a few months, to help stabilize the market, which would limit the upside to prices.

On the demand side, faster strategic stock building from 2027 on low end-2026 reserves and higher SPR targets in response to lingering uncertainty would pose an upside risk to prices. But slowing demand should high prices accelerate fuel efficiency gains and shifts to other fuels, or the energy shock weigh meaningfully on GDP growth, would pressure prices in the other direction.

On net, we think risks to oil prices are skewed to the upside in the short term and over the longer term. The persistence of several prior large supply shocks underscores the risk that oil prices may stay above \$100/bbl for longer in risk scenarios with lengthier disruptions and large persistent supply losses.

European natural gas: risk of a tighter longer-term outlook

Similar to oil, on the natural gas side the recent escalation of events and the associated increased risk that supply disruptions linger have increased the risk premium embedded in European natural gas prices, which are now more than 90% above pre-war levels, at €61/MWh (\$21/mmBtu). These price levels incentivize max gas-to-coal switching in power generation, but are not high enough to incentivize switching from gas into oil products.

Assuming Qatari LNG production begins to ramp up from early April, near-term prices would likely rise modestly, with the Apr/May TTF contract rising to €75/70MWh to reflect a higher risk premium and higher front-loaded demand destruction (gas-to-oil switching), until the market sees an actual increase in supply. As this early demand destruction creates a greater supply cushion, natural gas prices should be able to price gradually lower vs coal, with 2H26 TTF in the €30-40/MWh range.

However, the longer it takes for the ramp-up in Qatari LNG production to start, the longer European gas prices would likely stay in the oil switching range, above €70/MWh, to generate incremental demand destruction. A particularly longer disruption, with Qatari LNG production at zero for two months or longer, could lift prompt TTF contracts further still, likely close to €100/MWh (\$35/mmBtu), to broaden industrial demand destruction across Europe and Asia in order to offset these supply losses ahead of the winter. And the long-term (3-5 years) damage to Qatari production capacity inflicted by recent attacks, worth 3% of global LNG supply, along with announced delays to the construction of Qatar's new LNG trains, would likely slow the largest ever LNG supply growth wave from 2027 we've expected, and delay the resulting significant downside pressure on European gas prices.

Samantha Dart, Co-head of Global Commodities Research

Email: samantha.dart@gs.com
Tel: 212-357-9428

Goldman Sachs & Co. LLC

Daan Struyven, Co-head of Global Commodities Research

Email: daan.struyven@gs.com
Tel: 212-357-4172

Goldman Sachs & Co. LLC

Alexandra Paulus, Commodities Research Strategist

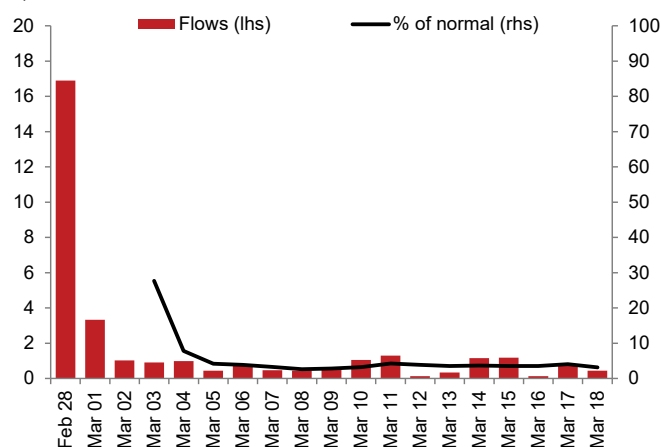
Email: alexandra.paulus@gs.com
Tel: 212-902-7111

Goldman Sachs & Co. LLC

Disruptions caused by the Iran conflict...

Estimated daily oil flows through the critical Strait of Hormuz are down 90%+ from normal levels...

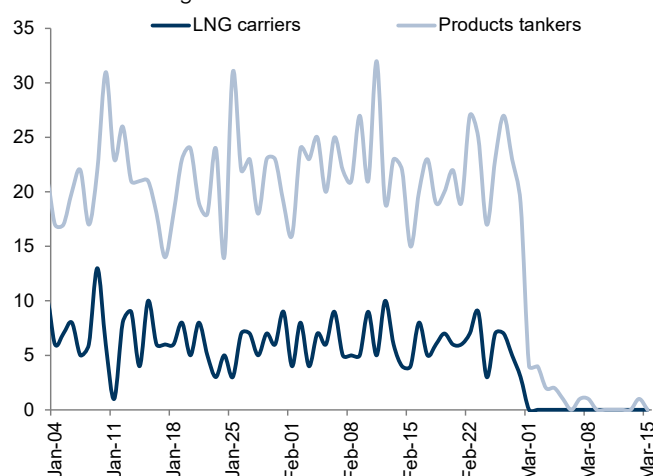
Estimated oil exports through the Strait of Hormuz based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Vessel count as a % of normal is based on an average of S&P and Kpler data on the daily # of oil tankers crossing the Strait of Hormuz relative to the Jan 1-Feb 14 average for S&P and Feb 28 for Kpler. Flows are estimated by applying this proportion to assumed normal flows of 20mb/d (average over last ~8 years). Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Gas and refined product tankers have also largely stopped transiting the Strait of Hormuz amid the conflict...

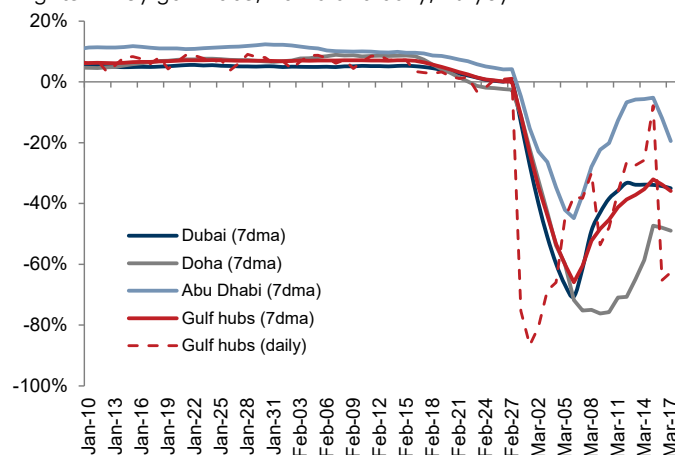
Number transiting the Strait of Hormuz



Source: Clarksons Research, Goldman Sachs GIR.

The conflict initially resulted in a significant reduction in flights to and from the Gulf, though some of that has now recovered

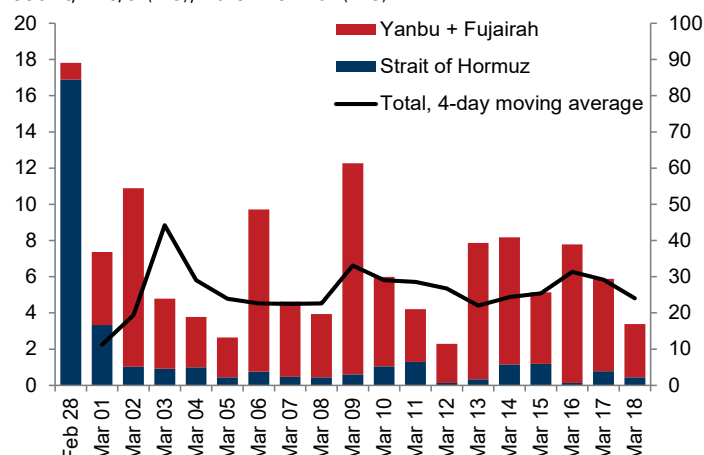
Flights in key gulf hubs, 7dma and daily, % yoy



Source: OAG, Flight Aware, Goldman Sachs GIR.

...leading to increased redirection via the ports in Yanbu (Red Sea, Saudi Arabia) and Fujairah (Gulf of Oman, UAE)

Estimated oil flows from Persian Gulf based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Normal flows are assumed to be 20mb/d for the Strait of Hormuz and 2025 average for Yanbu (1.4mb/d)+Fujairah (1.7mb/d).

Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

...and shipments of chemical products from the Middle East, where sizable amounts of such products are produced, have been disrupted according to major trade publications

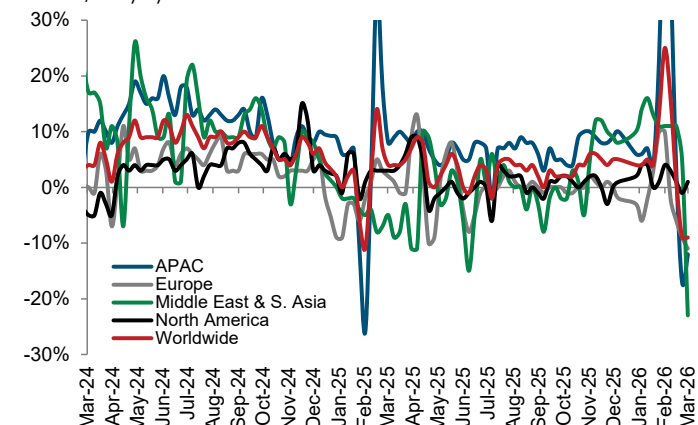
Regional exports by chemical chain as a share of global exports

Chemical product	Iran	UAE/Kuwait/Qatar	Eastern Saudi Arabia
Ammonia	~5% of global exports	~4% of global exports	>10% of global exports
Urea	>5% of global exports	>10% of global exports	>5% of global exports
Phosphate	N/A	N/A	>10% of global exports
Sulfur	<5% of global exports	>20% of global exports	>10% of global exports
Polyethylene	>5% of global exports	>5% of global exports	>5% of global exports
MDI	Small	N/A	>10% of global exports
Caustic	Small	<5% of global exports	Small
PVC	N/A	N/A	N/A
MTBE	<5% of global exports	<5% of global exports	>15% of global exports
TiO2	N/A	N/A	N/A
Methanol	>15% of global exports	Small	>15% of global exports

Source: ICIS, Argus, IFA, COMTrade, CMA, S&P Global, Goldman Sachs GIR.

Global air cargo volumes, which Middle East airlines account for 13% of, have also fallen amid the conflict

Global top-20 forwarders' chargeable air cargo weight by origin, 14dma, % yoy



Source: World ACD, Goldman Sachs GIR.

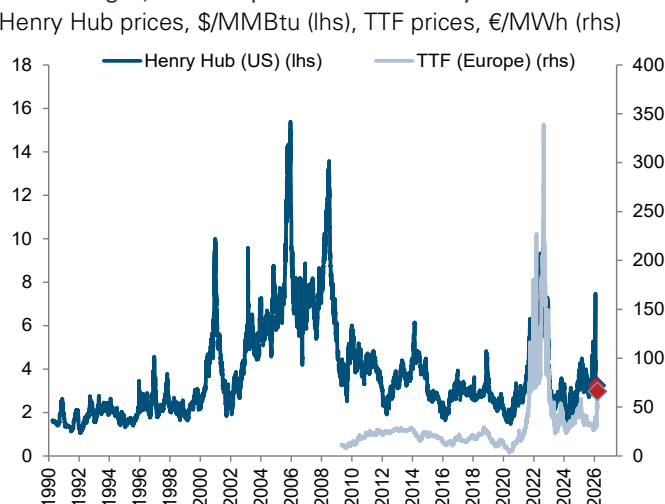
...have led to increased prices

Oil prices have significantly increased, now exceeding \$100/bbl, though they remain below the highs of the energy crisis
Brent crude oil prices, \$/bbl



Source: Bloomberg, Goldman Sachs GIR.

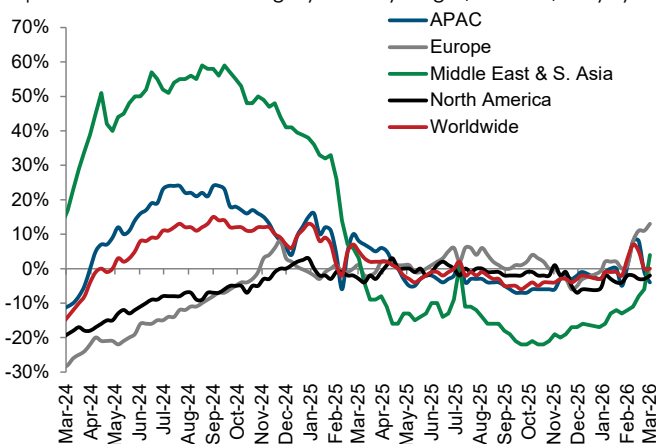
European natural gas prices have nearly doubled since the conflict began, while US prices have risen only ~11%
Henry Hub prices, \$/MMBtu (lhs), TTF prices, €/MWh (rhs)



Source: Bloomberg, Goldman Sachs GIR.

Air cargo rates have risen for cargo from Asia and Europe

Top-20 forwarders' air cargo yields by origin, 14dma, % yoy



Source: World ACD, Goldman Sachs GIR.

US gasoline prices have also risen, from around \$3/gallon before the conflict began to nearly \$4/gallon currently

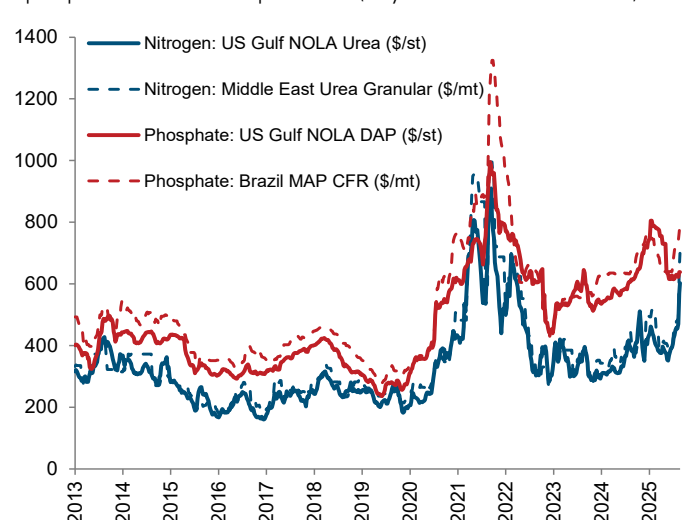
US all grades all formulations retail gasoline prices, \$/gallon



Source: EIA, Haver Analytics, Goldman Sachs GIR.

Fertilizer product prices have increased

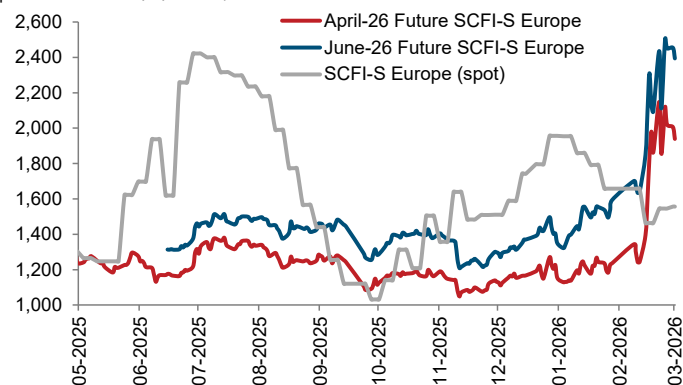
Spot prices of fertilizer products (key benchmarks for each)



Source: Bloomberg, Goldman Sachs GIR.

In container shipping, future freight rates have increased as carriers have implemented fuel and emergency surcharges

Future SCFI-S (Shanghai Containerized Freight Index based on Settled Rates) Europe (futures contracts) (index, base=1,000 points as of 6/1/2020)



Source: Clarksons Research, Bloomberg, Goldman Sachs GIR.

Special thanks to GS Commodities Research team (Alexandra Paulus), US Chemical Equity Research team (Duffy Fischer and Jordan Lee), and European Transportation Equity Research team (Patrick Creuset and Nathan Arnaud) for data.

Iran conflict: growing economic impacts

Joseph Briggs assesses the global economic impacts from the ongoing Iran conflict

As the Iran conflict ends its third week, we continue to think that the biggest economic impacts will come through higher energy prices, which will slow growth and boost inflation in most countries, with the risks skewed toward more growth downside and inflation upside. But we see limited risk that the large second-round inflation effects that were a prominent feature of the 2021-2022 pandemic shock will re-emerge, as—unlike in 2021-2022—the major risk to global supply and inflation today is mostly confined to energy.

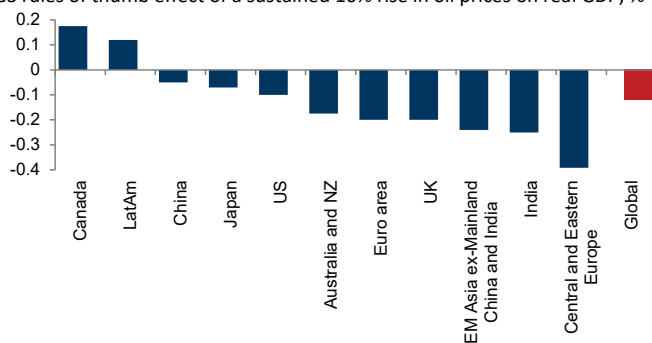
Economic impacts: rules of thumb

To gauge the economic impacts of the conflict, we rely on rules-of-thumb estimates of how higher oil prices affect GDP and consumer prices. On the growth front, our rules-of-thumb imply that each 10% increase in oil prices lowers global GDP by a bit more than 0.1%, reflecting real income and spending drags that are only partially offset by boosts to income and spending in oil exporters such as Canada and some LatAm economies.

On the inflation front, our rules-of-thumb imply that each 10% rise in oil prices raises global headline inflation by 0.2pp, with larger impacts in Asia ex-China and Europe, owing to increases in energy costs and indirect effects on downstream production. The core inflation impacts are smaller, ranging from 3-6bp for each 10% increase in oil prices depending on the country.

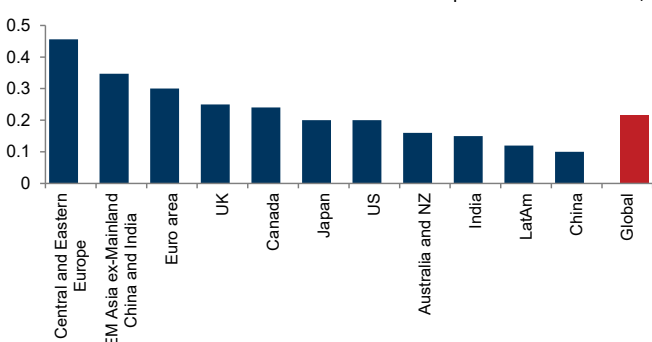
Each 10% increase in oil prices lowers global GDP by 0.1%...

GS rules of thumb effect of a sustained 10% rise in oil prices on real GDP, %



...and boosts global headline CPI by 0.2%

GS rules of thumb effect of a sustained 10% rise in oil prices on headline CPI, %



Source: Goldman Sachs GIR.

As such, a sustained rise in oil prices will likely have a sizable impact on the global economy. To level set, prior to the conflict, we expected global GDP growth of 2.9% this year on an annual average basis and headline inflation of 2.3% on a Q4/Q4 basis.

Our rules of thumb imply that the current three-week disruption will result in a 0.3% drag to global GDP and 0.5-0.6% boost to global prices. In a 30-day disruption scenario, we estimate a 0.5% headwind to global GDP and 0.9% boost to global prices. And in a 60-day disruption scenario, we estimate a 0.9% headwind to global GDP and 1.7% boost to global prices. And the longer the Strait of Hormuz disruptions extend, the greater the hit to growth and boost to inflation.

Even on top of these direct macro implications of higher oil prices, we see macro risks further skewed toward growth downside and inflation upside for two reasons. First, the disruption of Qatari gas supplies, which account for around 20% of global LNG supply, poses additional challenges for Asia and Europe, especially after the recent attacks on energy infrastructure. While the US remains relatively insulated owing to its status as the world's largest natural gas producer, Asia and Europe's heavy reliance on LNG leaves prices and industrial production more exposed. Second, the recent tightening in financial conditions—our global FCI has tightened by 51bp since the conflict began—could slow growth further if sustained.

Not your brother's supply disruption

The recent selloff in front-end rates signals an expectation that global central banks will be especially sensitive to higher inflation given the memory of post-pandemic supply disruptions and associated inflation surge. While the size of this energy shock will likely prompt some caution among policymakers on rate cuts, a key difference from 2021-2022 is that today's supply shock appears concentrated in the energy sector, whereas the energy price increases of 2022 were only one aspect of a much broader global supply chain crisis and inflation surge. We see three reasons why the impacts on non-energy supply will likely be less severe this time around.

First, non-energy trade with Gulf economies accounts for only 1% of global trade, with major DMs and EMs even less exposed. By contrast, the 2021-2022 China and East Asia trade disruptions affected 20% of global trade.

Second, while Gulf exporters account for a large share of global production for certain chemical and metal products—including helium, sulfur, methanol, polyethylene, aluminum, and fertilizers—Gulf production of these products accounts for less than 0.02% of global GDP. By contrast, disruptions to semiconductor production in 2022 affected trade worth ~1% of global GDP. Even more importantly, these product disruptions are unlikely to create shortages in critical, non-substitutable inputs that would lead to broader production shutdowns.

Third, the problematic shipping disruptions of 2021-2022 appear less likely to repeat. Gulf economies are not major re-exporters of most products (aside from boats), non-tanker ocean shipping costs have ticked down since the conflict began, and the recent rise in airfreight prices, while notable, implies only a modest uplift to global inflation (<5bp). Taken together, this limits the risk that the large second-round inflation effects observed in 2021-2022 will re-emerge.

Joseph Briggs, Senior Global Economist

Email: joseph.briggs@gs.com
Tel: 212-902-2163

Goldman Sachs & Co. LLC

Markets: growth, the next shoe to drop?

Kamakshya Trivedi argues that growth downside could be the next shoe to drop in markets if the war extends and escalates

The Iran war and resulting spike in energy prices have reverberated sharply across global markets. Two particular features of the price action stand out. First, apart from positioning unwinds, global markets have priced this as an inflationary shock across DM and EM economies, with hawkish policy implications that have pushed rates higher and to a more limited extent, equity prices lower. Second, energy price shocks produce winners and losers, and the relative performance of assets, including FX and regional equities, have reflected those terms-of-trade (ToT) differences. Markets have been slower to focus on the growth implications of the energy shock, suggesting that growth downside could be the next shoe to drop.

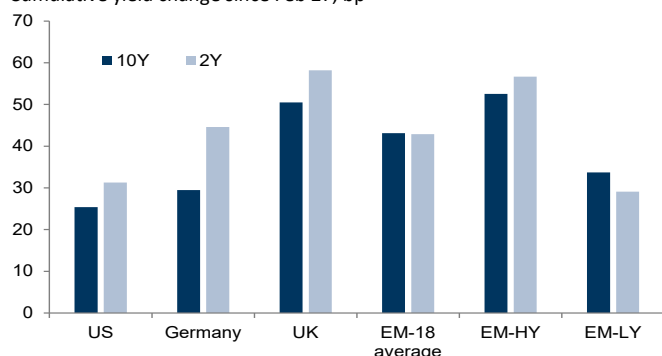
Interest rates: inflation fears drive a hawkish repricing

The ~50% increase in oil prices as a result of the effective shutdown of energy transit through the Strait of Hormuz was quickly reflected in inflation fears and interest rate pricing. That shift is understandable, and generally in line with the direction of revisions in our economic forecasts toward higher headline inflation, smaller increases in core inflation, and later interest rate cuts across the US, UK, and some EMs.

That said, the initial reactions across interest rates markets were even more acute, probably reflecting a couple of factors. First, coming into this shock, we and most investors expected a relatively benign inflation outlook for this year (vs. 2025) given soft wage inflation and structural considerations such as China's export disinflation and potential AI impacts. So, market participants were likely wrong-footed going into this shock. Indeed, the largest reactions occurred in markets that were priced for the most rate cuts, including UK front-ends in G10 and Hungary in EM. Second, the trauma of the 2022 inflation shock, during which interest rates rose much more than initially anticipated, is still fresh in market participants' minds. While it's certainly possible that the current energy shock can worsen, the sharp post-Covid economic uncoiling amid large fiscal support as well as the supply chain bottlenecks that led to higher and more sustained inflation in 2022 are absent today (see pg. 16).

Front-end yields have risen sharply as inflation fears erode rate cut pricing across G10 and EM

Cumulative yield change since Feb 27, bp



Source: Bloomberg, Goldman Sachs GIR.

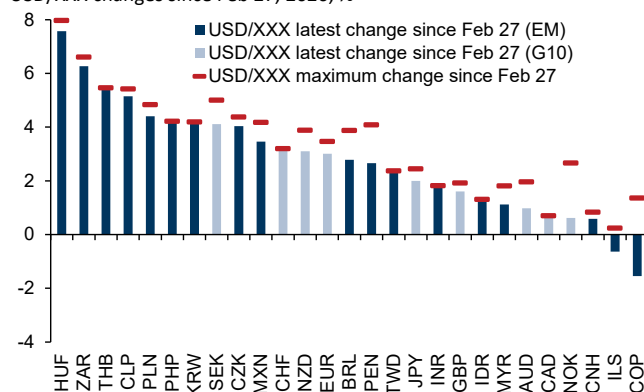
FX and equities: an energy ToT shock

In contrast to the broadly hawkish rates response, the FX response has been more differentiated. The Dollar has appreciated broadly, as expected during a global risk-off event that also fuels a sharp rise in energy prices, as the Dollar sits on the right side of the ToT shock. By contrast, most European and Asian currencies are on the wrong side of that shock, and as the war has progressed, global FX pairs have increasingly reflected those differences. Within G10, the currencies of energy exporters like Norway and Canada have outperformed. Likewise, within EM, the currencies of energy producers Malaysia and Brazil have outperformed, and the Colombian Peso has appreciated in nominal terms versus the Dollar. We expect the ToT shock to remain the dominant driver of FX markets for now and, as such, see these trends continuing over the near term.

In equities, positioning unwinds dominated the initial selloffs, with European and Asian markets that had seen the strongest performance in the past year underperforming the most. But the price action has since become more fundamentally driven, with markets such as India and CEEMEA, where earnings are more fundamentally vulnerable, experiencing only limited rebounds while North Asia and China markets, where earnings are more resilient to the shock, have modestly recovered. Looking forward, we expect this type of differentiation to extend as long as energy prices remain elevated.

Relative FX weakness versus the Dollar lines up along the ToT axis

USD/XXX changes since Feb 27, 2026, %



MENA economies: big hit, possible scars

Farouk Soussa lays out the near-term hit, and possible long-term scars, to MENA economies

The Iran war is unambiguously bad for business in the Gulf, resulting in sharp declines in oil and gas production and revenues as well as broader economic activity. Beyond the near-term hit, the war could also leave lasting scars on regional economies, though much will depend on the regional energy landscape and geopolitical order that emerges once the dust settles, which remains far from clear.

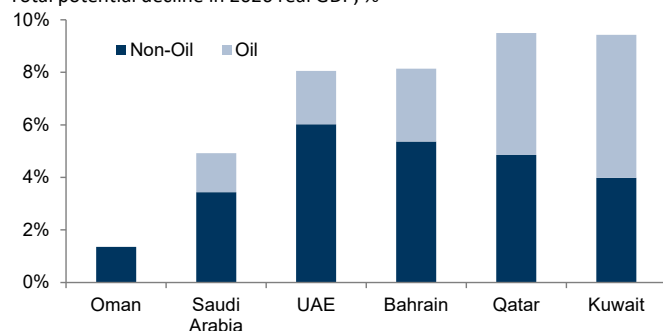
Hits to the oil and non-oil economies

The hits to oil and gas production are significant but uneven across the GCC. In a scenario in which the conflict depresses energy production at current levels until the end of April before gradually recovering to pre-war levels over a two-month period, we find that Kuwait and Bahrain would lose over a fifth of oil production this year. In Qatar, missile damage to the 77mtpa Ras Laffan LNG plant will shut in 17% of the country's LNG production for the next 2-3 years, according to Qatar Energy CEO Saad Al-Kaabi. We estimate that total production there could be down as much as one-third this year. Saudi Arabia and the UAE, which both have capacity to divert flows away from the Strait of Hormuz, would lose around 10% and 15% of production, respectively. Oman, which sits outside the Strait, is largely unaffected from an oil production perspective.

Beyond oil and gas, the war is significantly negatively impacting near-term economic activity, including tourism, aviation, logistics, hospitality, domestic trade, manufacturing, and financial services. While the final impact remains uncertain, we estimate that the economic hit could be as large as, if not larger, than that experienced during the Covid pandemic. The combined effect of oil and gas production losses and a broader decline in economic activity could see GCC economies contract this year by more than any other time in the past 30 years.

GCC economies will see large hits to oil and non-oil output

Total potential decline in 2026 real GDP, %



Source: Haver Analytics, Goldman Sachs GIR.

Serious financial implications

The financial implications of the war are also serious. We estimate that the GCC collectively is losing roughly \$700m in oil revenues alone each day as the conflict endures, bringing total losses so far to over \$15bn. Two months of disruption at current levels and losses would approach \$80bn. And losses in natural gas and other energy supplies will only add to these figures. While such a loss would be manageable for GCC economies given their wealth, it would necessitate even more borrowing than we had anticipated prior to the war, which

could have implications for credit spreads in the medium term. Much, though, will depend on how oil and gas prices evolve after the end of hostilities. Should prices adjust slowly to their pre-war path, then higher earnings may compensate Gulf producers for the losses endured during the war. If, on the other hand, prices correct sharply downward, these losses will crystallize over the course of the year.

More questions than answers over the longer term

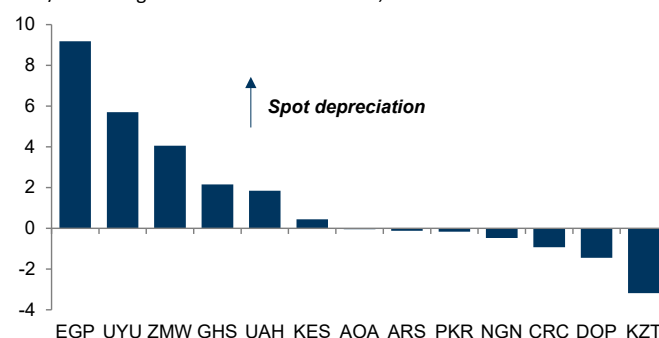
Beyond the immediate hit to energy production, revenues, and growth, markets are also mulling the longer-term implications of the war for GCC economic prospects. In particular, questions have arisen about potential scarring to consumer and investor confidence in the region; whether populations, the majority of which are expat workers, would recover to pre-war levels and over what time period; and whether plans to grow and diversify economies through increased investment in non-oil sectors will be affected, especially around AI/data centers which were one of the early casualties of Iran's retaliatory attacks against its GCC neighbors. The answers to these questions will shape the longer-term implications of the war for GCC economies, though much will depend on the regional geopolitical order that emerges once the dust settles, which remains highly uncertain.

Spillovers beyond the GCC

The impact of the conflict extends beyond the GCC, affecting other countries across the MENA region. The kinetic action is directly impacting Iraq, Jordan, and Syria, with each experiencing varying levels of economic disturbance. Ongoing conflict between Israel and Hezbollah in Lebanon as well as fears of further escalation casts a dark shadow over the country's economic future and the recovery prospects for Lebanon's defaulted Eurobonds. Egypt, a key focus for international investors given the popularity of the local market carry trade over the past two years, is experiencing considerable financial contagion. A combination of heavy pre-war positioning, generalized risk-off behavior, and overblown fears about the impact on the Egyptian economy have led to a significant portfolio retrenchment among investors, leaving the Egyptian Pound (EGP) as the worst performer among frontier FX since the start of the war. However, despite the short-term volatility, we remain constructive on EGP over the medium term as we believe that a combination of robust FX buffers and a (so far) refreshingly orthodox policy response will likely underpin investor and local confidence.

EGP has been the worst performer among frontier currencies

USD/XXX change since the start of the war, %



Source: Bloomberg, Goldman Sachs GIR.

Farouk Soussa, Senior MENA Economist

Email: farouk.soussa@gs.com
Tel: 44-20-7051-9349

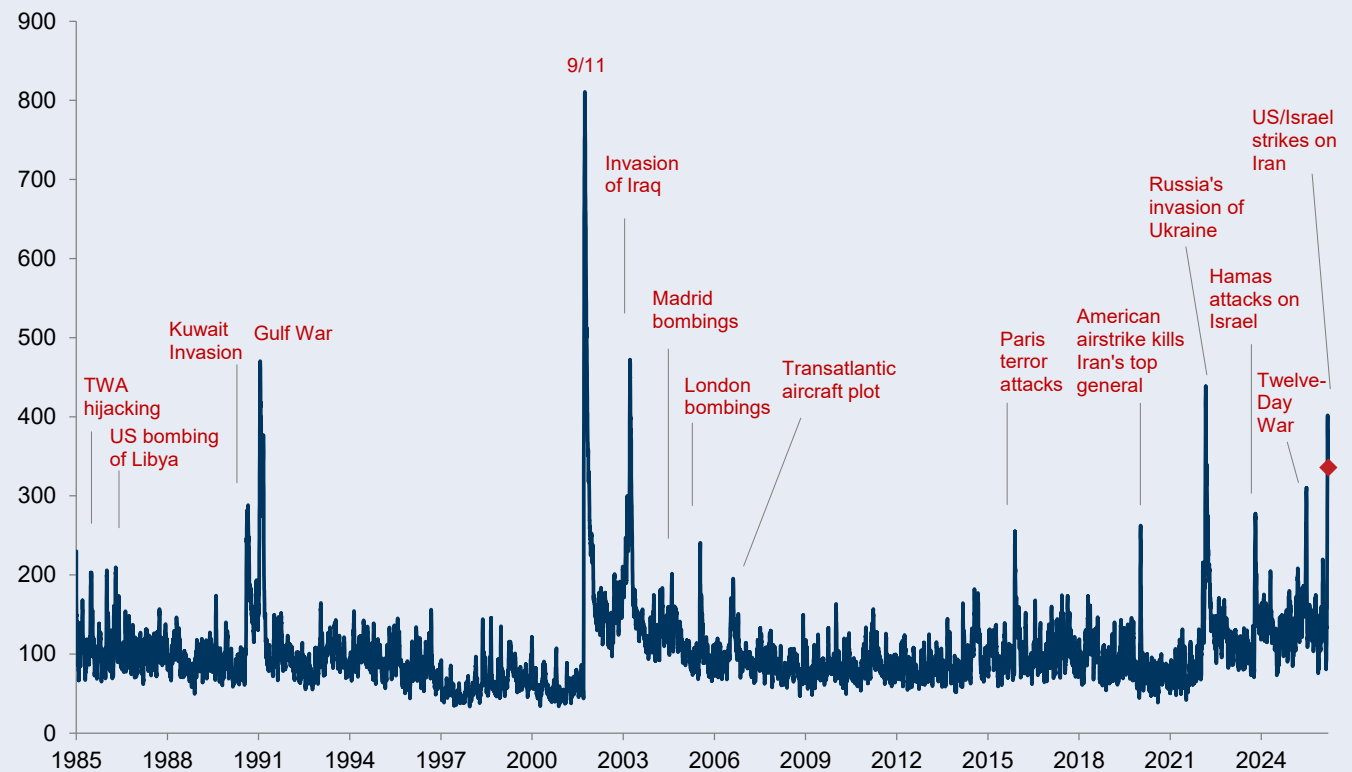
Goldman Sachs International

Geopolitical risk through the years

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

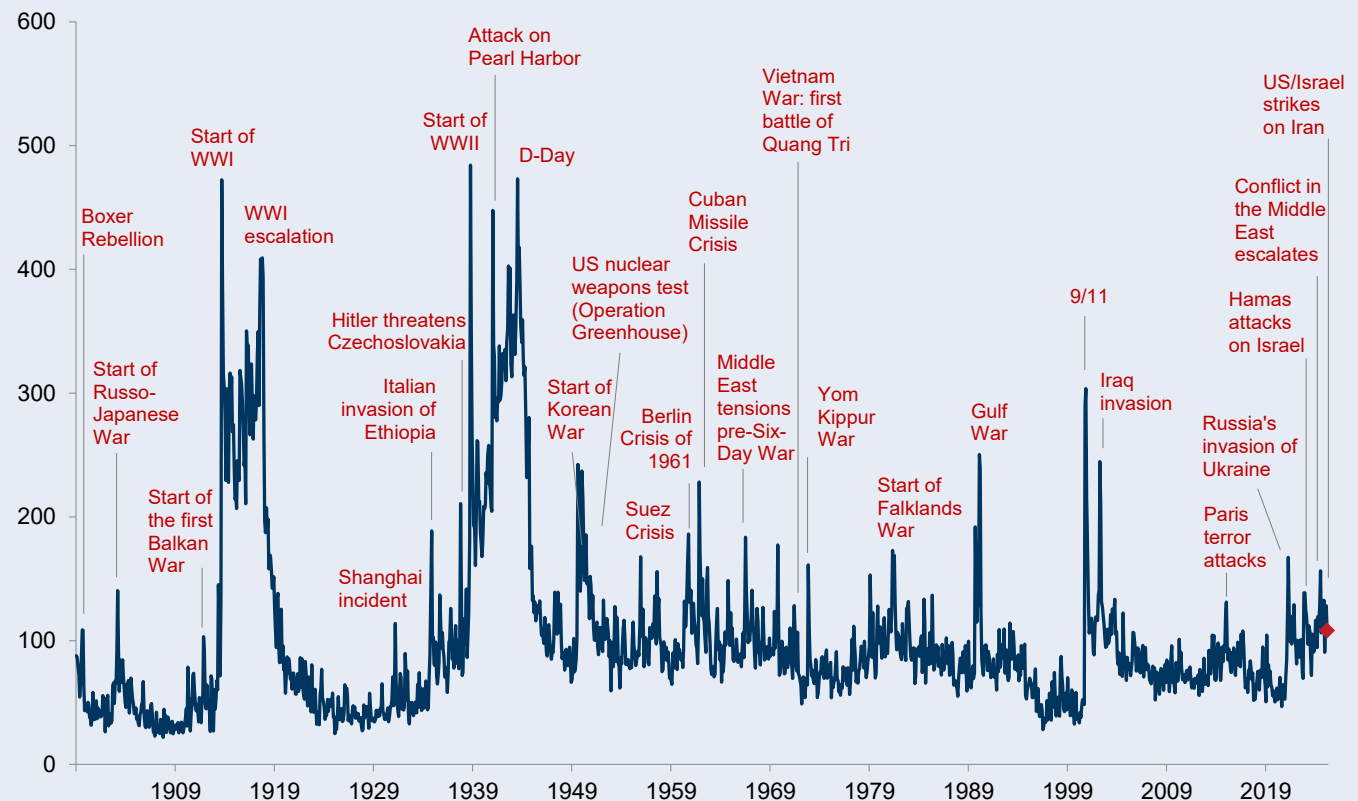
1985-Present

Daily Geopolitical Risk Index, 1985-2019=100



1900-Present

Monthly Geopolitical Risk Index, 1900-2019=100



Note: Daily index as of March 18 and monthly index as of March 1. The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See [here](#) for more information.

Source: [Dario Caldara and Matteo Iacoviello](#), Federal Reserve Board, Goldman Sachs GIR.

Summary of our key forecasts

- Goldman Sachs Global Investment Research.*



Forecasts

Economics										Markets						Equities						
GDP growth (%)		2026		2027		Interest rates 10Yr (%)		Last	E2026	E2027	FX	Last	3m	12m	E2026		E2027		Returns (%)	12m	YTD	E2026 P/E
		GS (Q4/Q4)	Cons. (CY)	GS (CY)	Cons. (CY)	GS	Cons.								GS	Cons.						
Global		2.4	--	2.5	2.7	2.7	2.6	US	4.25	4.10	4.15	EUR/US	1.15	1.14	1.20	Price	7.600		S&P 500	15.0	-3.5	21.4x
Japan		2.2	2.1	2.4	2.2	2.0	2.5	Germany	2.86	3.00	3.00	GBP/US	1.34	1.33	1.33	EPS	\$309	\$319	MXAPJ	18.1	5.6	13.8x
China		4.6	4.7	4.7	5.0	4.7	4.6	Japan	2.26	2.00	2.00	USJPY	158	160	155	Growth	12%	16%	Topix	19.1	5.9	17.9x
Euro area		1.0	1.3	1.0	1.4	1.2	1.2	UK	4.73	4.25	4.00	US/GBP	6.87	6.85	6.70				STOXX 600	6.8	-1.2	14.7x
Policy rates (%)		2026		2027		Commodities		Last	3m	12m	Credit (bp)	Last	2026	4Q26	2026		2027		Wage Tracker 2026 (%)			
		GS	Mkt.	GS	Mkt.	Crude Oil, Brent (\$/bbl)	CPI								Unemp. Rate	CPI	Unemp. Rate	Q1	Q2	Q3	Q4	
US		3.13	3.47	3.13	3.28	3.17	3.75	4.00	USD	IG	86	85	90	US	2.7	4.6	1.8	4.5	3.5	--	--	--
Euro area		2.00	2.45	2.00	2.47	61.98	43	29	Nat Gas, TTF (EUR/MWh)	HY	294	295	315	Euro area	2.5	6.3	1.6	6.2	--	--	--	--
China		1.30	1.40	1.20	--	12.046	13,000	12,400	EUR	IG	102	95	99	China	0.9	--	0.9	--	--	--	--	--
Japan		1.00	1.16	1.50	1.59	4,600	4,805	5,445	Gold (\$/troyoz)	HY	310	298	315									

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedvue.html.

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

Top of Mind archive



Issue 146
Will AI eat software?
March 12, 2026



Issue 145
Americas First
January 26, 2026



Special Issue
2025: 4 themes in charts
December 15, 2025



Issue 144
The US-China tech race
December 4, 2025



Issue 143
AI: in a bubble?
October 22, 2025



Issue 142
Data (un)reliability
September 11, 2025



Issue 141
Stablecoin summer
August 19, 2025



Issue 140
US fiscal worries: Is this time different?
June 12, 2025



Issue 139
Fed independence: how concerning?
May 15, 2025



Issue 138
Tariff-induced recession risk
April 17, 2025



Issue 137
Europe's shifting security landscape
March 13, 2025



Issue 136
Trump tariffs: mostly talk, or big action?
February 26, 2025



Issue 135
High bond yields: here to stay?
February 5, 2025



Special Issue
2024: 4 themes in charts
December 18, 2024



Issue 134
Will China's policy stimulus be enough?
December 11, 2024



Issue 133
Market concentration: how big a worry?
November 25, 2024



Issue 132
Post-election economic policies
October 21, 2024



Issue 131
Is the Fed behind the curve?
September 3, 2024



Issue 130
How investable is Europe?
August 1, 2024



Issue 129
Gen AI: too much spend, too little benefit?
June 25, 2024



Issue 128
Central bank divergence room to run?
May 21, 2024



Issue 127
Weighing the GLP-1 market
April 12, 2024



Issue 126
Global transit & trade: in rough waters
March 12, 2024



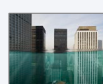
Issue 125
2024: The year of elections
February 1, 2024



Issue 124
Middle East risks
December 5, 2023



Issue 123
US outperformance: at a turning point?
October 30, 2023



Issue 122
Commercial real estate risks
October 9, 2023



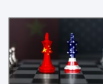
Issue 121
Corporate credit concerns
August 10, 2023



Issue 120
Generative AI: hype, or truly transformative?
July 5, 2023



Issue 119
Daunting debt limit dynamics
May 22, 2023



Issue 118
US-China: more decoupling ahead?
May 1, 2023



Issue 117
All about bank(panic)s
April 3, 2023



Issue 116
(Japanese) Bonds, Bonds, Bonds
February 23, 2023



Issue 115
The Bigger Worry: Growth or Inflation?
January 27, 2023

Source of photos: www.gettyimages.com.

Disclosure Appendix

Reg AC

We, Allison Nathan, Jenny Grimberg, Ashley Rhodes, Joseph Briggs, Samantha Dart, Alec Phillips, Farouk Soussa, Daan Struyven, Kamakshya Trivedi and Alexandra Paulus, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Other disclosure

Iran is subject to comprehensive sanctions by the United States.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from

Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report

broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

Goldman
Sachs
Research

TOP^{of} MIND



WILL AI EAT SOFTWARE?

For over a decade, software has come to “eat” the world. But the emergence of new agentic AI tools for software development in recent months has fueled concerns that AI could in turn “eat” software, triggering a sharp re-rating of the software sector as well as other sectors vulnerable to AI disruption. So are these fears about the future of software (and other industries) warranted, or overdone? GS’ Gabriela Borges, Sherlund Partners’ Rick Sherlund, and Cohesity’s Sanjay Poonen generally agree that AI won’t eat software but diverge on how radical a transformation legacy software firms must undergo to remain competitive and how durable incumbent moats are. With all this uncertainty, we then explore what it will take to stabilize share prices (earnings stability), whether stress in software-

exposed credit can catalyze a turn in the credit default cycle (unlikely), and how investors should be positioned, with Sherlund and GS analysts and strategists agreeing: investors should be selective, in software and beyond.

“

Legacy software companies aren’t standing still. They are innovating as fast followers... That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective.

- Gabriela Borges

Moats buy incumbents time to adapt, as seen in prior software disruption cycles... But these moats aren’t insurmountable—AI will very likely erode them over time.

- Rick Sherlund

Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you.

- Sanjay Poonen

”

WHAT’S INSIDE

INTERVIEWS WITH:

Sanjay Poonen, CEO and President, Cohesity

Rick Sherlund, Founder and CEO, Sherlund Partners LLC

Gabriela Borges, US Software Equity Research Analyst, Goldman Sachs

AI DISRUPTION: SOFTWARE SUBSECTOR IMPACTS

Adam Hotchkiss and Matthew Martino, GS Equity Research

SOFTWARE STOCKS: EARNINGS STABILITY NEEDED

Ben Snider, GS US Portfolio Strategy Research

MANAGING AI RISK IN PORTFOLIOS

Christian Mueller-Glissmann, GS Asset Allocation Research

SOFTWARE DISRUPTION & CREDIT: NO HARD FALL

Shamshad Ali, GS Credit Strategy Research

ALTS MANAGERS: LIMITED AI RISK SO FAR

Alexander Blostein and Michael Vinci, GS Equity Research

AI DISRUPTION: EQUITY SECTOR IMPACTS

GS Equity Research

...AND MORE

Allison Nathan | allison.nathan@gs.com

Jenny Grimberg | jenny.grimberg@gs.com

Ashley Rhodes | ashley.rhodes@gs.com

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S. This report is intended for distribution to GS institutional clients only.

The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

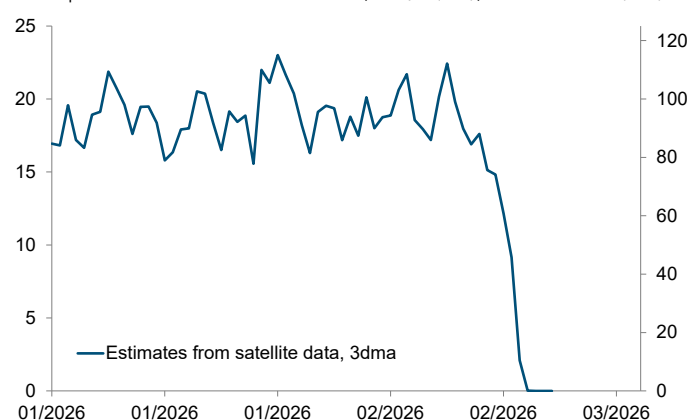
- No major changes in views.

Datapoints/trends we're focused on

- Oil disruptions related to the US-Iran conflict, which pose downside risks to growth and upside risks to inflation, depending on the longevity and severity of the conflict.
- Tariffs; we expect the global tariff that has replaced the IEEPA tariffs to lower the increase in the effective tariff rate since the start of 2025 to 9pp from 10pp.
- Labor market; the Feb jobs report suggests a higher risk that labor demand remains too weak to support the labor market stabilization we had seen in previous months.
- Concerns about nonbank lending, which have risen.

Strait of Hormuz oil transport grinds to a near halt

Oil exports via the Strait of Hormuz, mb/d (lhs), % of normal (rhs)



Source: Kpler, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

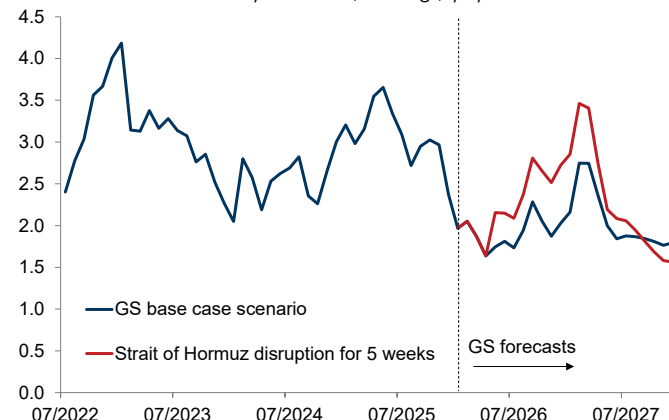
- We recently raised our FY26 average core CPI inflation forecast to 2.1% (from 1.6%) to reflect our higher energy price and USD/JPY forecasts.

Datapoints/trends we're focused on

- LDP's landslide victory in Japan's Lower House elections, which we think will make it easier for the Takaichi Administration to implement its proactive fiscal agenda, posing increased risk to Japan's fiscal position.
- BoJ policy; we continue to expect the next BoJ rate hike in July, and think two recent board member nominations have lowered the probability of an earlier rate hike.
- Shunto negotiations; we expect a base pay rise of 3.2-3.5%.

Rising Japanese inflation

Core CPI inflation rate by scenario, % chg., yoy



Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

- We lowered our 2026 EA growth forecast and raised our 2026 EA headline inflation forecast to reflect higher energy prices amid the US-Iran conflict, with growth risks skewed further to the downside and inflation risks to the upside.
- We lowered our 2026 UK growth forecast to 1.0% yoy (from 1.1%) and raised our headline inflation forecast to 2.4% yoy (from 2.3%) to reflect higher energy prices; as such, we now expect the next BoE cut in April (vs. March before).

Datapoints/trends we're focused on

- European structural reforms; recent political momentum behind further reform steps to counter Europe's waning international competitiveness is promising, in our view.

Europe: still losing competitiveness

Real exchange rate with China (lhs, index, 2019=100) vs. survey-based assessment of int'l competitiveness (rhs, z-score)



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

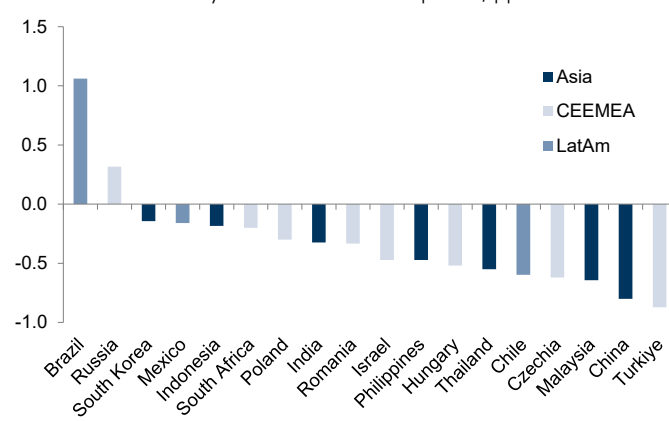
- We lowered our 2026 GDP growth forecasts for Saudi Arabia, Oman, Qatar, Bahrain, Kuwait, and UAE to reflect Strait of Hormuz disruptions and a fall in non-oil economic activity amid the US-Iran conflict, with risks skewed to the downside.

Datapoints/trends we're focused on

- US-Iran conflict impacts beyond the ME; we find that a 10% rise in oil prices would be especially negative for Türkiye and oil importers across Asia, while oil exporters Brazil and Russia are likely to be the biggest beneficiaries.
- China growth; we expect real GDP growth to fall to 4.8% yoy this year (from 5% in 2025), in line with the recently-announced official growth target of 4.5-5% this year.

Higher oil prices: negative for most EMs

Real GDP sensitivity to a 10% rise in oil prices, pp



Source: Goldman Sachs GIR.

Will AI eat software?

For over a decade, software has come to “eat” the world, dominating an ever-wider range of industries. But the emergence of new agentic AI tools for software development in recent months (see pg. 11) has fueled concerns that AI could in turn “eat” software, triggering a sharp re-rating of the software sector (see pg. 9) as well as other sectors vulnerable to disruption from these increasingly powerful and accessible AI tools. So are these fears about the future of software (and other industries) warranted, or overdone?

We ask three long-time software industry watchers: Gabriela Borges, GS US Software analyst, Rick Sherlund, Founder and CEO of Sherlund Partners, and Sanjay Poonen, CEO and President of data security and management firm Cohesity.

Borges explains that “AI is software” because it is code designed to perform tasks. So, rather than eat software, she expects AI to *expand* the software market. But she also expects competition to intensify as AI lowers the cost of code. The main risk for legacy software firms, she says, is that AI-native entrants capture the new AI opportunity while the incumbents are left with the old opportunity—namely, being the “system of record”—that shrinks as a share of the software stack. But Borges notes that legacy firms aren’t standing still but rather “innovating as fast followers.” This, together with moats such as domain experience that new entrants can’t quickly replicate, could ultimately lead incumbents to deliver better outcomes for customers, in her view. But she stresses that they must prove they can leverage their moats to do so, which hasn’t happened yet.

Sherlund shares Borges’ conviction that AI won’t eat software, viewing the current moment akin to past transformational technology shifts that were highly disruptive but ultimately propelled the software industry onto a new growth curve. In his view, software “is being reborn around AI”, necessitating a fundamental re-architecture of legacy systems around large language models (LLMs) and AI agents. While Sherlund acknowledges that incumbent moats can slow disruption, he is skeptical of their long-term durability, arguing that they will buy legacy software firms time rather than guarantee their survival.

And Poonen, for his part, agrees that new AI tools for software development mark a transformative moment for the industry, warning that “just like any technology wave, you must surf this tsunami, or it will demolish you.” In his view, how legacy software firms fare in the AI era will depend crucially on whether they can sustain their current price per user. Firms unable to defend pricing in their core offerings will need to expand and improve their products. But, like Sherlund, Poonen is explicit that just bolting AI agents onto legacy systems may not be sufficient—incumbents must ultimately be willing to rebuild, in some cases from scratch, to remain competitive.

So, with this uncertainty, what will it take to stabilize the sector and the firms most exposed to it? Ben Snider, GS Chief US Equity Strategist, argues that stabilization is unlikely to occur anytime soon, with history suggesting that share prices in industries facing structural disruption (e.g. newspapers) and an associated deep rethink of firms’ terminal value tend to stabilize only when earnings stabilize, underscoring Borges’ point that “the numbers need to contradict the narrative.”

Shamshad Ali, GS senior credit strategist, also expects pressure to persist in the broadly syndicated loan (BSL) market—which software comprises a meaningful share of—as impairment in software-exposed loans will likely be difficult to avoid. But he doesn’t see this as sufficient to catalyze a turn in the credit default cycle, assuming the benign macro backdrop and lower funding cost environment continue.

GS US Financials analysts Alexander Blostein and Michael Vinci are somewhat more positive on alternative asset managers, who have been hit hard given the meaningful software exposure in their private equity and direct lending portfolios. Blostein and Vinci argue that despite this exposure, firmwide risk remains manageable as software lending represents a relatively small share of management fees, software loans are largely senior secured and short duration, and managers have been highly selective in their software investments. But they caution that sentiment and headline risk will likely persist, with slower retail fundraising and accelerating redemption requests potentially weighing on growth despite the durability of underlying earnings streams.

So, how should investors be positioned? Sherlund and Borges agree: this is not an environment for binary bets on software’s survival or collapse, but one that demands selectivity. Sherlund argues that “investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon” and views the most promising companies as “those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks.” Borges, for her part, is focused on fast followers and firms with defensible moats, seeing select opportunities in applications, infrastructure, and cybersecurity. And GS US Software analysts Adam Hotchkiss and Matthew Martino see value in Vertical Software, Data Infrastructure, and Physical-to-Digital firms.

GS Head of Asset Allocation Research Christian Mueller-Glissmann then digs into how to manage AI risk in portfolios more broadly given the outsized role the technology sector plays in driving US equity valuations and returns. To achieve more balance between innovation and inflation risks, he believes investors may need to continue rotating from capital-light sectors, like technology, to capital-heavy sectors that AI is less likely to disrupt (think telecoms, industrials, and utilities), though he favors being more selective given the aggressive rotations that have already occurred.

Finally, with the “AI scare trade” having moved beyond just software to many other sectors—from consumer internet and information services to insurance, CRE services, financials, and transportation—GS equity analysts assess how new AI tools could impact these industries (see pgs. 21-23). Their key message: while AI could disrupt some firms, many may also benefit, reinforcing the idea that investors should be selective as AI technology continues to evolve.

Allison Nathan, Editor

Email: allison.nathan@gs.com
Tel: 212-357-7504
Goldman Sachs & Co. LLC



Interview with Gabriela Borges

Gabriela Borges is Senior Equity Research Analyst at Goldman Sachs covering US Software. Below, she argues that AI won't eat software and some of the recent price action has gone too far, but legacy software firms will need to prove themselves through better fundamentals.



Jenny Grimberg: How does the current software correction compare to past corrections?

Gabriela Borges: The YTD correction is squarely anchored to terminal value debates: instead of analyzing a change in the near-term demand trendline or the mathematical impact of higher interest rates—as was the case in the 2022 correction—the market is

questioning software moats and business models. While this debate is centered primarily on the application software firms, investors are also questioning whether the infrastructure /security stack could be impacted, and the ROI tied to hyperscaler capex. This is sparking a return to first principles questions about the value the industry provides and the nature of competition. And this is all happening against a backdrop of elevated uncertainty given the rapidly changing technology. So, the end state is difficult, if not impossible, to predict.

Jenny Grimberg: Generative AI technology has been around for several years. So, what's changed recently?

Gabriela Borges: The technology is now beginning to impact enterprises. Most AI chatbot applications that have emerged since ChatGPT's debut have primarily impacted the consumer-facing world. Concerns about how data is maintained, cleaned, and validated have been a huge limiting factor for enterprise AI adoption, and regulatory, compliance, and security frameworks are in the process of adapting to make enterprises comfortable adopting these mission-critical yet invasive technologies. At the same time that progress has been made on these fronts, tools more relevant for enterprise use cases have emerged.

For example, Anthropic's recently-released Claude Cowork product makes its Claude Code product accessible to all knowledge workers, not just coders. And its new large language model (LLM) Claude Opus 4.6 has several domain-specific applications, including in cybersecurity, legal, and financial services. OpenAI debuted Frontier, an enterprise platform for building, deploying, and managing AI agents and is strategically leaning into enterprise monetization. And personal AI assistant OpenClaw launched with capabilities much more powerful than a typical AI chatbot's; usage today is primarily limited to technically-savvy consumers, but this may expand as security and usability challenges are solved. So, AI use cases and the power to execute them has progressed, leading the technology to intersect more with the traditional software world and creating a competitive bake-off dynamic between new entrants and today's software leaders at enterprises now better equipped to utilize the technology.

Jenny Grimberg: So, what's your response to those who argue that AI will "eat" software?

Gabriela Borges: The key question is, what do you have to believe to be bullish on software? You have to believe that the

total addressable market (TAM) will grow more than enough to offset the rise in competition. We feel comfortable that the TAM will grow because, despite the current nomenclature, AI *is* software—it is code designed to perform a set of tasks, which is in many ways simply an extension of the software category. But the competition part of the question is currently the bigger unknown as the rapid pace of technological change is attracting new entrants and a wave of VC money. At the same time, the cost of code is falling, which also catalyzes more competition. In our view, the biggest risk for today's software leaders is that the new opportunity AI creates accrues to these new competitors, and the incumbents are left with outsized exposure to the old opportunity that compresses as a percentage of the total software stack.

One way to illustrate this is with the concept of a "system of record." Many of today's application software leaders are systems of record. These systems, such as Enterprise Resource Planning, Customer Relationship Management, and Human Resources, are the digital backbone of the business, storing the foundational data upon which it operates. AI may find it hard to disrupt these for two reasons. First, the data and historical context sitting in the system of record cannot be extracted holistically and in a way that captures ongoing data updates owing to nuances with meta context and data labeling as well as with stream processing that are proprietary to the system of record and break when extracted, even when using sophisticated APIs. Second, using generative AI technology to replace deterministic logic is expensive and ineffective. Put simply, it makes more strategic and economic sense for new AI entrants to leverage existing systems of record for high quality data and context than to replace them. The question then becomes whether the incumbents can also capture the new opportunity because of this domain experience and context.

Jenny Grimberg: So, is it possible to be bullish on agentic AI and *not* bearish on legacy software?

Gabriela Borges: Yes, because some legacy software firms will likely capture the upside opportunity of AI. We expect some incumbents will be able to prove to customers that their domain experience drives better AI outcomes. It's worth highlighting that incumbents are not static—many have world-class engineering teams that benefit from the same AI productivity gains as start-ups, founder involvement, and strong cash balances to respond to new competitive threats.

But the playbook to capture AI upside is three-fold, and companies must succeed on all three fronts, which won't be easy but is possible. First, companies must minimize any technical debt in their tech stack. That means ensuring all acquisitions integrate nicely into the company's core technology backbone and data flows across different modules and is centralized and uniformly cleaned. Second, companies must understand the holes in their roadmaps. One misconception is that a software leader must be a leader in technology. It can instead be a fast follower, letting the VC

ecosystem fund next-gen technologies, seeing which have durability and the best product market fit, and then acquiring them or using internal R&D to capture that product market fit, which is arguably a better use of capital than spending heavily to develop every new technology in-house.

Third, companies must figure out how to monetize AI, which may require a change in pricing model. For example, instead of charging per customer service representative, a software company could instead charge per successfully completed customer service call. Another type of pricing model is enterprise license agreements (ELAs), which allow customers to access many different products for one price. ELAs have existed in software for some time. Salesforce recently shifted to agentic ELAs (AELAs), giving customers access to their AI and other products. This is important because it shifts the software pricing model more holistically to include agentic value and allows customers to have fluidity between seats and outcomes. Importantly, we don't think customers can reduce seat count and maintain/enhance productivity unless software tools are picking up the slack and delivering more value.

Jenny Grimberg: What will allow the sector to stabilize?

Gabriela Borges: The numbers need to contradict the narrative. The software sector has been in a four-year-long depression; for 14 consecutive quarters, revenue growth has decelerated and the cost of acquiring new customers has risen. So, the numbers need to stabilize. It would be hard to argue that software companies are being disintermediated if they put up better fundamentals, because the only way they can do so in the AI era is by having exposure to the new opportunities and budgets that AI creates. Evidence of this would be clean beat-and-raised quarters and improved unit economics, either on an industry or, more likely, company-specific basis.

Jenny Grimberg: But couldn't that just push out the bear case for software disruption rather than prove it wrong?

Gabriela Borges: This is where moats become important. A company must prove its fundamentals are holding up because it has a defensible moat. One example is data architecture and collection moats, which are a feature of some security software companies and are outside the realm of AI's current and likely future capabilities. Cloudflare, for example, has a physical edge compute network co-located in telecom data centers, meaning that it owns physical computing infrastructure deployed close to end-users. Palo Alto Networks and CrowdStrike directly collect data, either through physical hardware deployed at the network edge or, in CrowdStrike's case, through an agent installed on an end-user device. These are classic examples of network effects that have compounded over 10+ years.

Another defensible moat is domain experience. Software firms with domain experience understand how knowledge workers like salespeople, accountants, and financial analysts work and use their tools owing to years of being embedded in an enterprise. That usage also allows these firms to collect proprietary data, some of which is very nuanced and difficult to abstract. Beneath the surface of Microsoft Copilot, for example, is a highly nuanced graph showing how knowledge workers in a specific enterprise collaborate, which Microsoft has the intelligence to create because it has observed how these workers interact with its software for many years. That is a powerful type of domain experience. Microsoft is also vertically integrated, operating across all three layers of software—the platform, infrastructure, and application layers—which allows it to observe and optimize how enterprise work gets done.

Jenny Grimberg: But couldn't AI native companies replicate those moats over time?

Gabriela Borges: Yes, but time matters, as does whether it makes strategic sense to do so. If everything were frozen in place right now and AI-native firms were given two years to figure out how to replicate domain experience, for example, they probably would. But, again, legacy software companies aren't standing still. They are innovating as fast followers. The incumbents have access to the same AI technology as the AI firms themselves—the LLMs, the coding productivity tools, etc.—and so are benefitting from the same productivity improvements. That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective. They already have the system of record, domain experience, and user interface. Layering AI on top of that gives incumbents everything they need to build a successful agentic solution more powerful than any one new competitor can offer. But ultimately, incumbents will have to demonstrate they can leverage their moats to deliver such an outcome, and so far, that hasn't happened in a broad enough way. So, the pressure is on for these companies.

Jenny Grimberg: Given everything we've discussed, has the recent sharp selloff in software gone too far?

Gabriela Borges: The pace of competition and the level of uncertainty is undoubtedly high. However, two areas of discovery value allow us to be bullish on many software assets. First is the dynamic we've discussed on incumbents being able to leverage domain experience to drive better AI outcomes. Second is the potential for more companies to manage to GAAP earnings, which provide a level of valuation support for the stocks. The level of competitive noise is likely near a local maximum, and the perception vs. reality gap should narrow ahead for the majority of software firms. Looking at it from that lens, some of the recent price action has likely gone too far.

Interview with Rick Sherlund

Rick Sherlund is Founder and CEO of Sherlund Partners LLC, an investment advisory firm focused on the rapidly evolving AI software sector and traditional enterprise SaaS space and is a Senior Advisor at Wedbush Securities. Below, he argues that the software industry is being reborn around AI, necessitating a complete re-architecture of existing software systems.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Does the emergence of new agentic AI tools for software development mark a transformative moment for the traditional software industry?

Rick Sherlund: Very much so. The software industry undergoes a major platform shift roughly every 10-20 years as new technologies emerge.

While these shifts can be highly disruptive, they also serve as powerful catalysts for growth, propelling the entire software industry onto new growth curves. We are at such a juncture today, with over 10,000 venture-backed AI-native companies poised to encroach on and disrupt the ~\$400bn global Software as a Service (SaaS) market.

Allison Nathan: How does this compare to past periods of transformation?

Rick Sherlund: Some technology shifts, such as the move to mobile, allowed companies to adapt their existing architecture without fundamentally altering business processes. However, other shifts created true discontinuities that forced companies to completely rebuild their systems from the ground up. The move to relational databases in the 1980s/1990s is a prime example. Companies could not simply tweak their existing frameworks. Instead, they had to reconstruct their architectures entirely—a task akin to demolishing and rebuilding a skyscraper from its foundation. Once-dominant players that were unable to reinvent their platforms to compete with Oracle, which was a leader in promoting relational database architecture, were ultimately left behind. Similar upheavals occurred during the transitions to client-server models and later to cloud computing.

The rise of AI marks a similarly transformative shift, enabling processes that legacy system architectures cannot readily accommodate. Faced with falling valuations, incumbent SaaS providers must overhaul their systems. Remaining competitive in the AI era requires software companies to do much more than add incremental AI capabilities to automate narrow legacy human tasks. While such bolt-on functions may be quicker and look good in the marketing literature, they are superficial and risk missing the more critical and transformative opportunity to rearchitect systems around large language model (LLM) intelligence with reasoning at the core to orchestrate evolving autonomous agents, which can power a new generation of business processes. This shift is not only disruptive for the software industry, but for businesses and sectors globally, representing a fundamental change in technology platforms.

Allison Nathan: What new processes will such AI-enabled platforms be able to achieve?

Rick Sherlund: Traditional SaaS platforms rely on hard-coded workflows and human-centric, menu-driven interfaces. By contrast, AI-based systems feature more conversational interfaces, embedding LLM intelligence at their core to orchestrate agents capable of reasoning, decision-making, and executing tasks previously handled by humans. This enables the integration of entirely new processes. For example, legacy customer support consists of menu-based interactive voice responses (IVR) and irritating cascading menus. Users may frequently scream “representative” in frustration to exit the menu system to connect with a human to address their issue, but even that representative may not be able to do so. And customer support remains costly.

So, companies are turning to AI-powered virtual agents that are available 24/7, multilingual, and trained on comprehensive support materials. These agents run on hyperscaler systems that avoid problems of availability and scalability in periods of peak demand. Agents may also identify that a customer has, say, an older version of a product that may no longer be supported. The agent can then pivot from a cost center support function to a revenue function by recommending a new product. So, AI agents can seamlessly shift from support to sales and billing, orchestrating complex, multistep workflows across disparate systems with minimal human intervention.

As infrastructure improves to provide governance and guardrails, AI agents will increasingly handle autonomous, agent-to-agent processes, with humans overseeing compliance and exceptions. This shift will unlock even more processes and expand automation as systems become more intelligent and capable of advanced reasoning.

Allison Nathan: So, are concerns that AI could “eat” software, which have sparked a major selloff in legacy software companies, warranted, or overblown?

Rick Sherlund: At the end of the day, AI is expressed in software, so declaring software “dead” is misguided. Software will endure, even if the industry looks different than it does today. Certain segments of the software market—particularly applications—will experience considerable disruption, but the broader industry will be reincarnated, adapting to new processes and reaping substantial benefits from the widespread integration of AI capabilities across all business functions. This will lead to a step function increase in productivity that was not possible with traditional SaaS architectures.

Allison Nathan: Won't the rise of "vibe coding" eventually make software replaceable, as some argue?

Rick Sherlund: Concerns that "vibe coding" will make it so simple to write software that software itself becomes trivial are largely unfounded. Coding accounts for only around 15–20% of writing enterprise systems. So, while automating code generation is a major efficiency gain, it is just one aspect of a far more complex process. Designing effective workflows, defining data structures, and establishing secure access controls are all critical and complex tasks. Developers must decide how databases are organized, what data is stored, and how sensitive information, like payroll data, is protected. These challenges extend well beyond basic coding. So, easier coding won't sound the industry's death knell. On the contrary, the industry will likely become more vibrant than ever as firms leverage the new technology to remain competitive. So, the software market is not dead; it is being reborn around AI.

Allison Nathan: How defensible are legacy software moats amid the AI transition?

Rick Sherlund: Complex workflows are a powerful moat, demanding deep domain expertise to design effective interactions with vendors and suppliers in the supply chain. This expertise extends not only to vertical markets but also across horizontal complexities. Data access is another powerful moat, as software companies fiercely protect their customers' data and resist third-party access, recognizing its tremendous value. Scale and distribution moats are powerful as well—building and maintaining an enterprise sales force is extremely costly, so established distribution networks further protect incumbents.

Together, these moats buy incumbents time to adapt, as seen in prior software disruption cycles. For example, larger companies with complex workflows like SAP and Oracle were slower to adopt cloud architecture but eventually did so, leveraging their moats to buy time. So, firms with these protective moats, like the large SaaS incumbents, have more time to adjust amid the AI transition. But these moats aren't insurmountable—AI will very likely erode them over time.

Allison Nathan: How are incumbent software companies responding?

Rick Sherlund: Incumbent software companies are working hard to fundamentally re-architect their systems. Large firms like Salesforce, Workday, and ServiceNow—who were disruptors in the shift from client-server to the cloud—are now integrating LLMs into the core of their systems and developing agentic frameworks to support intelligent, autonomous workflows. Having been disruptors themselves, these vendors recognize that major technological shifts require comprehensive transformation, not just superficial bolt-ons.

These companies are also making significant efforts to address broader issues such as governance, security, and the need for advanced data structures like knowledge and context graphs. Such measures are vital for understanding data and workflow intent, ensuring that systems can operate autonomously while maintaining compliance and security. While these efforts do not guarantee that the incumbent SaaS companies will "win" against emerging AI-native competitors, they represent significant steps toward adapting and maintaining relevance in

an evolving market. Whether they can move fast enough and reposition to leverage their market presence, distribution, and scale advantages to be net beneficiaries of the AI era remains to be seen.

Allison Nathan: By contrast, how can AI-native firms compete and win in the space?

Rick Sherlund: Disruptors have historically succeeded by entering at the lower end of the market, where there is less invested in incumbent systems and companies are more open to adopting innovative solutions. This was the case with Salesforce and Workday, who began by serving small- and medium-sized businesses rather than directly competing with SAP.

Over time, as new entrants enhance their offerings and scale their distribution, they can move gradually upmarket, competing with larger incumbents—adopting a "land and expand" strategy. Salesforce, for example, began with sales automation before expanding into marketing and customer service, while Workday started in Human Resources before adding financial capabilities. Ultimately, the path to market disruption starts at the bottom where the protective moats are fewer and fresh ideas can gain traction, and scales upward.

Allison Nathan: Given all that, how do you foresee the competitive landscape for software evolving?

Rick Sherlund: As more AI-native companies enter the lower end of the market, the software industry will likely become increasingly fragmented. We saw this with the move from client-server systems to the cloud/SaaS model, where specialized "best-of-breed" software solutions were embraced over traditional monolithic suites from providers like SAP and Oracle. Now, with the rise of agentic AI systems, we will likely see a further disaggregation as agents enable a fragmentation of functions and an unbundling of existing SaaS functions. This is already evident in the customer support segment, with AI agents transforming Contact Center as a Service (CCaaS) and IVR systems, as well as in the emergence of dedicated sales and marketing agents.

Allison Nathan: How can investors distinguish between software companies that will likely remain resilient in the face of AI-driven disruption, and those that may be at risk of becoming obsolete?

Rick Sherlund: Investing in software has now become riskier, making it important to distinguish between layers of the software stack—applications are probably more at risk than infrastructure software, while areas such as governance, security, and data management stand to benefit most from AI advances. Investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon, even if these firms will eventually need to undergo significant transformation to remain competitive.

Ultimately, the most promising companies are those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks. At the end of the day, only those with the right architecture will be able to adapt and capitalize as AI capabilities rapidly advance.

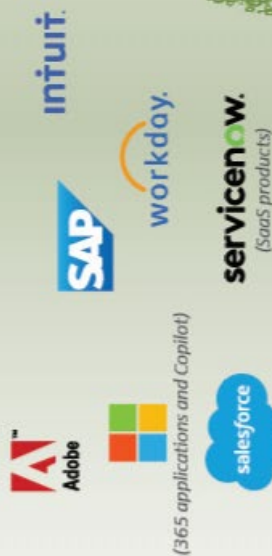
The software ecosystem, at a glance

The software industry can be thought of in three layers: Applications, Platforms, and Infrastructure. Companies operating within each layer are considered software companies because their core offering is the creation, deployment, or sale of software. A company can appear in multiple layers of the software stack depending on its product offerings, in a strategy known as “vertical integration”.

Applications

The application layer represents the software products that end-users directly interact with to perform specific tasks.

Applications include an array of software such as enterprise resource planning (ERP) systems, customer relationship management (CRM) tools, productivity suites, and industry-specific applications.



Platforms

The platform layer provides the software environment, tools, and services that enable developers to build, deploy, and manage applications. Platforms include operating systems, databases, development frameworks, cloud Platform-as-a-Service (PaaS) offerings, and API platforms.



Infrastructure

The infrastructure layer comprises the core computing resources, networking capabilities, and security and storage solutions that underpin all software.

Infrastructure includes physical and virtual servers, data centers, cloud

Infrastructure-as-a-Service (IaaS)

offerings, databases, and cybersecurity essentials.



Note: List of companies in each layer is not exhaustive.

Source: Goldman Sachs GIR.

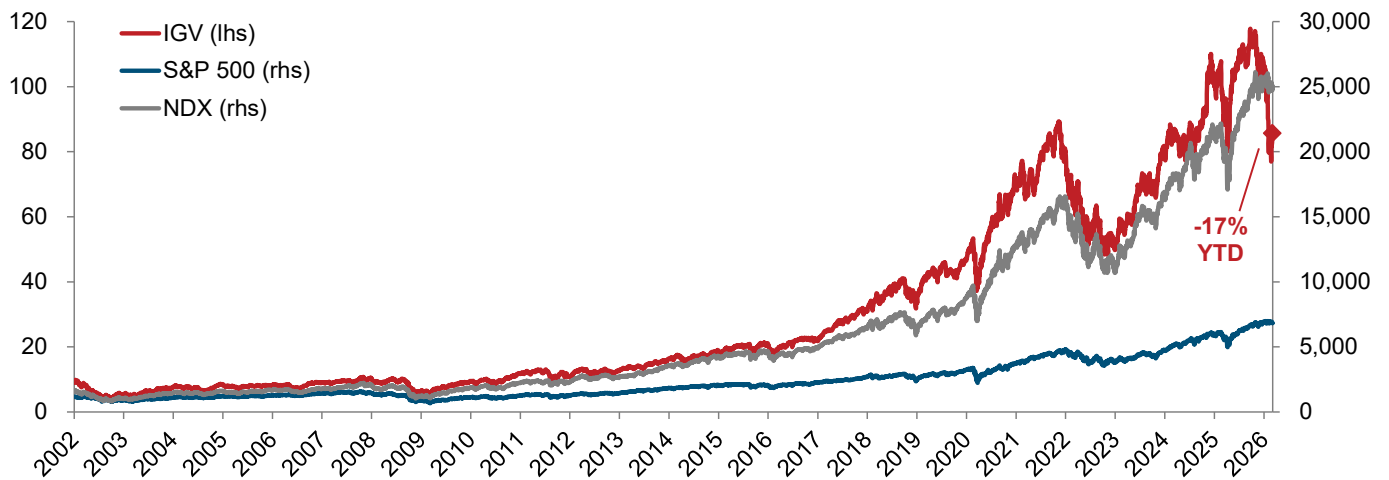
Special thanks to GS US Software Analyst Matthew Martino for data and guidance.

Illustration created using random AI generated code.

The software selloff, visualized

The iShares Expanded Tech-Software Sector ETF (IGV), which tracks the performance of around 120 North American stocks in the software industry, has declined by around 17% since the beginning of the year and 26% from its fall 2025 highs

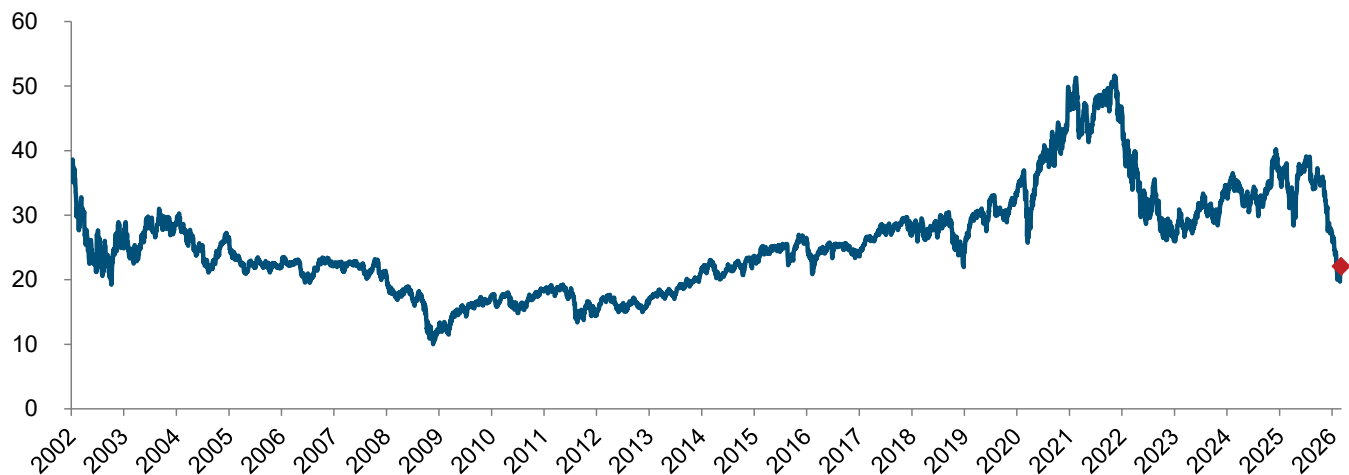
Price



See [here](#) for more details about the IGV ETF.

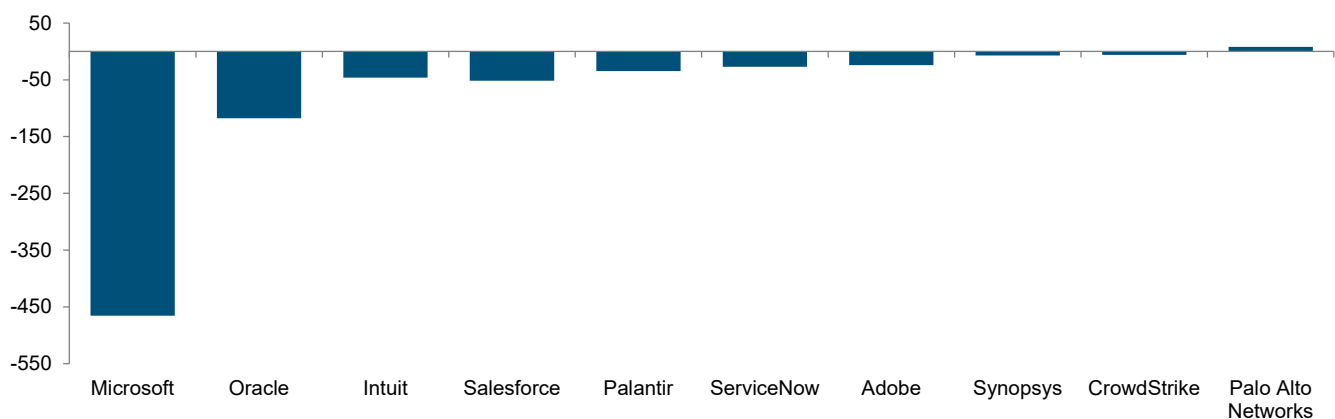
Source: FactSet, Goldman Sachs GIR.

Valuations have also fallen sharply, with the companies in the ETF trading at roughly 22x, well below their Covid-era peak of 51x IGV NTM P/E



Source: FactSet, Goldman Sachs GIR.

The 10 largest stocks in the IGV ETF have in total lost nearly \$800 billion in market capitalization since the beginning of the year. Market cap change for each of the 10 largest stocks in IGV since January 2, 2026, \$bn



Source: Bloomberg, Goldman Sachs GIR.

Special thanks to GS US Portfolio Strategist Christophe Sung for data.

AI disruption: software subsector impacts

Q: Vertical Software, Data Infrastructure, and Physical-to-Digital are key subsectors of the software space. To what extent are legacy firms in these subsectors vulnerable to AI disruption?

Vertical Software

A: Vertical software refers to products built for specific industries rather than general-purpose systems like email or accounting tools. These systems store critical data, reflect industry-specific workflows, and are deeply embedded in companies' daily operations. Vertical vendors include firms like Guidewire Software (insurance), Tyler Technologies (public sector), and Veeva Systems (life sciences). As AI technology continues to rapidly evolve, the key question is whether new agentic technologies will make vertical systems less valuable or even replace them. Our view is nuanced but ultimately constructive. First, we believe vertical software systems will remain the core "systems of record" (see pg. 4-5) for their industries. The data and context embedded in these platforms are both unique to complex, industry-specific networks and essential for AI to generate meaningful, reliable outputs. Second, AI gives vertical software vendors an opportunity to enhance connectivity between complicated industry workflows and get customers more value out of their data. Over time, this will likely drive synergies between products and result in higher revenue, even if competition grows.

The important nuance here is execution. Management teams that are fast followers—those that actively integrate new AI capabilities and maximize the value of available LLM technology—are better positioned to limit value abstraction. Conversely, less innovative vendors, rather than be valued as an immersive software ecosystem, could see their value erode to that of a database, which would likely weigh on pricing power and growth.

Competition in this layer will likely intensify. We are already seeing early signs of this with businesses that deploy engineers directly within a customer's environment to implement AI (often called forward deployed engineering), as well as early-stage tools launched by LLM companies themselves (i.e., Claude plugins). That said, incumbents that already own the underlying data and industry context layer start from a position of strength. Customers are also likely to be relatively patient with these vendors given the strength of their relationships and deep embeddedness of the software.

These dynamics could ultimately drive more M&A. Incumbents seeking to remain competitive may increasingly turn to M&A to keep pace with AI-native innovation, while private equity firms may find vertical businesses with clear data and context moats increasingly attractive given the compression of public market valuations.



*Adam Hotchkiss,
US Emerging Software Analyst*

Data Infrastructure and Physical-to-Digital

A: Data infrastructure refers to platforms that store, govern, and secure data so it can be accessed and used across an organization, such as Snowflake, MongoDB, and Rubrik. We view AI as a force multiplier in this space. As organizations deploy AI copilots and agents, they will create and query more data, demand faster response times, and tolerate less downtime. This raises the value of this foundational layer, because it determines whether AI can run reliably and securely in production.

The primary risk this subsector faces is that, over time, workloads could shift to newer or cheaper platforms—a familiar pattern in technology transitions—and more integrated AI stacks could reshape enterprises' spending priorities. However, we see these as competitive dynamics to monitor rather than immediate existential threats.

The disruption debate is more pointed for physical-to-digital companies. These are software firms that sell digital products to physical asset-intensive industries like construction and transportation. On the surface, these products can look like everyday applications. In reality, they sit inside mission-critical workflows with governance and compliance baked in, which raises the bar for AI-native challengers.

The long-term risk is not that AI replaces these platforms outright, but that a general-purpose agent layer could sit above them and reduce their standalone importance. But our view is that AI adoption will gravitate toward systems that control proprietary data and link directly to real-world outcomes. For such systems, AI will likely become a source of reinforcement rather than disruption.



*Matthew Martino,
US Software Analyst*

AI tools for enterprise

Tool	Summary
Core enterprise AI platforms	
Anthropic Claude Cowork	An AI productivity tool that operates as an AI agent that can perform tasks directly on a user's computer, including automating repetitive tasks such as organizing files, extracting data from PDFs into spreadsheets, and generating presentations.
OpenAI Frontier	An enterprise-grade platform designed to build, deploy, and manage autonomous AI agents that act as "digital co-workers." AI agents can perform complex, multi-step tasks across business systems like CRM and data warehouses.
Microsoft Copilot	An AI digital assistant integrated across Microsoft 365 apps (Word, Excel, Teams), Windows, and web browsers, designed to boost productivity and streamline workflows.
Google Gemini for Workspace	An AI digital assistant integrated directly into Gmail, Docs, Drive, Slides, Sheets, and Meet that helps users boost productivity through drafting, organizing, summarizing, and brainstorming.
OpenClaw	An open-source autonomous AI agent that acts as a personal digital assistant capable of managing files, running terminal commands, and automating tasks across platforms like Discord, Slack, and WhatsApp.
Adept AI	Uses autonomous AI agents that can execute manual, repetitive, end-to-end workflow tasks across existing software applications.
AI for customer service and sales	
Sierra AI	An AI-powered customer service agent that can handle complex conversations across digital channels including around tracking orders or resolving issues.
Ada CX	An AI customer service platform designed for enterprises to automate support across chat, email, and voice channels.
Gong AI	A revenue intelligence platform using AI to analyze customer interactions to improve sales performance and identify risks.
Clari AI	Provides AI-driven revenue forecasting and deal intelligence.
AI for HR	
HR Plugin for Claude Cowork	Allows HR professionals to deploy specialized agents that operate across local files and enterprise apps. It can draft offer letters, build onboarding plans, and run compensation analyses by pulling data from multiple spreadsheets.
Paradox	A tool for high-volume recruitment that handles candidate screening, answers FAQs, and automatically schedules interviews based on recruiter availability.
AI for financial services	
AlphaSense	An AI-powered market intelligence platform that aggregates millions of data points from equity research, filings, and news to provide real-time insights for analysts.
Alteryx	Automates repetitive data workflows and analytical modeling, allowing finance teams to run various risk scenarios and consolidate disparate data sources without manual effort.
Prophix One	A finance-native platform embedding AI agents directly into budgeting, forecasting, and "continuous close" workflows to reduce human error in reporting.
Trullion AI	An AI accounting platform that automates lease accounting and revenue recognition.
Anaplan (PlanIQ)	An AI platform that detects anomalies and generates predictive forecasts across finance, sales, and supply chain operations.
AI for manufacturing and supply chains	
Siemens & NVIDIA (Industrial AI OS)	A joint operating system that transforms "passive" digital twins into active intelligence, allowing real-time simulation and optimization of factory floors.
Blue Yonder	An AI-driven supply chain platform that provides cognitive analytics to predict disruptions and automatically adjust production and logistics schedules accordingly.
Cognex Vision AI	Leverages deep learning for high-speed machine vision quality control, detecting subtle defects that traditional rule-based systems miss.
Interos	An AI risk management platform that monitors global supply chains for geopolitical, financial, and ESG-related disruptions.
Phaidra	Provides autonomous AI control systems for industrial manufacturing and mission-critical facilities, such as data centers.
AI for legal and professional services	
Harvey AI	A domain-specific AI platform used by major law firms for regulatory research, document analysis, and drafting documents.
Luminance	An AI platform specialized in M&A due diligence and autonomous NDA negotiation, capable of identifying anomalies across large document sets.
Ironclad	A contract lifecycle management tool that uses AI to automate the drafting, negotiation, and tracking of legal obligations at scale.
Casetext (CoCounsel)	An AI legal assistant that performs case law research, summarizes depositions, and reviews documents for relevance using advanced reasoning.
AI for retail and e-commerce	
Salesforce Einstein	An integrated AI layer for CRM that generates personalized product descriptions, predicts customer churn, and provides merchandising insights.
Shopify Magic	Suite of AI tools that automates inventory management, generates marketing copy, and personalizes product recommendations.
ConversionBox	An AI shopping assistant that uses conversational AI to guide customers through product discovery, acting as a digital salesperson on e-commerce sites.

Note: Table does not constitute an exhaustive list of all AI tools for enterprise.

Source: Company data, various news sources, compiled by Goldman Sachs GIR.

Interview with Sanjay Poonen

Sanjay Poonen is CEO and President of Cohesity, an AI-powered data security and management firm, and serves on the boards of Philips and Snyk. Previously, Poonen was COO at VMware and President of SAP. Below, he argues that agentic AI has the potential to be enormously disruptive for the software industry, and legacy firms must adapt by leveraging the technology to improve their product offerings or risk being demolished by it.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Based on your vast experience in the software industry, does the emergence of new agentic AI tools for software development mark a transformative moment for the industry?

Sanjay Poonen: Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive. As an engineer who began my career writing code, this is a truly exciting and transformative moment—back in the 1990s, I often stayed up late trying to debug code, hoping to find another engineer also working late whom I could consult. Today, that “person” is an AI assistant.

However, just like any technology wave, you must surf this tsunami, or it will demolish you. If software engineers aren’t leveraging AI technology to write code and improve productivity today, it’s a crime. The best software engineers I know, both at Cohesity and beyond, say that the technology is revolutionizing the way they code. So, we’ve thoroughly embraced it at Cohesity. But every business should be thinking about how these tools can support and improve not only their software engineers, sales reps, and human resources personnel, but also their products and services. I see limitless potential to leverage generative AI in almost every industry, from medtech to aeronautics and even space.

“Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you.”

Allison Nathan: You paint a very positive picture of the benefits of AI for software and productivity. But the whole software sector has re-rated substantially lower because of concerns that legacy software companies won’t be able to compete even if they are embracing AI technology themselves. So, are these concerns warranted?

Sanjay Poonen: With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI-native or other competitors leveraging the technology. At the core of every Software as a Service (SaaS) application is a CRUD—Create, Read, Update, Delete—database that maintains data, application logic, and workflows, which AI tools have the potential to radically disrupt. If a company can sustain its price

per user amid this disruption, it will likely be fine. But some companies may need to lower their price per user to remain competitive. That doesn’t necessarily mean the company will go bankrupt or disappear, but it does have important implications for revenue growth and terminal value. So, every CEO, analyst, and investor is rightly assessing this risk.

“With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI native or other competitors leveraging the technology.”

Allison Nathan: Is it realistic to think that these companies will be able to effectively respond to this pricing pressure and stay competitive?

Sanjay Poonen: Yes, but it requires innovation. Firms whose core products are under pressure may need to add a second or third module with enhanced capabilities that garners a higher price per user to compensate for the lost revenue. For example, a company could offer a small, medium, and large packaging of its product, with the price per user of the small, base product coming down, but the price per user of the larger product offerings higher.

Alternatively, firms can expand into new areas to grow their total addressable market (TAM) and sustain revenue growth even as price per user declines. When I was at SAP, we recognized that a price per user ceiling existed for our ERP software and responded by adding a system of engagement on top of the system of record by acquiring analytics companies and building new business objects. This allowed us to raise the effective price per user for the combined product offering. Every company facing disruption of their core product will need to make these types of adjustments.

SaaS companies can also attempt to shift from a price per user to a price per outcome model, where fees are based on the measurable value they provide to the customer. That pricing model is harder to sustain because tracking outcomes can be challenging and an IT buyer typically has a fixed budget and is often unable or unwilling to enter into a vendor contract with fees that depend on the broader company’s performance. Because of that, most companies price per user. The only company I am aware of that uses a price per outcome model is Palantir, which often sells its product at the CEO level, where the power to make these decisions resides.

Allison Nathan: So, is the indiscriminate selloff across the software space overdone?

Sanjay Poonen: The answer to that will ultimately be determined by the revenue growth of these companies. As both a student of software and the CEO of a software company, I will be closely watching whether growth is holding up or slowing down. That said, in the security world Cohesity lives in, the large market reaction to Anthropic's announcement that it had built a code scanning tool for security was akin to Anthropic sneezing and everyone coming down with pneumonia. But the market seemed to quickly realize that this was an overreaction and that this tool won't enable users to build a firewall or a data or endpoint security product overnight.

Allison Nathan: Even if that's true today, are you concerned that it won't be in the future given the rapid rate at which generative/agent AI technology is advancing?

Sanjay Poonen: Former Intel CEO Andy Grove said it best: "only the paranoid survive." Nobody in this space can afford to be cavalier about the potential for disruption. As a CEO, I have to assume that someone could use AI tools to build a competitor. In fact, I recently asked an AI assistant to build a Cohesity competitor to see what it would do. Unsurprisingly, its first response was that this was a "tall order." But it was able to think about the challenge in logical steps, which revealed an understanding of how to build a competitor. So, it's only safe to assume that, given time, these tools will eventually be able to do just that. For that reason, it is incumbent upon myself and every other CEO in a similar position to take advantage of these same tools to expand and improve our own product offerings. Again, we must ride the tsunami to our benefit rather than let it demolish us.

“Former Intel CEO Andy Grove said it best: “only the paranoid survive.” Nobody in this space can afford to be cavalier about the potential for disruption.”

Allison Nathan: But is just bolting AI agents onto legacy layers ever going to be as successful and powerful as native AI product offerings, or will legacy companies eventually have to start from scratch?

Sanjay Poonen: Software is typically built in architectural layers, with the top layers easiest to refresh and the core, bottom layers much harder to. But even timeless software evolves every decade or two, so engineers must be comfortable with the idea of completely rewriting part of or even the whole architectural stack. The good news is that using new tools and approaches to rewrite code from scratch will probably take a small fraction of the time it took to write the original code. Even then, disrupting the whole stack is undoubtedly hard, even if necessary. In such instances, companies might decide to shelter in place and keep the

business and cashflow running while simultaneously building or buying the next generation of the company. I see this as akin to living in your old house while building a new one next door and only tearing down the old house once the new one is ready.

We dealt with this at every organization I've worked at. At SAP, as new companies that were building a SaaS version of ERP in the cloud threatened to disrupt our on-prem product, we had to consider whether we could build a cloud-native product fast enough without disrupting our core platform. The best companies find a way to constantly evolve their stack to remain competitive.

Allison Nathan: While large players like a Microsoft or Salesforce may be able rebuild, is there an inherent disadvantage for the smaller companies that have less resources to do so?

Sanjay Poonen: It's wrong to assume that reinvention is easier at a Microsoft or Salesforce, and in fact the opposite may be true. It took new leadership for Microsoft to transform from a Windows company to a cloud company and now an AI company. In my experience, the bigger the company, the greater the molasses, dinosaur mindsets, and resistance to change. Smaller companies may not have as many resources, but they have the advantage of speed. It's a bit like a speedboat vs. an aircraft carrier—small firms can usually make decisions faster and are less likely to have a large legacy install base that they must worry about transforming. Microsoft pivoting away from Windows to Azure was a big and highly risky shift. A company starting completely from scratch has no legacy to deal with. I'm reminded of the joke that God created the world in seven days because he didn't have an install base. The bigger the install base, the tougher it is to pivot and change.

All that said, whether small, medium, or large, as long as a company maintains a mindset and culture that is open to innovation, none of these challenges are too daunting to achieve. Companies with such a mindset can start things from scratch and build on them without destroying the house they live in, which allows them to create S-curve upon S-curve of continued innovation.

Allison Nathan: So, what could the software industry look like in five or 10 years?

Sanjay Poonen: I expect more vertical industry software capabilities in industries like space, healthcare, material sciences, and robotics. Consider SpaceX's vision of data centers in space, where cooling isn't a problem. That endeavor will undoubtedly require software. The same could be said for biotech, medtech, and robotics, which will almost certainly increasingly attract top software talent that previously went to the Googles and Salesforces of the world, driving innovation that could help people live healthier and longer lives. So, my hope is that the software industry spreads its skillset into other vertical industries with positive implications for society and the world. And AI will undoubtedly be a part of every one of those industries.

Software stocks: earnings stability needed

Ben Snider argues that a rebound in the stocks most exposed to AI disruption risk likely won't come until earnings stabilize

The AI narrative has shifted. After three years of equity investor enthusiasm regarding the tailwinds from AI, recent equity market rotations reflect a growing focus on the risk of AI disruption. Software stocks have been at the center of this shift, falling by 18% from their YTD highs and now sitting roughly 26% below late 2025 levels. Stocks in a range of data-intensive industries beyond software, including insurance, media, and business services, have also declined sharply in recent weeks. While the moves have been swift and severe, we think a rebound in the stocks most exposed to AI disruption risk will take time, as disproving the disruption narrative won't be easy or quick.

Re-rating: a decomposition

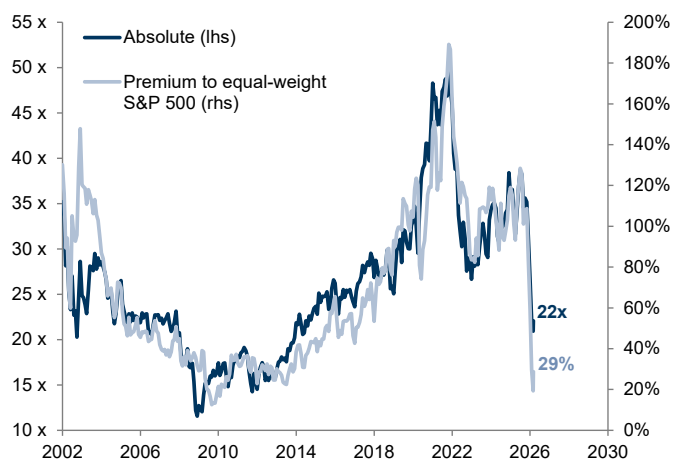
After entering 2026 at extremely elevated valuations, growth expectations, and weights in investor portfolios, the software industry has sharply re-rated to reflect declining expectations for future earnings growth as AI disruption risk grows.

The industry's forward P/E multiple has fallen from roughly 35x in late 2025—consistent with 2028 consensus revenue growth in the 15-20% range, over twice the roughly 6% growth expected for the median S&P 500 stock and the highest expected growth in at least two decades—to around 22x currently, the low end of the multiple's recent range and the lowest absolute level since 2014, consistent with 5-10% earnings growth.

Given the high starting point of valuations, software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff, and its 9% S&P 500 weight is roughly double its weight a decade ago.

Software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff...

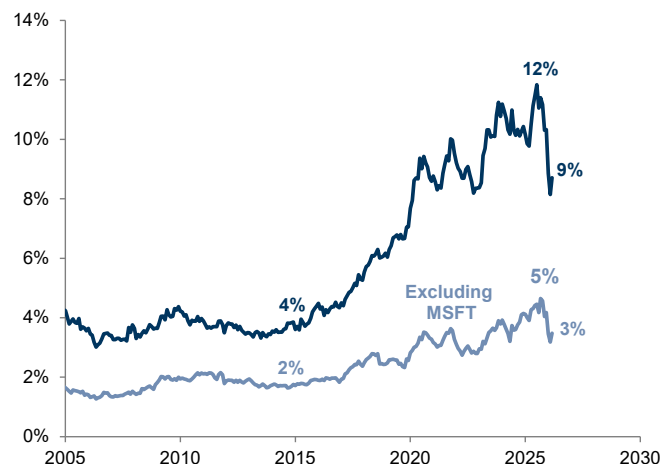
Software forward P/E, X (lhs), % (rhs)



Source: FactSet, Goldman Sachs GIR.

...with its weight in the S&P 500 roughly double that of a decade ago

Software weight in the S&P 500, %



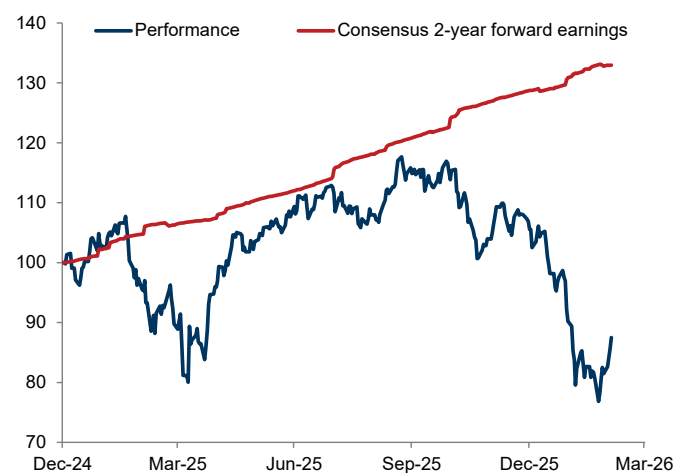
Source: FactSet, Goldman Sachs GIR.

In addition to lower growth expectations, the recent re-rating reflects rising uncertainty around the long-term trajectory of the industry and the “terminal value” of stocks previously viewed as long-duration growth assets. In other words, share prices now reflect not just slower expected growth, but also doubts about the durability of business models as AI disruption raises questions about pricing power and competitive barriers.

The impact of this long-term uncertainty can be seen in the juxtaposition of recent share price volatility and 4Q25 earnings results. In contrast with falling share prices, recent earnings results have reflected strong profitability—including double-digit earnings growth last quarter, with profit margins for the industry sitting at record highs and roughly twice those of the rest of the market—and have driven upward revisions to near-term analyst earnings estimates.

The performance and earnings of software stocks have diverged

Indexed performance and earnings of software stocks



Source: FactSet, Goldman Sachs GIR.

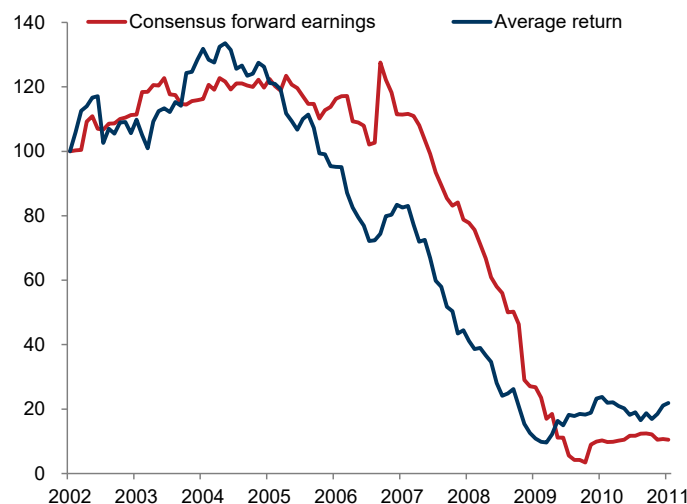
Lessons from history: don't expect a V-shaped rebound

Historical episodes of disruption risk suggest that share prices in affected industries will likely stabilize only when earnings stabilize, and that these periods of uncertainty can last for years. In the early 2000s, US newspaper stocks fell by an average of around 95% over the course of five years, and the group bottomed out only slightly ahead of the stabilization in

forward earnings estimates. Tobacco stocks in the late 1990s experienced disruption risk driven by litigation, rather than technology, but the pattern was similar. In that episode, share prices declined by over 50% in 1997-1998 and troughed only as the litigation settlement reduced uncertainty and the near-term earnings outlook stabilized. Given the rapid pace of AI advancement today, disproving the narrative of disruption risk is difficult when near-term earnings outlooks in the vulnerable industry have barely begun to weaken, much less stabilized.

US newspaper stock prices stabilized only when earnings stabilized

Indexed avg return and consensus forward earnings of US newspaper stocks



Source: FactSet, Goldman Sachs GIR.

For investors, the lesson is not to expect a V-shaped rebound in the parts of the equity market most affected by recent disruption fears. For these industries, near-term earnings can be an important signal of business resilience, but in many cases will be insufficient to disprove long-term downside risk. For the group to re-rate higher, investors will likely require more clarity on AI's eventual business impact, which will take time, or a substantially lower starting point for valuations and ownership. This may be an easier lift for the parts of the software and business services industries with particularly resilient business moats, but those moats, too, will need to be proven.

Of course, in any broad-based selloff there will likely be some proverbial "babies thrown out with the bathwater" that represent value opportunities for stock pickers. In the current episode, the key is the speed with which companies can disprove disruption risk.

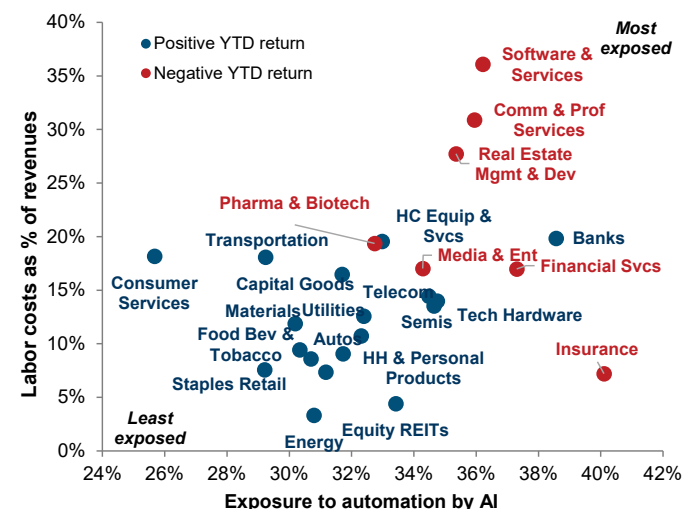
Avoiding AI disruption risk: look to "real world" assets

Another consequence of the recent focus on disruption risk is a rotation of investor portfolios toward "real world" assets with perceived insulation from AI disruption risk. Recent outperformers in this vein include a handful of cyclical industries like industrials and materials that have also enjoyed tailwinds from an improving economic growth backdrop. GS forecasts for above-trend economic growth have lent fundamental support to cyclicals' recent strength, and we entered 2026 recommending investors own many of these stocks. However, the tailwind from cyclical economic acceleration also highlights the broader equity market vulnerability if the growth backdrop proves less robust than

expected or industry-specific AI disruption fears spread to broader concerns about labor market disruption. This risk helps explain the recent outperformance of more defensive "AI-insulated" sectors, including consumer staples and biopharma, and we think the appropriate strategy is to balance cyclical exposures with some relatively defensive equity positions.

The recent focus on AI disruption risk has triggered a rotation toward "real world" assets

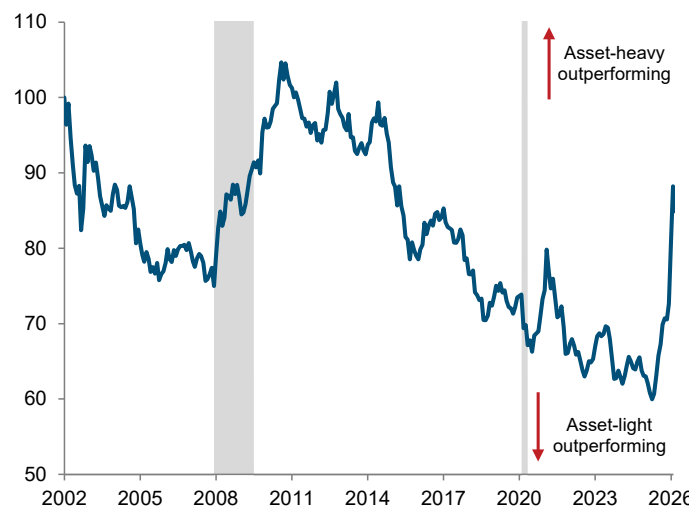
Russell 1000 industry median



Source: Revelio, company filings, Goldman Sachs GIR.

Increased investor desire for "AI insulation" has led asset-heavy stocks to sharply outperform

Asset-heavy vs. asset-light indexed performance



Note: Grey bars indicate recessions.

Source: Goldman Sachs GIR.

Ben Snider, Chief US Equity Strategist

Email: ben.snider@gs.com
Tel: 212-357-1744

Goldman Sachs & Co. LLC

Managing AI risk in portfolios

Christian Mueller-Glissmann discusses the why and how of managing AI risk in portfolios amid growing AI disruption fears

Investor concerns about AI risks for the global technology sector have grown amid ongoing worries about hyperscalers' massive AI capex spend and, most recently, rising fears that new AI tools will disrupt tech incumbents, especially legacy software firms. Such concerns matter because global portfolios are highly exposed to the technology sector, especially within US equities, where software stocks have a relatively large weight, with substantial exposures to technology and software specifically also a feature of private markets.

But the fate of the tech sector also matters more broadly given how closely tied structural cycles in US equity valuations and outperformance have been to the performance of tech stocks. To achieve more balance between innovation and inflation risks within portfolios, we think investors may need to continue rotating from capital-light to capital-heavy businesses that AI won't easily disrupt, though the aggressive rotations we've already seen warrant being more selective.

A tech-dependent equity market...

What happens to the tech sector has broader implications for the equity market for several reasons. The sector has historically been a key driver of structural cycles for US equities, with major waves of technological innovation often significantly impacting the broader market. During the 1960s PC boom, 1990s Internet boom, and the post-Global Financial Crisis (GFC) ascent of the Magnificent 7, tech's share of total market cap rose sharply, and significant market corrections often followed these periods. Today, the TMT sector once again commands the largest weight within US equities, comparable to Tech Bubble levels, but the sector's contribution to earnings is now substantially greater compared to then.

Tech's weight in the US equity market has risen sharply during major waves of technological innovation

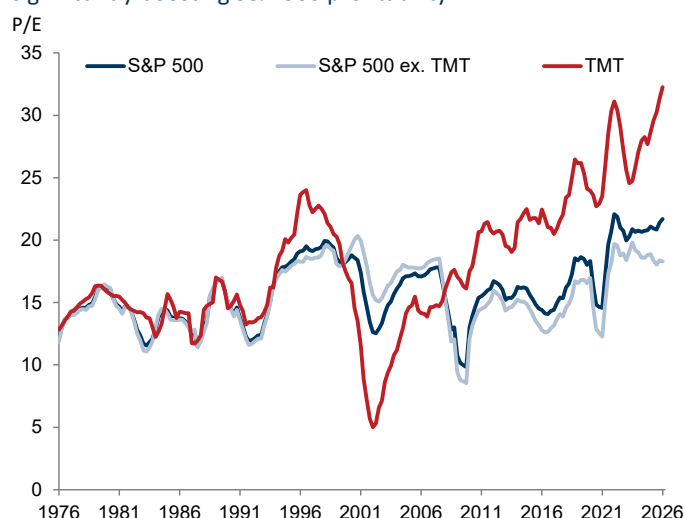
Tech market cap and earnings weight within US equities, %



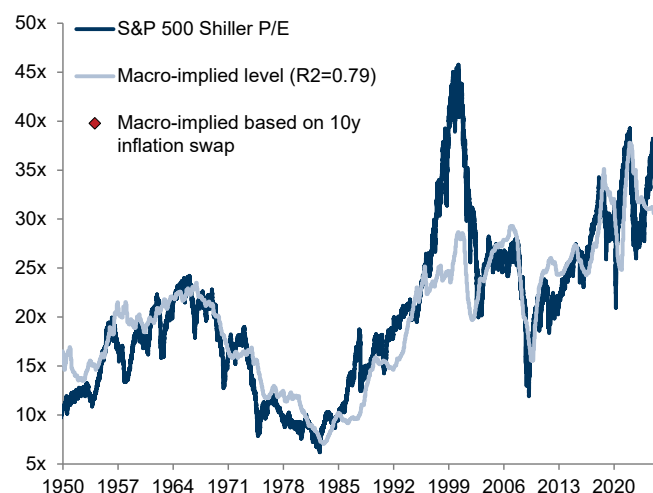
Tech has also played a key role in driving structurally higher S&P 500 profitability and valuations. Since the GFC, the return on equity (ROE) across the TMT sector has nearly doubled,

significantly lifting the S&P 500's overall profitability. This trend accelerated during the pandemic as technology adoption surged. Higher profitability has, in turn, supported higher equity valuations. Indeed, the rise in the S&P 500's Shiller P/E over the past two decades reflects two interconnected elements: improved corporate profitability—driven by the growing weight of the highly profitable, capital-light US tech sector—and persistently low and stable inflation. And the strong performance of large cap tech stocks has driven a large amount of the US equity outperformance since the GFC.

The ROE for the TMT sector has nearly doubled since the GFC, significantly boosting S&P 500 profitability



High profitability, coupled with low inflation expectations, have fueled a rise in US equity valuations



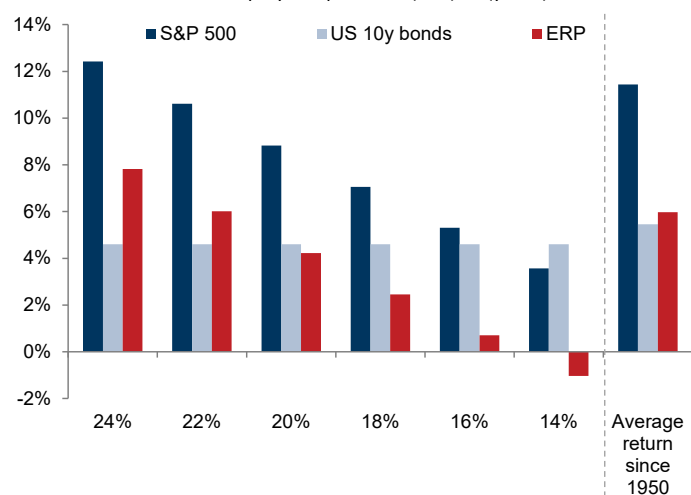
...meets AI disruption fears

Investors have extrapolated the TMT sector's high ROE, even as these firms pay out relatively little to shareholders, essentially betting that major tech conglomerates will maintain their profitability as they make substantial reinvestments. However, last year anxieties around AI capex began to impact some of the largest spenders in the sector, particularly the hyperscalers. Beyond concerns around potential future AI revenues, the big spenders are also becoming more capital-heavy, which might weigh on ROE in the medium term.

More recently, rising concerns about AI disruption have weighed on parts of the TMT sector, especially software stocks, with investors increasingly worried that the tech sector may end up disrupting itself. Advances in AI-powered task automation, for example, could undermine the pricing power of some of the most profitable large-cap software firms. While AI adoption may also boost other businesses' ROEs, a decline in the profitability of the market's largest and most lucrative segment could adversely affect the S&P 500's overall ROE and dampen long-term return expectations, which currently rely on the compounding of those ROEs. And markets have been pricing negative AI disruption faster than potential tailwinds from AI adoption.

Lower corporate profitability could portend lower long-term equity returns and less attractive equity risk premia

S&P 500 ROE in 10 years scenarios, % (x-axis) vs. annualized nominal returns for stocks and bonds and equity risk premium (ERP), % (y-axis)*



*See this [note](#) for more detail about the long-term return calculations.

Source: Haver Analytics, Goldman Sachs GIR.

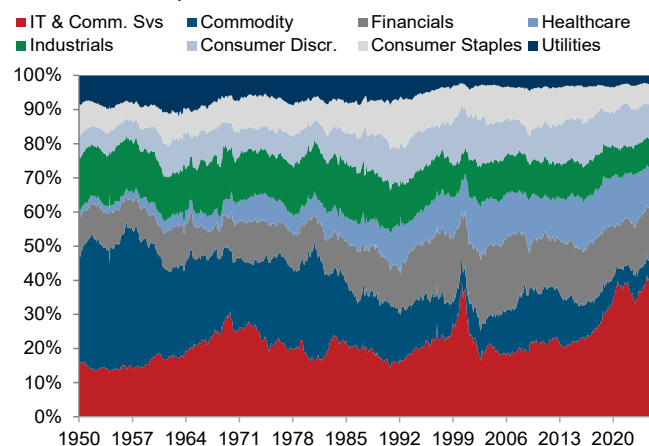
Balancing innovation and inflation risks

We expect an AI-driven rise in productivity to eventually support corporate profitability. But concerns about AI disruption may continue to be priced faster than the benefits of AI adoption. On top of that, the AI application firms driving disruption are mostly private companies and, as such, not easily investable, which makes it difficult to diversify AI disruption risk. Investors may therefore demand more risk premium for at-risk businesses, which has already driven a rotation and value convergence between capital-light growth sectors like TMT and capital-heavy value sectors like telecoms, industrials, and utilities that AI is unlikely to disrupt.

Owing to a boost from innovation, the TMT sector has become a much larger part of the S&P 500 since the GFC, while capital-heavy sectors have underperformed, partly owing to a lack of inflation. As investors continue to worry about AI disruption, this may lead them to demand a higher equity risk premium, which could result in some de-rating of equities vs. bonds, with bonds potentially also benefitting from disinflation due to AI disruption.

The TMT sector holds a dominant position within the US equity market, with capital-heavy sectors holding a relatively small one

Share of S&P 500 by sector, %



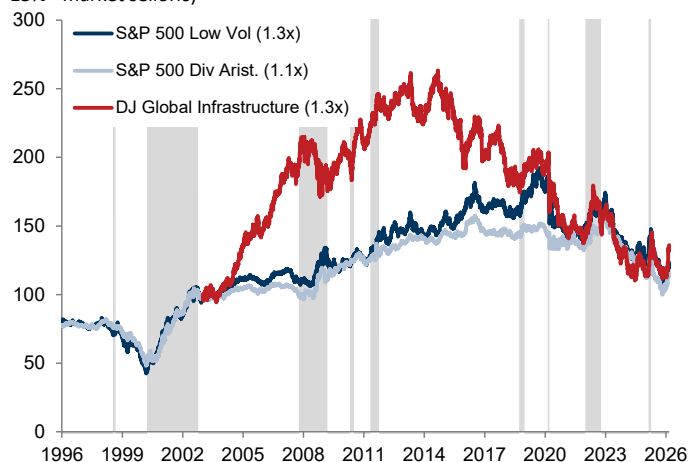
Source: Bloomberg, Datastream, Goldman Sachs GIR.

To manage innovation and inflation risks within portfolios, investors may need to continue rotating further from growth to value and capital-light to capital-heavy businesses, both in equity and credit. However, after the aggressive rotations since the beginning of the year, we favor being more selective. Within equities, we continue to see diversification benefits from defensive styles such as low volatility stocks, dividend aristocrats, and infrastructure, which materially outperformed on a risk-adjusted basis during the Tech bubble burst.

We also expect more dispersion as AI will likely continue to produce more winners and losers. This should support strategies that buy individual stock and sell index options and long/short hedge funds in both equity and credit. And we continue to view geographical diversification as an attractive strategy. TMT exposure is typically lower in non-US equity markets and tends to be more hardware-focused, with such markets also offering more capital-heavy sector opportunities.

Defensive styles have lagged in recent years but outperformed over the long run

Total return relative performance vs. S&P 500 (grey bars indicate periods of 15%+ market selloffs)*



Source: Datastream, Bloomberg, Goldman Sachs GIR.

Christian Mueller-Glissmann, Head of Asset Allocation Research

Email: christian.mueller-glissmann@gs.com
Tel: 44-20-7774-1714

Goldman Sachs International

Software disruption & credit: no hard fall

Shamshad Ali argues that software loan losses owing to AI disruption fears are unlikely to catalyze a turn in the credit default cycle

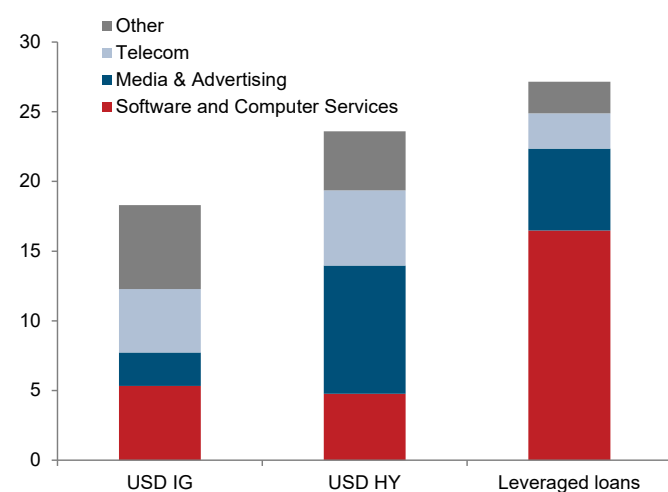
Recent headlines around AI-driven disruption—and the associated sharp drawdown in software stocks—have weighed on software-exposed credit, most notably in the broadly syndicated loan (BSL) market. As sectors like insurance, brokers, and other services exposed to AI disintermediation have also come under pressure as more new AI tools have emerged and investors worry about downstream implications, we believe the BSL market faces the greatest risk should disruption concerns translate into weaker earnings trajectories for traditional software companies, which would erode interest coverage ratios for these already levered borrowers. Given that software accounts for a meaningful share of the BSL market, some degree of index-level stress would be difficult to avoid if fundamentals continue to deteriorate. That said, the benign macro backdrop and lower funding costs should help limit the risk of a turn in the credit default cycle.

A more exposed BSL market

Software accounts for only a small slice of the USD IG and HY bond indices (roughly 5%) but a meaningfully larger share of the BSL market (16%). While many IG software issuers are large incumbents (including the AI hyperscalers), software-exposed issuers in the HY bond and leveraged loan markets have underperformed broader benchmarks as investors have repriced disruption risk. HY software has shifted from trading tighter than the average spread level of the overall HY index to roughly double index spreads, while secondary prices for software loans have fallen to the lowest levels since April 2025.

Software accounts for only a small slice of IG and HY but a meaningfully larger share of the BSL market

TMT subindex exposure in USD credit, %



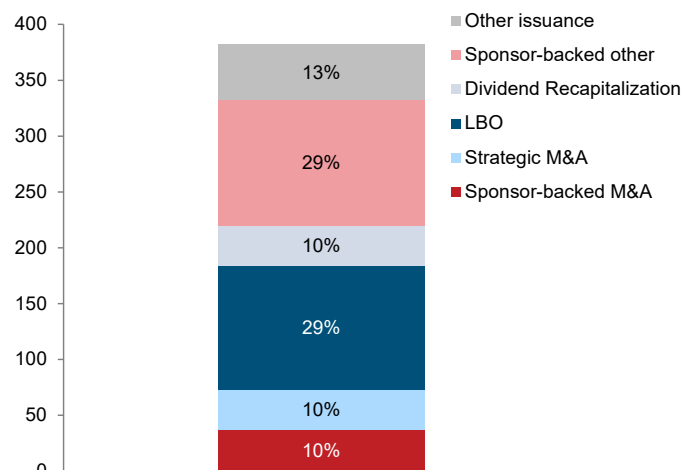
Source: Bloomberg, Pitchbook LCD, Goldman Sachs GIR.

A defining feature of software-related loans is the linkage to sponsor-backed (typically private equity) activity. Since 2021, nearly 40% of technology leveraged loan supply has been tied to sponsor-driven M&A and LBO financing. Another 10% of issuance stemmed from dividend recapitalization—firms taking on new debt to pay dividends—as sponsors sought liquidity amid a less favorable exit environment. The upshot is that

underwriting is no longer just about borrower fundamentals; investors also need to carefully consider sponsor incentives, time horizon, and sponsors' willingness to support a firm's capital structure through difficult periods as the industry navigates a changing paradigm.

80% of tech loan issuance since 2021 has been sponsor-backed

Share of total gross Computer and Electronics loan issuance since 2021 by category, \$bn



Source: Pitchbook LCD, Goldman Sachs GIR.

Some impairment likely, but not a turn in the default cycle

The potential persistence of AI disruption pressures would likely most clearly manifest via downgrades and lower secondary prices in the software-heavy portion of the loan index. Particularly important to watch would be the implications of such shifts for the CLO community, the main buyer of leveraged loans. CLOs have built-in rules or covenants designed to ensure that the overall diversification and quality of the underlying loans in their portfolios remain within acceptable limits (concentration, over-collateralization and collateral-quality tests). The downgrading of too many loans to a CCC rating (BSL CLOs typically have a limit of 7.5% for CCC exposure) could trigger various penalties. As a result, prices for these loans could move beyond fundamentals as the investor base potentially shifts from CCC-averse CLO buyers.

That said, bifurcation in loans isn't new. Even in periods when lower-quality loans cheapen, the broader index can remain resilient when higher-quality borrowers repeatedly reprice their loans at lower spreads and new loan supply is muted, as has been the case this year. Overall, while some impairment in software-exposed loans is hard to avoid, we don't think this impairment is sufficient to catalyze a credit default cycle, for two reasons. First, the macro backdrop remains benign, and history suggests that defaults typically coincide with rather than predict the broader cycle. Second, lower interest rates and improved funding conditions should continue to provide incremental relief to floating-rate borrowers.

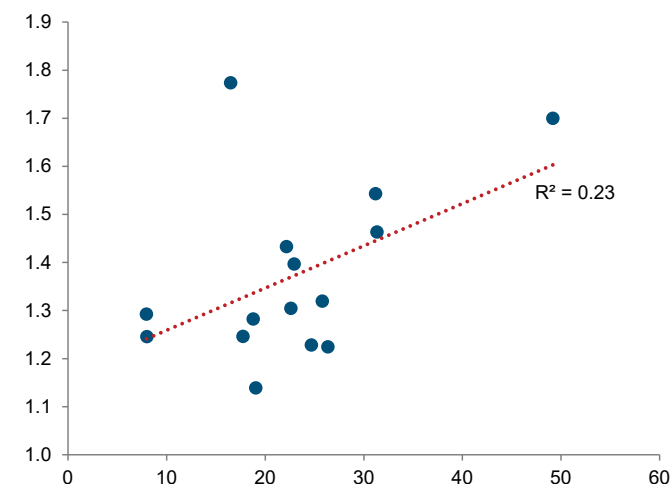
Pain in direct lending, but de-leveraging risk looks contained

Beyond the BSL market, direct lending—and by extension business development companies (BDCs) that invest in small and mid-sized, often private, firms—has been in the crosshairs given meaningful allocations to software. Indeed, our equity analysts estimate that software portfolio companies account for

22% of BDC loans (see pg. 20). Consistent with this, BDC equities have tended to trade at wider discounts to net asset value (NAV) while BDC credit spreads generally have widened in accordance with software exposure.

BDC credit spread widening correlates with software exposure

Software exposure as of 3Q25, % (x-axis) vs. on-the-run BDC senior unsecured bond spread change, ratio relative to Jan. 23 level (y-axis)



Source: Bloomberg, Goldman Sachs GIR.

We are closely monitoring signs of stress in BDC portfolios and have long argued that some writedown in asset valuations is likely. Recent liquidations of assets in BDC portfolios have also fueled speculation about asset health despite the loans selling at high, near-par, dollar prices. While full transparency isn't available on the sales, our equity analysts continue to think that credit trends in BDC portfolio data seem benign for now, but near-term portfolio data will remain under scrutiny.

While the rising rate of redemptions in BDCs is worth monitoring, we continue to think that the probability of a liquidity event or forced de-leveraging in direct lending is low owing to generally moderate leverage and substantial dry powder that can re-enter the complex and provide support, especially at more attractive valuations.

More dispersion from the next leg of the AI trade

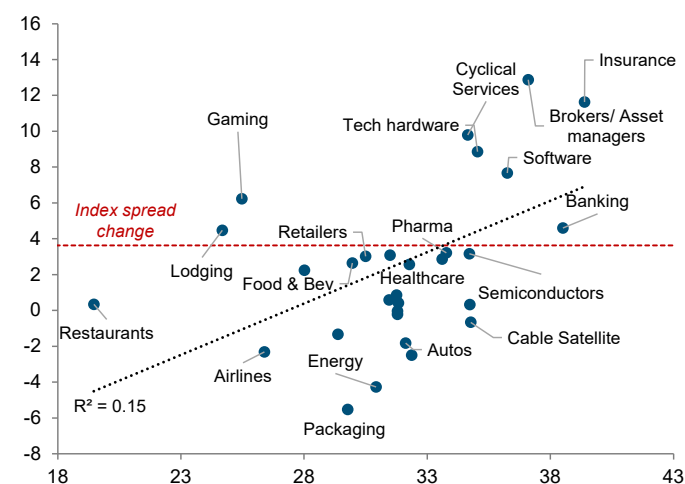
While AI disruption fears remain centered on software, they have now spread to several other sectors, including insurance, brokers, and broader financial services (see pgs. 21-23), as the foundational model providers continue to roll out new domain-specific products. Amid this, IG and HY sector performance has become correlated with exposure to AI automation, mirroring the broader risk asset rotation toward "real economy" sectors that are less exposed at the expense of white-collar services that could face more disruption risk.

Looking ahead, we think this higher dispersion dynamic will persist, for two reasons. First, continued improvements in AI capabilities will likely impact more industries. Second, dispersion within industries should rise as company outcomes diverge based on the execution of AI adoption strategies. This dispersion will make it more difficult for investors to calibrate fair value for affected sectors, and issuer selection will become increasingly important. But we still see limited scope for serious impairment, especially for higher quality firms. Beyond their incumbent status and low starting points for leverage, we

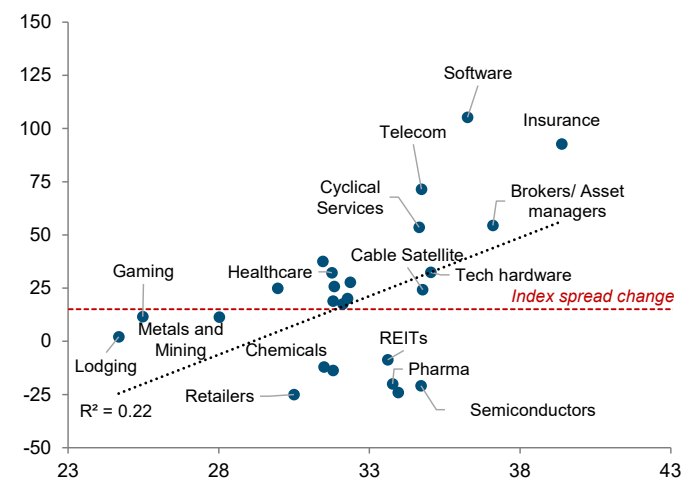
would note that the concerns around terminal values that have fueled much of the equity volatility are less salient for fixed duration credit assets, which are concerned with repayment of interest and principal.

Credit sector performance has become correlated with AI exposure

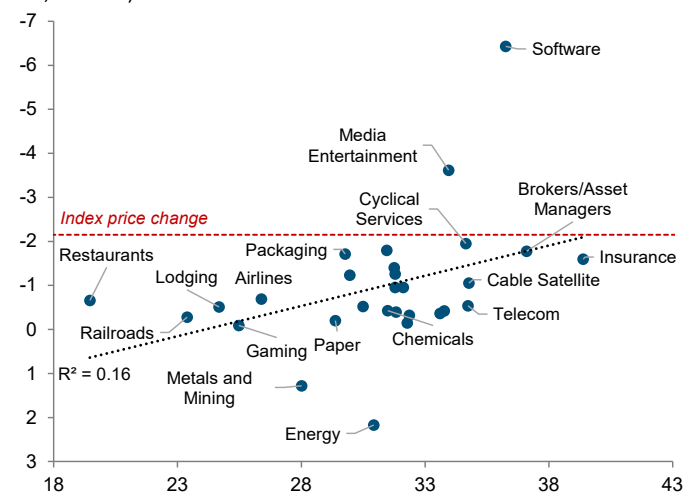
AI automation exposure, % (x-axis) vs. IG spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. HY spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. leveraged loan price change YTD, \$ (y-axis, inverted)



Source: Revelio, Bloomberg, Goldman GIR.

Shamshad Ali, Senior Credit Strategist

Email: shamshad.ali@gs.com
Tel: 212-902-6712

Goldman Sachs & Co. LLC

Alts managers: limited AI risk so far

Alexander Blostein and Michael Vinci argue that Alternative Asset Managers have limited exposure to AI-related software disruptions

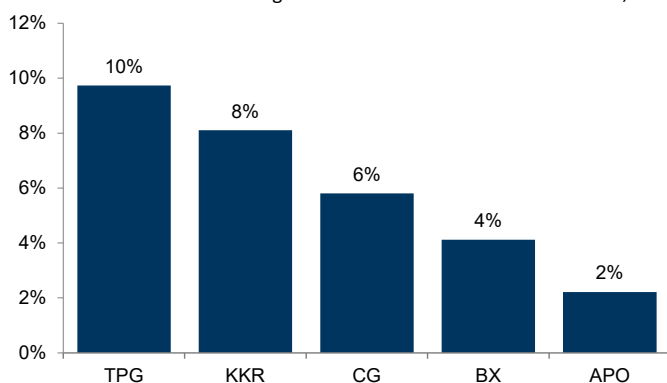
Concerns about AI-driven disruption risk have once again weighed on Alternative Asset Managers' stocks, which have declined ~27% YTD, bringing share prices and valuations below post-Liberation Day lows (NTM P/E net of stock-based compensation (SBC) at ~15X currently). This decline largely reflects concerns about software exposure across both private equity and private credit as well as fears that weaker investment performance could hamper growth. However, firmwide software exposure remains limited—averaging ~7% of total AUM—with software lending only modestly contributing to management fees, and managers have also been highly selective in their software investments, choosing those with more robust moats. While we think the recent selloff is overdone and continue to see significant upside across the sector, we remain cautious around names with outsized Retail Credit exposure given accelerating redemption requests.

Modest software exposure

Overall, we view Alternative Asset Managers' exposure to software investments as manageable. On the private equity (PE) front, while software has accounted for ~25% of industry-wide deployment volumes in recent years, exposure at Public Alternative Managers is more modest, with software activity among PE firms averaging only ~6% of total management fees. These estimates may even prove conservative as actual net exposure could be lower after managers monetized many investments in prior years.

PE software investments average ~6% of total management fees

2025 estimated firmwide management fees from Info Tech investments, %



Source: Preqin, Goldman Sachs GIR.

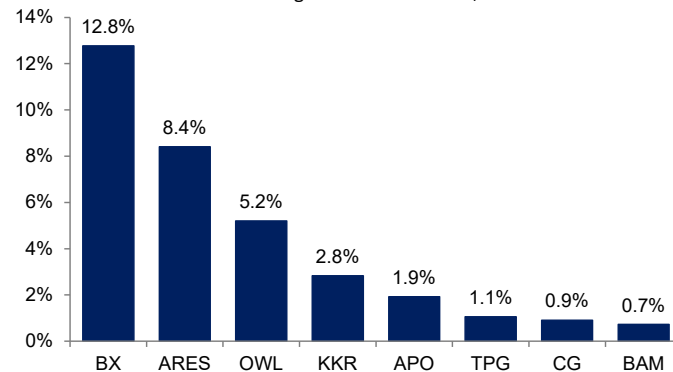
On the credit front, across BDCs (where public data is available), we estimate that software loans represent ~22% of gross loans at an industry level, with a similar share among Public Alternative Managers—~20% of Direct Lending FPAUM on average. This translates to around 25% of Direct Lending base management fees.

But software lending is a relatively small portion of firmwide management fees when accounting for the broader spectrum of Alt asset classes that GPs manage. We estimate that Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average. Even in

the most adverse scenario of a significant markdown in software loans, our stress analysis (5%-15% mark-to-market) suggests the effect on firmwide base fees would remain in the low single-digits. And, importantly, these software loans are mostly senior secured and short duration, with the underlying companies generally showing strong EBITDA and cash flow.

Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average

Estimated software direct lending firmwide base fees, %



Source: Company data, Goldman Sachs GIR.

Selective software investments

We also note that not all AI-related risk is created equal. Software holdings of most managers skew toward vertical, mission-critical businesses with proprietary data and platforms. These businesses are more likely to benefit from AI as opposed to the more disruption-prone horizontal software applications. AI-related risk has also become a central focus of investment processes, with one senior leader recently noting that AI is now among the first and most critical topics examined when bringing an investment before the investment committee. Evaluating AI-related risk has become a deeply embedded practice within GP investment committee deliberations and portfolio management frameworks, not just a reaction to recent market concerns.

We remain selective looking forward

While heightened market volatility and greater uncertainty related to AI disruption risks could dampen near-term PE exit activity, our conversations with management teams have given us more confidence around drawdown fund portfolio construction. That said, we continue to believe that sentiment/headlines will weigh on retail/wealth flows. While the group's earnings streams remain highly durable, we see a slower pace of retail fundraising/accelerating Credit redemption trends potentially weighing on growth. Despite the disruption, deployment outlooks are constructive, and for Opportunistic Credit managers, most see opportunity in the current backdrop. We continue to think the group's earnings base and management fee growth are considerably more durable than current valuations suggest and see significant upside across the space.

Alexander Blostein, US Financials Equity Research Analyst

Email: alexander.blostein@gs.com
Tel: 212-357-9976

Goldman Sachs & Co. LLC

Michael Vinci, US Financials Equity Research Analyst

Email: michael.vinci@gs.com
Tel: 212-357-8239

Goldman Sachs & Co. LLC

AI disruption: equity sector impacts

What types of companies in your coverage universe could be disrupted by and, conversely, benefit from new AI tools, and how much of this is already priced into stocks?

Consumer Internet

Eric Sheridan and team, GS Equity Research

- AI tools are currently having a democratizing effect on product development and innovation within the consumer internet sector, presenting disintermediation risk while also driving efficiency gains. Within our TMT coverage universe, digital advertising and gaming are the most exposed subsectors to AI disintermediation risk and have also seen the most advanced adoption of AI tools—a trend that the recent declines in several gaming and advertising stocks reflects.
- Digital advertising, in particular, has advanced the furthest in AI-driven product development and end-user adoption (outside of cloud computing). Companies within the subsector are currently integrating AI into four key workflows: campaign planning and strategy, creative development, media buying and optimization, and CRM and personalization. GOOGL's Performance Max and META's Advantage+ have been exceptionally successful in integrating multiple AI use cases into a single product. These AI-powered tools have the potential to both "abstract away"/automate many advertising services as well as improve the overall performance and efficiency of ad spend. This should result in shifting profit dynamics including: 1) accelerating the shift of ad spend from traditional to digital channels (an estimated \$170bn addressable opportunity), 2) automating ad creative generation (~\$114bn), 3) consolidating ad tech "intermediaries" (~\$25bn), and 4) disrupting the traditional role of ad agencies (~\$161bn).
- Beyond online advertising, the specter of AI disruption has caused outsized volatility across many other subsectors—including media & advertising, online marketplaces, online dating, travel, online education, and online gaming & gambling—which we expect to persist. Incumbent consumer internet platforms are playing a mix of "offense" (launching AI-driven consumer applications that enhance their existing offerings and revenue) and "defense" (developing AI capabilities to counter competitive applications that could be disruptive to existing legacy applications) to counter AI disruption risk. Several companies are also partnering directly with leading AI platforms to drive direct traffic and monetization. Examples include ChatGPT in-app integrations, agentic commerce, in-app chatbot placement, and data/content licensing for model training.

US Business and Information Services

George Tong, GS Equity Research

- Concerns over AI-driven disintermediation have sparked a sharp de-rating of the Information Services sector since mid-2025, with the sector experiencing a particularly notable valuation step-down in early Feb 2026. The median NTM P/E multiple in our Info Services coverage has fallen from 32x in June 2025 to 20x currently as investors worry that free or cheaper AI tools could diminish or eliminate demand for existing Info Services offerings. However, we believe some companies in the sector are better positioned to withstand AI disruption than others.
- Companies with a sustainable moat against AI are those with proprietary data, entrenched workflows with customers, scale and brand advantages, and/or a regulatory moat, and have made meaningful AI investments. These firms are not only insulated from AI risks but also are positioned to benefit from AI integration via product upgrades, accelerated pricing increases, increased cross-selling, higher retention rates, and new business growth. From a stock perspective, the multiples of many of these firms have recently contracted, making them attractive buying opportunities. The market may be overestimating the AI-related hit to these firms' organic revenue growth (with some estimates in the mid-to-high single digits) and we believe earnings at these firms remain stable and are not in a negative revision cycle. Upcoming earnings results that demonstrate beat-and-raise patterns and sustained momentum in organic revenue growth should act as a catalyst for valuation upside, in our view.
- By contrast, firms most vulnerable to AI disruption are those that have limited proprietary content, weak integration with client workflows, limited scale or brand advantages, no regulatory moat, and/or haven't invested enough in AI themselves. They also face potential headwinds from lower demand for white collar staffing roles due to AI automation.

North American Insurance

Robert Cox and team, GS Equity Research

- Recent reports that OpenAI has incorporated insurance services directly into ChatGPT sparked a significant selloff in insurance providers. We believe Private and Commercial (P&C) insurance companies will need to adapt to keep pace with these emerging AI tools. Certain personal lines & micro business insurance brokers are the most vulnerable to disruption as competitors increasingly leverage AI to simplify the purchase of lower complexity products like auto/home insurance and disrupt various consulting services. Shifts in distribution could also impact auto/home insurers' customer acquisition models, with auto insurers also facing long-term TAM concerns from increasing autonomous vehicle (AV) penetration. Commercial (re)insurers are well positioned as trusted, efficient, regulated capital allocators, especially if they quickly move to embrace AI efficiencies. Larger insurance brokers are also well positioned due to the data and structural moats associated with complex consulting and brokerage, however certain small account broking and less complex consulting could be exposed to AI disruption.
- We think both insurers and brokers can achieve significant efficiency gains through AI automation. Larger market cap firms with big technology budgets will likely benefit first. Insurers are using AI to automate claims processing and speed up policy handling, while brokers aim to automate routine processes/services. First movers are likely to see near- to medium- term margin/growth benefits, although whether these benefits will ultimately lead to lower premiums for customers remains an open question.
- The market has primarily focused on the risk of revenue disruptions from AI rather than on potential efficiency gains. While valuations for P&C insurers are under pressure from concerns about a softer pricing cycle, making it more difficult to isolate AI impacts, the market is clearly penalizing insurance brokers for potential revenue risks—a move which we think is overdone. Auto insurers have faced multiple contraction from AVs and could see revenue disruption from AI-based distribution platforms, but we think these headwinds will ultimately be manageable for underwriters with historically strong direct underwriting platforms. By contrast, commercial (re)insurers are in some cases benefiting from reported AI efficiency gains—CB has reported a ~20% headcount reduction over 3-4 years—and are seen as relatively safe from revenue disruption.

CRE Services

Julien Blouin, GS Equity Research

- We believe AI disruption concerns across CRE services businesses are significantly overstated. The physical nature of the CRE ecosystem provides meaningful insulation versus digitally-native industries. Historical precedents—listings platforms, online property management, virtual tours, automated valuation models—show that CRE technology adoption unfolds over years and decades, not quarters, and most importantly these innovations improved transparency, efficiency, and productivity but did not structurally disintermediate CRE brokerage and service providers.
- Transactional brokerage businesses face the most theoretical AI exposure, particularly for standardized mid-market mandates where AI can drive underwriting automation, OM generation, lease abstraction, and buyer matching. Yet wholesale disintermediation remains unlikely given the relationship-driven, localized, negotiated, and liability-bearing nature of CRE transactions (particularly for larger deals). Facilities, property, project, and investment management businesses are even more insulated as they are operationally complex, and often require on-premises workers/technicians. We believe AI will enhance functions like scheduling, cost control, and predictive maintenance rather than eliminate demand.
- The valuation opportunity is compelling. CRE services companies are benefiting from a transaction recovery that remains mid-cycle, with volumes still 44% below 2021–2022 peaks and 20% below 2017–2019 levels. Yet sector multiples have contracted sharply on AI fears, reverting to 2023 discounts versus the S&P Equal Weight—levels last seen when transactions were under extreme pressure. We view this selloff as an attractive entry point into a sector with powerful structural and cyclical tailwinds, supporting double-digit earnings CAGRs through 2028.

European Financials

Oliver Carruthers and team, GS Equity Research

- Within our European Financials coverage universe, AI disruption concerns have centered on market structure firms such as LSEG and DB1, though their AI exposure varies. LSEG's stock has declined since mid-2025 amid concerns that AI-powered analysis could disrupt its data business.
- More recently, focus has also turned to private markets and their exposure to software-based service business models. A de-rating of public software comparables off the back of AI-disruption concerns could lead to negative mark-to-market adjustments for portfolio companies and limit near-term monetization. That said, many private equity firms are benefitting from rising AI demand, including through digital infrastructure investments. And traditional asset managers have largely been shielded from AI disruptions with managers actively leveraging AI to boost efficiency and alleviate margin pressure as well as enhance efficiency across client servicing, risk management, and investment intelligence.
- AI disruption poses greater risk for private credit managers as fund performance is more sensitive to dispersion and, unlike buyout funds, private credit managers cannot offset losses with investments into "winners". We've seen significant underperformance among Alternative Asset Managers, with private credit-focused firms most impacted. Wealth managers and investment platforms also experienced an AI-related selloff after Altruist launched an AI-powered tax planning tool, which raised concerns that automated advice tools could increase competition and compress fees. However, we note that these tools also have the potential to enhance adviser productivity and increase efficiency for firms able to strategically integrate AI into their offerings.

European transport, infrastructure, and construction

Patrick Creuset and team, GS Equity Research

- When it comes to AI risk, the main debates in the European transport, infrastructure, and construction sector center around 1) whether the incumbents will be disrupted by AI technological progress, and 2) the extent to which AI productivity gains will be deflated away.
- We view the AI disruption risk as relatively low for the sector as incumbents benefit from large physical networks (e.g., sizable blue-collar workforces as well as numerous physical gateways and warehouses), have higher value-add proposition (e.g., end-to-end supply chain management), and benefit from network effects—all of which will be difficult for new AI-native entrants to match. Incumbents also stand to unlock significant advantages from integrating AI technology into their existing operations, further supporting their resilience.
- We are also less concerned about the prospect of AI-related productivity gains being deflated away. While some productivity gains will likely be passed on to customers—as has historically been the case within the industry—we expect the magnitude of productivity gains to be meaningfully larger than the potential yield erosion.
- Asset-light logistics companies are particularly well-positioned to benefit from integrating AI tools into their ongoing digitization push. The sector's SG&A operations have historically involved many repetitive, documentation-intensive tasks, including quoting, booking, carrier payment tracking, and customs clearance. We believe that further digitization and AI deployment will eventually enable industry players to automate some of these workflows and improve efficiency.

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

Top of Mind archive



Issue 145
Americas First
January 26, 2026



Issue 130
How investable is Europe?
August 1, 2024



Special Issue
2025: 4 themes in charts
December 15, 2025



Issue 129
Gen AI: too much spend, too little benefit?
June 25, 2024



Issue 144
The US-China tech race
December 4, 2025



Issue 128
Central bank divergence room to run?
May 21, 2024



Issue 143
AI: in a bubble?
October 22, 2025



Issue 127
Weighing the GLP-1 market
April 12, 2024



Issue 142
Data (un)reliability
September 11, 2025



Issue 126
Global transit & trade: in rough waters
March 12, 2024



Issue 141
Stablecoin summer
August 19, 2025



Issue 125
2024: The year of elections
February 1, 2024



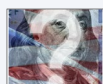
Issue 140
US fiscal worries: Is this time different?
June 12, 2025



Issue 124
Middle East risks
December 5, 2023



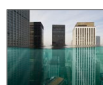
Issue 139
Fed independence: how concerning?
May 15, 2025



Issue 123
US outperformance: at a turning point?
October 30, 2023



Issue 138
Tariff-induced recession risk
April 17, 2025



Issue 122
Commercial real estate risks
October 9, 2023



Issue 137
Europe's shifting security landscape
March 13, 2025



Issue 121
Corporate credit concerns
August 10, 2023



Issue 136
Trump tariffs: mostly talk, or big action?
February 26, 2025



Issue 120
Generative AI: hype, or truly transformative?
July 5, 2023



Issue 135
High bond yields: here to stay?
February 5, 2025



Issue 119
Daunting debt limit dynamics
May 22, 2023



Special Issue
2024: 4 themes in charts
December 18, 2024



Issue 118
US-China: more decoupling ahead?
May 1, 2023



Issue 134
Will China's policy stimulus be enough?
December 11, 2024



Issue 117
All about bank(panic)s
April 3, 2023



Issue 133
Market concentration: how big a worry?
November 25, 2024



Issue 116
(Japanese) Bonds, Bonds, Bonds
February 23, 2023



Issue 132
Post-election economic policies
October 21, 2024



Issue 115
The Bigger Worry: Growth or Inflation?
January 27, 2023



Issue 131
Is the Fed behind the curve?
September 3, 2024



Special Issue
2022: 3 themes in charts
December 15, 2022

Source of photos: www.gettyimages.com.

Disclosure Appendix

Reg AC

We, Allison Nathan, Jenny Grimberg, Ashley Rhodes, Christian Mueller-Glissmann, CFA, Ben Snider and Shamshad Ali, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

We, Eric Sheridan, Alexander Blostein, CFA, Gabriela Borges, CFA, Patrick Creuset, Julien Blouin, Oliver Carruthers, CFA, Robert Cox, Adam Hotchkiss, Matthew Martino, George K. Tong, CFA and Michael Vinci, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Rating and pricing disclosures

Aon Plc (Buy, \$338.77), Ares Management Corp. (Buy, \$117.04), Arthur J. Gallagher & Co. (Buy, \$229.65), CBRE Group (Buy, \$141.19), CVC Capital Partners (Buy, €12.01), Cloudflare (Buy, \$192.31), CrowdStrike (Buy, \$426.16), Cushman & Wakefield Ltd. (Buy, \$13.25), DoorDash Inc. (Buy, \$183.48), HubSpot Inc. (Buy, \$291.47), Jones Lang LaSalle Inc. (Buy, \$309.36), KKR & Co. (Buy, \$95.59), London Stock Exchange Group (Buy, 8,744p), Microsoft Corp. (Buy, \$410.68), Newmark Group (Buy, \$14.71), Palo Alto Networks (Buy, \$163.16), Ryan Specialty Group (Buy, \$39.34), Salesforce Inc. (Buy, \$201.39), ServiceNow Inc. (Buy, \$120.38), Snowflake Inc. (Buy, \$177.45), Spotify Technology S.A. (Buy, \$552.12), StepStone Group (Buy, \$47.62), TPG Inc. (Buy, \$44.96) and Willis Towers Watson (Buy, \$300.52)

Logo disclosure

Please note: Third party brands used in this report are the property of their respective owners, and are used here for informational purposes only. The use of such brands should not be viewed as an endorsement, affiliation or sponsorship by or for Goldman Sachs or any of its products/services.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A

component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	61%	46%

As of January 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,055 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Distribution of ratings: See the distribution of ratings disclosure above.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure

of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, The Investment Trusts Association, Japan, and Japan Investment Advisers Association. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific

disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company neither has a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such.

The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report

broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.